

BOUNDARY STRUCTURES, SPECIALIZATION, AND EXACT CERTIFICATES TECHNICAL SUPPLEMENT TO INTRINSIC RECONSTRUCTION OF WEBS

QIAN QI

ABSTRACT. This supplement contains the complete primary-boundary, relative-specialization and exact-coordinate theory accompanying the reconstruction article. It also gives the ambient component-pencil classification and its scheme-theoretic residual involution. These results are not hypotheses of the smooth-locus inverse theorem. References with ordinary numerical section numbers refer to the reconstruction article; sections of this supplement are prefixed S.

CONTENTS

S1.	Ordinary powers and relative flat flags	2
S2.	The intrinsic nilpotent-depth filtration	3
S3.	Reye geometry, infinitesimal moduli, and the novelty boundary	4
S3.1.	The classical nine-dimensional quotient	5
S3.2.	What the finite ambiguity means	6
S3.3.	Comparison with nonreduced reconstruction theory	7
S3.4.	Ballico's failure-locus framework	7
S3.5.	Residue of the main theorem	8
S3.6.	Comparison of the inverse data	8
S3.7.	Role of the relative primary theorem	9
S3.8.	Extent of the primary tables	9
S4.	The ambient component pencil	9
S4.1.	Secants, tangent fibres, and flag lines	9
S4.2.	Algebraic rank strata and Fitting ramification	11
S4.3.	An alternative coordinate character calculation	13
S5.	Boundary structure and family theorems	13
S5.1.	Sharp global laws	13
S5.2.	The intrinsic graded boundary packet	14
S5.3.	Primary specialization	15
S5.4.	The simple-contact locus	15
S5.5.	Variation and finite ambiguity	16
S6.	Universal determinant completion and the intrinsic boundary atlas	16
S7.	Fixed tensors across projection corank two	22
S8.	Residual presentations and the rank-two contact boundary	25
S9.	The stratified nilpotent algebra at projection corank two	29
S10.	Proofs for the intrinsic boundary atlas	34
S10.1.	Universal determinant completion	34
S10.2.	The depth bound and graded multiplication	35
S10.3.	The first layer at projection corank two	35

S10.4.	Rank-drop depth and the finite coefficient atlas	38
S10.5.	Projection corank three	39
S10.6.	Projection corank four	40
S10.7.	Specialization and associated supports	42
S11.	Relative primary signatures and specialization laws	42
S11.1.	Stabilizer slices and generic Groebner bases	42
S11.2.	Geometric primary signatures for the nine mixed-kernel types	43
S11.3.	Pure-block rank drop at projection corank two	44
S11.4.	Relative assassins and base change for the colon tower	46
S11.5.	Explicit orbit closures and graded specialization	53
S12.	Sharp global depth and geometric boundary laws	54
S12.1.	A normalized determinant projection	55
S12.2.	Two global minimal supports and a resolution	56
S12.3.	A cross-corank family with explicit components	57
S12.4.	An explicit infinite family of proper quotient maps	59
S13.	Deformation to the normal cone: standard mechanism and the computed special fibre	60
S14.	Exact separating pencils	61
S15.	Weighted determinantal flags and the classical order profile	62
S16.	An explicit system and exact certificate specifications	63
	References	65

S1. ORDINARY POWERS AND RELATIVE FLAT FLAGS

Corollary S1.1 (Ordinary powers). *For $n \geq 1$, write $D^{[n]} = V(\mathcal{I}_D^n)$ and $E^{[n]} = V(\mathcal{I}_E^n)$. Then*

$$\mathcal{I}_D^n = \mathcal{I}_\Delta^n \mathcal{I}_E^n = \mathcal{I}_\Delta^n \cap \mathcal{I}_E^{2n} \quad (\text{S1.1})$$

is an irredundant primary decomposition on U . With $\mathcal{T}_n = \text{tor}_{\eta_\Delta}(\mathcal{O}_{D^{[n]}})$,

$$0 \longrightarrow \mathcal{T}_n = \mathcal{O}_{E^{[n]}}(-n\Delta) \longrightarrow \mathcal{O}_{D^{[n]}} \longrightarrow \mathcal{O}_{n\Delta} \longrightarrow 0. \quad (\text{S1.2})$$

Its annihilator defines $E^{[n]}$. The generic multiplicity on Δ is n , the nilradical index along E is $2n$, and the length of $\mathcal{T}_n$ at η_E is $n(n+1)/2$.

Proof. In the local ring let $P = (t, f)$. The associated graded ring of P is a polynomial ring in two variables over $\mathcal{O}_E$, and the initial form of t is one of those variables. Hence $(P^{2n} : t^n) = P^n$, proving $(t^n) \cap P^{2n} = t^n P^n$. The same graded argument proves that these powers are primary. The witnesses t^n and f^{2n} at η_E prove irredundancy. Localizing at (t) contracts the ideal to (t^n) , so the generic torsion is $(t^n)/t^n P^n$, and cancellation gives its annihilator P^n . The class of t has index $2n$ at E , since t^{2n} belongs to the ideal but t^{2n-1} does not. At the generic point of Δ , f is a unit. Finally $\mathcal{O}_{U, \eta_E}/(t, f)^n$ has the $n(n+1)/2$ monomials of total degree less than n as successive graded basis elements. $\square$

Lemma S1.2 (Flat flags and arbitrary base change). *Let $U \rightarrow T$ be smooth, and let $E \subset \Delta \subset U$ be relative smooth closed subschemes, each inclusion a Cartier divisor. Put $L = \mathcal{I}_\Delta$ and $P = \mathcal{I}_E$. Then $\mathcal{O}_U/P^n$ and $\mathcal{O}_U/L^n$ are flat over T . For $D_n = V(L^n P^n)$ the sequence*

$$0 \rightarrow L^n \otimes \mathcal{O}_{E^{[n]}} \rightarrow \mathcal{O}_{D_n} \rightarrow \mathcal{O}_U/L^n \rightarrow 0 \quad (\text{S1.3})$$

is exact and remains exact after any base change. Its base-changed terms are the indicated ordinary powers and products. In particular D_n is flat over T .

Proof. The immersion $E \subset U$ is regular of codimension two and has locally free conormal module. The conormal filtration of $\mathcal{O}_U/P^n$ has quotients $\text{Sym}^j(P/P^2)$ on E , for $0 \leq j < n$; these are base-flat. Induction proves flatness. The analogous principal filtration proves flatness of $\mathcal{O}_U/L^n$. Multiplication by a local equation t^n of L^n identifies the kernel in (S1.3) with $L^n \otimes \mathcal{O}_{E^{[n]}}$, since t is a non-zero-divisor. Both outer terms are base-flat, so the middle term is base-flat. For a base algebra $A \rightarrow A'$, the only possible left obstruction in tensoring this short exact sequence is $\text{Tor}_1^A(\mathcal{O}_U/L^n, A')$, which is zero. The sequences defining L , P , and their powers stay exact for the same reason. Thus their images after base change are the actual ideals generated by the base-changed equations, and multiplication gives their stated products. No assertion about specialization of selected primary components is used. $\square$

Corollary S1.3. *Over G_e° , the schemes E , Δ , and U form the smooth relative flag of Lemma S1.2. All $D^{[n]}$ and $E^{[n]}$ are flat, and (S1.2) commutes with arbitrary base change. Each complex fibre has (S1.1).*

Proof. Use the universal quotient $\mathcal{S}$ and the graph bundles (4.2)–(4.3) over the smooth family of Proposition 3.3. The Fitting elimination works over the parameter ring, so it gives $t(t, f)$ relatively. Apply the lemma, and apply Corollary S1.1 separately on each fibre for the primary assertion. $\square$

S2. THE INTRINSIC NILPOTENT-DEPTH FILTRATION

We isolate the functorial mechanism that unifies the two local models. Let Z be any noetherian scheme and let $\mathcal{N}_Z$ denote the nilradical of $\mathcal{O}_Z$. Since Z is noetherian, $\mathcal{N}_Z$ is a coherent nilpotent ideal. For $j \geq 1$ define

$$\mathcal{Z}_j(Z) = V\left(\text{Ann}_{\mathcal{O}_Z}(\mathcal{N}_Z^j)\right). \quad (\text{S2.1})$$

Lemma S2.1 (Functoriality). *Every isomorphism $f : Z \rightarrow Z'$ carries $\mathcal{Z}_j(Z)$ isomorphically to $\mathcal{Z}_j(Z')$ for every j . The construction commutes with restriction to open subsets.*

Proof. An isomorphism carries the nilradical to the nilradical, hence carries its j -th power and its annihilator ideal to the corresponding objects. Both formation of powers and annihilators commute with localization. $\square$

Apply this to $Z = \widehat{D}_R$. On the intrinsic open obtained by restricting to the smooth locus of Z_{red} , Proposition 4.1 identifies Z with D_R . Theorem 4.2 gives the local ring

$$A = \mathcal{O}/(t(t, f))$$

with nilradical generated by the class of t . Therefore

$$\text{Ann}_A((t)) = (t, f)/(t(t, f)),$$

so (S2.1) recovers E_R scheme-theoretically.

Proposition S2.2 (Depth one is the polarized ramification support). *On the smooth-reduction locus of $\widehat{D}_R$,*

$$\mathcal{Z}_1(\widehat{D}_R) = E_R, \quad \widehat{\mathcal{N}}_R|_{E_R} = \mathcal{O}_{E_R}(-\widehat{\Delta}_R).$$

Consequently the pair

$$\left(\mathcal{Z}_1(\widehat{D}_R), \omega_{\mathcal{Z}_1(\widehat{D}_R)} \otimes (\widehat{\mathcal{N}}_R|_{\mathcal{Z}_1(\widehat{D}_R)})^{-9}\right)$$

is intrinsic when $e = 4$, and its graded section ring is the fourth Veronese of the polarized quartic section ring.

Proof. Only the last assertion is new notation. The annihilator calculation above is Theorem 4.2; the line-bundle identity and section-ring reconstruction are Proposition 7.2 and Theorem 1.2. $\square$

The point of (S2.1) is that it makes no reference to a projection chart. The closed schemes are automatically nested:

$$\widehat{\mathcal{N}}_R^{j+1} \subset \widehat{\mathcal{N}}_R^j \implies \text{Ann}(\widehat{\mathcal{N}}_R^j) \subset \text{Ann}(\widehat{\mathcal{N}}_R^{j+1}), \quad \mathcal{Z}_{j+1} \subset \mathcal{Z}_j.$$

The new issue is therefore not formal functoriality but the scheme-theoretic computation of the layers as one crosses projection rank. Theorem S9.1 solves this globally: on the whole Grassmannian the residual colon

$$\mathcal{J}_R^{\text{res}} = (\mathcal{I}_{\widehat{D}_R} : \mathcal{I}_{\widehat{\Delta}_R})$$

determines every annihilator and every associated-graded piece, including all projection coranks and rank-drop loci. The remaining task is geometric identification of those canonical colon schemes.

On the first higher-corank chart the next power is nonzero. The calculation in §S7 gives, on the simple-contact open,

$$\widehat{\mathcal{N}}_R^3 = 0, \quad \text{Ann}(\widehat{\mathcal{N}}_R^2) = \mathcal{I}_{\Sigma_R} / \mathcal{I}_{\widehat{D}_R}.$$

Section S8 removes the simple-root assumption. Section S9 computes the two layers simultaneously across the corank-one/corank-two boundary. On the mixed-regular neighbourhood,

$$\text{gr}_{\widehat{\mathcal{N}}_R} \mathcal{O}_{\widehat{D}_R} = \mathcal{O}_{\widehat{\Delta}_R} \oplus (\mathcal{L} \otimes \mathcal{O}_{\mathcal{C}_R}) \oplus (\mathcal{L}^{\otimes 2} \otimes \mathcal{O}_{\Sigma_R}),$$

where $\mathcal{C}_R$ is the unsplit contact cone and $\mathcal{L}$ is the Schubert conormal line. In particular $\mathcal{Z}_1$ specializes from the ramification support to its canonical contact closure with the vertex thickening forced by the Fitting algebra, while $\mathcal{Z}_2$ is the rank-two vertex. On the first tensor-rank boundary of the mixed map, $\mathcal{Z}_2$ acquires a canonical length-three nonreduced structure (Theorem S9.5).

Thus

$$\mathcal{Z}_1 \rightsquigarrow \mathcal{Z}_2$$

is not merely a comparison of two calculations on disjoint strata: it is the associated-graded specialization of one intrinsic nilpotent algebra.

Remark S2.3 (Relation with canonical filtrations on multiple structures). Nilpotent and multiple structures have long been studied through canonical filtrations and associated graded modules; ribbons are the multiplicity-two prototype, and primitive multiple schemes carry a canonical chain of lower multiplicities. See [4, 15, 11]. The construction (S2.1) is deliberately more elementary: it is available for the present Fitting schemes without assuming that they are primitive, locally complete-intersection multiple structures, or Cohen–Macaulay along every higher-corank stratum. The additional content here is that consecutive annihilator layers and their associated graded modules have a specific geometric meaning forced by multiplication: the first carries the polarized ramification K3, the second detects the rank-two Schubert support, and their specialization through contact collisions is controlled by the unsplit contact Cartier ideal rather than by a choice of primary roots.

S3. REYE GEOMETRY, INFINITESIMAL MODULI, AND THE NOVELTY BOUNDARY

We separate the classical moduli of webs and Reye congruences, the exact differential calculation for the polarized-K3 map, and the two nonreduced reconstruction theorems. Generic web reconstruction uses the deepest normal cone; it is not a consequence of the classical nine-dimensional count.

S3.1. The classical nine-dimensional quotient. For $e = 4$, a relation space $R \subset \text{Sym}^2 V$ is a web of quadrics. The bilinear incidence surface Z_R of §9 is a K3 surface with its fixed-point-free factor-exchange involution; its quotient is the classical Reye Enriques surface associated with the web. This construction and the resulting congruence of lines are classical; see [7, 1, 10, 13, 9, 16].

Arrondo–Sols compute that the Hilbert scheme of Reye congruences is smooth of dimension 24 and rational, and describe it by an open Grassmannian parameter of the corresponding data. After quotienting by the 15-dimensional projective coordinate group, the expected moduli dimension is therefore nine. In characteristic zero the nodal Enriques locus is the classical nine-dimensional divisor in the ten-dimensional Enriques moduli space; the modern theorem [16] identifies every classical nodal Enriques surface as a Reye congruence in their Fano-polarized sense. The distinction from a regular classical web is recalled below.

Thus the equality

$$24 - 15 = 9$$

is not a novelty claim of this paper.

Proposition S3.1 (The classical tangent quotient). *At a general regular web with finite projective stabilizer, the tangent space to the Reye moduli locus is the quotient of the 24-dimensional web tangent space by the 15-dimensional infinitesimal $\text{PGL}(V)$ -orbit:*

$$T_{[R]}\mathcal{M}_{\text{Reye}} \simeq T_R \text{Gr}(4, \text{Sym}^2 V) / \mathfrak{pgl}(V), \quad \dim T_{[R]}\mathcal{M}_{\text{Reye}} = 9.$$

The differential of the polarized-K3 map of Theorem S5.3 factors through this quotient.

Proof. The first statement is the tangent-space form of the smooth 24-dimensional Hilbert calculation of Arrondo–Sols on the regular web locus, followed by the infinitesimal quotient by projective coordinates. The stabilizer is finite at a general point, so the orbit tangent has dimension 15.

The construction $R \mapsto Z_R \simeq Y_R$ is equivariant for $\text{PGL}(V)$. Passing from an embedded quartic to its polarized isomorphism class kills the infinitesimal projective orbit; hence the differential factors through the displayed quotient. $\square$

Proposition S3.2 (Web parameter and Reye Hilbert point). *Let*

$$\mathcal{G} = \text{Gr}(4, \text{Sym}^2 V)$$

and let $\mathcal{G}^{\text{reg}}$ be the open on which the classical Reye construction has the expected Hilbert polynomial. The assignment of a web to its Reye congruence is a $\text{PGL}(V)$ -equivariant morphism

$$\Phi : \mathcal{G}^{\text{reg}} \longrightarrow \text{Hilb}_{\text{Reye}}.$$

Under the Grassmannian parameterization of the regular Reye locus in Arrondo–Sols, Φ identifies a dense open of $\mathcal{G}^{\text{reg}}$ with the corresponding 24-dimensional Hilbert component. In particular, at a general regular web,

$$d\Phi_R : \text{Hom}(R, \text{Sym}^2 V/R) \xrightarrow{\sim} T_{[\text{Reye}(R)]} \text{Hilb}_{\text{Reye}}.$$

After quotienting by the infinitesimal projective orbit this gives the tangent quotient in Proposition S3.1.

Proof. Over $\mathcal{G}$ the universal rank-four subbundle of $\text{Sym}^2 V$ gives the universal web. The incidence equations defining the Reye congruence are polynomial in the universal Pluecker coordinates. Restrict to the open on which their Hilbert polynomial is the regular Reye polynomial. Flatness there and the Hilbert functor give the morphism Φ ; equivariance is immediate from the construction.

Arrondo–Sols describe the regular Reye Hilbert component by the open Grassmannian parameter of the defining four-dimensional system of quadrics. With the present convention this parameter is

precisely a dense open of $\mathrm{Gr}(4, \mathrm{Sym}^2 V)$ (duality only changes whether one writes the system or its annihilator). Their smoothness and dimension-24 statement therefore identifies the tangent space of the Hilbert point with the tangent space of this Grassmannian. The latter is $\mathrm{Hom}(R, \mathrm{Sym}^2 V/R)$, proving the displayed isomorphism. Since the construction is $\mathrm{PGL}(V)$ -equivariant, the orbit tangent maps to the orbit tangent and the quotient gives Proposition S3.1. $\square$

The role of Lemma 8.1 is now precise. It is not used to discover the number nine. Its exact 25×25 integral minor proves that the particular polarized K3 attached to our multiplication problem does not collapse any of the 24 embedded web directions at the explicit rational point. After projective quotient, it proves rank nine for the map to polarized K3 moduli. Since maximal rank is open, the induced map from the classical nine-dimensional Reye deformation space to our polarized K3 image is generically unramified. This is the conceptual tangent statement behind Theorem S5.3; the determinant is an exact arithmetic witness for it.

S3.2. What the finite ambiguity means. The preceding paragraph locates the Torelli ambiguity of the polarized-K3 invariant alone. The failure scheme first reconstructs the polarized K3 $(Y_R, \mathcal{O}_{Y_R}(1))$. A relation system additionally supplies the free involution coming from factor exchange on the bilinear incidence model and hence an Enriques quotient with its Reye data. Classical Reye geometry then has substantial rigidity. In particular, Dolgachev–Reider prove that for a Reye polarization on an Enriques surface the stable rank-two bundle with c_1 equal to that polarization and $c_2 = 3$ is the Reye bundle and is unique up to isomorphism; it is the restriction of the universal quotient bundle from $G(2, 4)$.

Consequently the generic finite fibre in Theorem S5.3 should not be read as an uncontrolled family of quadratic tensors. Once a compatible Enriques quotient and Reye polarization are fixed, the Grassmannian Reye bundle is rigid in the preceding sense. The remaining finite packet is concentrated in the finite choices of compatible Reye/Enriques data over the recovered polarized K3, together with the passage back to the web. For the map defined by the polarized K3 alone, we prove finiteness but do not assert a numerical degree or degree one; that enumeration is distinct from generic reconstruction using the full failure scheme. The additional normal-cone invariant of Theorem 5.1 separates this packet without identifying its K3-only degree.

There is a second useful localization. On the mixed-regular projection-corank-two locus, the unsplit contact quartic of §S8 is the restriction of the already recovered quartic Y_R to the pencil of hyperplanes containing H . Thus the generic corank-two contact cover itself is determined by the polarized K3. The following theorem turns the finite Torelli packet into a precise finite-cover statement and identifies its monodromy.

Theorem S3.3 (Generic Torelli packet as a finite étale monodromy cover). *Let $\mathcal{W}$ be the irreducible nine-dimensional coarse quotient of the regular web locus by projective equivalence on the finite-stabilizer open, and let $\mathcal{K}$ be the nine-dimensional polarized-K3 image of Theorem S5.3. There are nonempty opens*

$$\mathcal{W}^\circ \subset \mathcal{W}, \quad \mathcal{K}^\circ \subset \mathcal{K}$$

such that

$$\tau : \mathcal{W}^\circ \longrightarrow \mathcal{K}^\circ$$

is finite étale of a constant degree

$$d_{\mathrm{Tor}} = [\mathbb{C}(\mathcal{W}) : \mathbb{C}(\mathcal{K})].$$

Its geometric monodromy is transitive on the d_{Tor} points of a fibre. Equivalently, if L is the normal closure of $\mathbb{C}(\mathcal{W})/\mathbb{C}(\mathcal{K})$, the monodromy group is the permutation image of

$$\mathrm{Gal}(L/\mathbb{C}(\mathcal{K}))$$

on the embeddings of $\mathbb{C}(\mathcal{W})$ in L .

The degree-four contact incidence and its discriminant are pulled back from the polarized quartic on $\mathcal{K}^\circ$. Hence they are constant on a Torelli fibre and do not account for d_{Tor} ; any further separation of the finite packet must use web-dependent structure beyond the polarized quartic. Theorem 5.1 supplies such structure in the deepest nonreduced normal cone, and Corollary 5.17 proves separation on a nonempty open of this finite cover.

Proof. The source $\mathcal{W}$ is irreducible because it is obtained from a nonempty open of the irreducible Grassmannian $\text{Gr}(4, \text{Sym}^2 V)$ by the projective-coordinate quotient on the finite-stabilizer locus. Theorem S5.3 says that the dominant map $\tau : \mathcal{W} \dashrightarrow \mathcal{K}$ is generically finite. Thus the induced extension of function fields is finite. In characteristic zero it is separable.

Normalize a nonempty affine open of $\mathcal{K}$ in $\mathbb{C}(\mathcal{W})$. After deleting the complement of the locus on which the original rational map is defined, and then deleting the branch and nonflat loci, one obtains a finite étale morphism from a nonempty open of $\mathcal{W}$. Its degree is the function-field degree displayed above. Connectedness of the generic source, or equivalently irreducibility of $\mathcal{W}$, gives transitivity of geometric monodromy. The description by the normal closure is the standard field-theoretic realization of that action.

The contact incidence is defined solely from the recovered embedded quartic $Y \subset \mathbb{P}^3$ by

$$(H, [\alpha]) \quad \text{with} \quad H \subset \ker \alpha,$$

and the discriminant is the discriminant of its restricted binary quartic. Both constructions commute with isomorphism of the polarized quartic. They therefore descend to $\mathcal{K}^\circ$ and pull back along τ , proving the last assertion. $\square$

S3.3. Comparison with nonreduced reconstruction theory. The use of nilpotent layers has classical antecedents. A ribbon is a square-zero double structure with an invertible conormal module, and Bayer–Eisenbud developed its intrinsic and canonical geometry [4]. Manolache surveys Cohen–Macaulay nilpotent structures and their filtrations [15]. Primitive multiple schemes carry a canonical filtration by lower multiplicities in the sense of Drézet [11].

Our local corank-one model

$$t(t, f) = (t) \cap (t, f)^2$$

indeed has a square-zero nilradical line on the embedded support, and the corank-two model has one further nonzero nilpotent power. We do not claim that the existence of a nilpotent filtration is new. The specific contribution is the following chain of implications, which does not occur in the cited multiple-structure theory:

$$\text{multiplication Fitting scheme} \implies (\mathcal{Z}_1, \widehat{\mathcal{N}}|_{\mathcal{Z}_1}) \implies \omega_{\mathcal{Z}_1} \otimes \widehat{\mathcal{N}}^{-9} \implies (Y_R, \mathcal{O}_{Y_R}(1)),$$

together with

$$\mathcal{Z}_2 = \text{rank-two Schubert support}$$

on the mixed-regular higher-corank locus. The ambient failure scheme is not assumed to be a primitive multiple scheme, and the filtration is defined by annihilators of powers of its intrinsic nilradical rather than by choosing an ambient thickening.

S3.4. Ballico’s failure-locus framework. The complete theorem text of Ballico’s 1993 article [2] has not been obtained for theorem-level comparison. We therefore do not infer either anticipation or non-anticipation from its title or metadata. A theorem-level comparison is possible with Ballico’s accessible 1996 continuation [3]. There the basic data are an integral projective variety X , a very ample line bundle L , the complete embedding by $H^0(X, L)$, and multiplication maps such as

$$H^0(L) \otimes H^0(L^k) \longrightarrow H^0(L^{k+1}).$$

The principal results propagate failure of quadratic or higher normality to suitable linear sections and finite subschemes; the parameter Grassmannian is a Grassmannian of projective linear sections.

The present construction has a different input and a different scheme-theoretic output. The fixed object is the finite local cube-zero algebra B_R ; a point of X_R is a unital subspace $\mathbb{C}1 \oplus K \subset B_R$; and the failure scheme is the zeroth Fitting scheme of

$$\mathrm{Sym}^m(\mathcal{O} \oplus \mathcal{K}) \longrightarrow B_R \otimes \mathcal{O}.$$

Its embedded associated supports, nilradical powers, and the polarized K3 reconstructed from their line bundles are part of the statement. Ballico's 1996 theorems are therefore an important precedent for organizing multiplication failure over a Grassmannian, but they do not by themselves supply the Fitting normal forms or the nilpotent-depth reconstruction proved here. This comparison is deliberately limited to the theorem text actually inspected; the separate 1993 comparison remains documentary work rather than a premise in a novelty claim.

S3.5. Residue of the main theorem. After these classical comparisons, the claims that remain specific to this paper are concise.

First, an abstract multiplication-failure scheme reconstructs the quartic polarization by the intrinsic line $\omega_{E_R} \otimes \mathcal{N}_R^{-9}$ and its section ring. Second, the same nonreduced scheme has a functorial annihilator filtration whose next nonzero stage recovers the first higher-corank Schubert support. Third, on the mixed-regular rank-two locus the unsplit contact ideal has a complete collision specialization law, the fifth-power vertex component remains genuinely primary, and the associated graded nilpotent algebra identifies both the contact cone and the rank-two Schubert support. The first decomposable mixed-kernel wall is also computed scheme-theoretically. Finally, Theorem S3.3 identifies the inverse problem for the K3 alone with a finite étale monodromy cover, while Theorem 5.1 proves that the actual nonreduced failure cone generically recovers the web and separates that packet. These statements are independent of whether the nine-dimensional Reye family itself is classical.

S3.6. Comparison of the inverse data. Classical inverse constructions already recover a web from suitable Reye data. In [16, Proposition 3.6], a Fano polarization H determines its Reye bundle, when that bundle exists. Theorem 3.7 of that work realizes the polarized Enriques surface in a Grassmannian. In characteristic different from two, Remark 3.8 uses the canonical K3 cover, its involution, and two exchanged line bundles L^+, L^- . A four-dimensional multiplication kernel defines symmetric $(1, 1)$ -equations and hence a web. Thus reconstruction from an Enriques polarization with its Reye structure is not asserted here as new. Nor is every nodal Enriques surface a regular classical Reye web: the distinction is explicit in [16, Remark 1.2 and Section 6].

These inputs differ from a single quartic polarization. The first-layer invariant in Theorem 1.2 recovers $(Y_R, \mathcal{O}_{Y_R}(1))$, without assuming the Reye involution or the second exchanged polarization as marked data. Theorem S3.3 therefore retains the function-field degree of this particular unmarked map; the classical reconstruction with additional data does not compute that degree. Theorem 5.1 starts from neither a marked Reye surface nor a specified web. It starts from the abstract nonreduced Fitting scheme, recovers an oriented deepest normal cone, and extracts two specified coefficient lines through Lemma 5.7. Lemma 5.10 proves that an abstract isomorphism acts on both lines through the same projective transformation. The exterior-support theorem then makes recombination unique on the entire smooth Jacobian locus, without an additional genericity restriction. This input and this extraction mechanism are the inverse assertions proved here; classical web-to-Reye and web-to-Jacobian constructions are used as comparisons, not rebranded as new results.

S3.7. Role of the relative primary theorem. The relative algebra supplies a simultaneous model for a bounded principal-colon tower. Generic freeness, flattening, associated-point technology, and effective descent are standard ingredients [27, 20, 21, 29]. Our use of them is a finite-diagram construction, not a new principle of generic freeness or primary existence. The outputs needed here are simultaneous formation of the colons and their quotients after base change, exhaustion of the geometric associated points, closed support packets on the original strata, and relative torsion modules realizing the packet lengths. The diagonal injections, their cokernels, and the regular-element witnesses are kept in the same diagram so that these outputs hold together. Individual flattening of prime supports would not imply this combination, and no theorem about canonical labelled primary decompositions is asserted. The bounded-principal-colon theorem is therefore a relative-algebra tool for the geometric results, rather than the basis for an independent foundational-priority claim. The determinant identities, calculated boundary schemes, and intrinsic inverse theorem carry the geometric content.

S3.8. Extent of the primary tables. The primary decompositions computed in full are the stated generic corank-two coefficient strata, their specified collisions, and the quotient-induced benchmark families. The first two minimal-support laws and the sharp index-five nilpotence theorem apply globally on G_4° . All their proofs and coefficient tables are retained. The seven-component two-parameter incidence calculation describes the specified pullback of an incidence resolution; it does not identify all embedded primary components of W_3 and W_4 at coranks three and four. The inverse theorem does not require such an atlas: it reads the degree-twelve maximal-minor space itself from the degree-sixteen ideal, before choosing a primary decomposition. Accordingly its hypothesis and conclusion are unchanged by this separation of the two questions.

S4. THE AMBIENT COMPONENT PENCIL

S4.1. Secants, tangent fibres, and flag lines. We describe the scheme-theoretic fibres of the two-component invariant, including those not covered by the rank-two certificate. This is a geometric description of the residual ambiguity of that invariant, not an assertion that every pair in such a fibre has isomorphic failure schemes. Put $X = \text{Gr}(4, W)$, and let $X^\times$ be the open where both projections are nonzero. Consider the morphism

$$\pi : X^\times \longrightarrow \mathbb{P}(E_{175}) \times \mathbb{P}(E_{35}), \quad [w] \longmapsto ([Pw], [Qw]).$$

For $[w] \in X^\times$, set $A = Pw$, $B = Qw$, and $H_w = \langle \ell_{F,w} : F \in \mathcal{I}_{\text{Pl},2} \rangle$. The rank $r_w = \dim H_w$ is independent of a representative.

Theorem S4.1 (Exceptional fibres of the component invariant). *The fibre of π through $[w]$, as a scheme, is the intersection of X with $\mathbb{P}(\mathbb{C}A \oplus \mathbb{C}B)$, with its two endpoints removed. In coordinates $[a : b]$ on this line its closed intersection with X has the following exhaustive forms.*

- (1) *If $r_w = 2$, its saturated ideal is $(b - a)$; it is the single reduced point $[w]$.*
- (2) *If $r_w = 1$, choose a generator $\ell = ca + db$ of H_w . Its saturated ideal is $((b - a)(ca + db))$. It is the length-two divisor $[1 : 1] + [d : -c]$, with multiplicity. The residual point equals $[w]$ exactly when $c + d = 0$; it is an excluded endpoint exactly when $cd = 0$. Otherwise it gives a second, distinct reduced point of the fibre of π .*
- (3) *If $r_w = 0$, the whole line lies in X , and the fibre is $\mathbf{G}_m$. Such a line consists precisely of the four-planes R in a fixed flag*

$$A_3 \subset R \subset B_5 \subset W, \quad \dim A_3 = 3, \quad \dim B_5 = 5.$$

Equivalently, the pure endpoints are decomposable and the two four-planes they represent intersect in dimension three.

On the locally closed reduced locus of case (2) with $cd(c+d) \neq 0$, passage to the residual point is a regular involution without fixed points. The differential of π has a nonzero kernel at $[w_R]$ precisely when

$$Qw_R \in \bigwedge^3 R \wedge W. \quad (\text{S4.1})$$

This is case (3), or the double-point subcase of case (2). Its kernel is then one-dimensional. In particular, a distinct secant pair does not force infinitesimal failure of reconstruction.

Proof. The ambient projection from $\mathbb{P}(E_{175} \oplus E_{35})$ to the product of projective spaces has fibre $\mathbb{P}(\mathbb{C}A \oplus \mathbb{C}B) \setminus \{[A], [B]\}$. Pullback along the closed immersion of X proves the first assertion scheme-theoretically. The Grassmannian ideal is generated by its quadratic Pluecker relations. Their restrictions generate $(b-a)H_w$. If H_w is the full two-dimensional space of linear forms, these forms have no common projective zero and the ideal sheaf is $(b-a)$. If H_w is one-dimensional, the nonzero quadratic $(b-a)\ell$ generates an already saturated principal ideal in $\mathbb{C}[a, b]$, giving its two zeros with their multiplicities. If $H_w = 0$, every Pluecker equation vanishes identically on the line. This proves the three scheme structures, including the endpoint and double-point distinctions.

We recall the flag description with its proof. Choose a point R_0 on a line in the Grassmannian and a complement to R_0 in W . On the Pluecker affine chart of R_0 , the line is parametrized by graphs of $tM : R_0 \rightarrow W/R_0$. Indeed the one-replacement Pluecker coordinates are its matrix entries and are linear in t . The two-replacement coordinates are the two-by-two minors of tM . They are also linear functions on the line, so their quadratic coefficients vanish. Thus $\text{rank } M = 1$, since the line is nonconstant. Its members share $\ker M$ of dimension three and lie in $R_0 + \text{im } M$ of dimension five; closing the affine line gives the asserted flag line. Conversely the four-planes between such a flag form a Pluecker line by wedging a fixed basis of A_3 with a varying vector of B_5/A_3 . Applied to the endpoints, this also proves the equivalent formulation.

On a chart where one coefficient of a nonzero $\ell_{F,w}$ is invertible, its coefficient pair gives the regular map

$$[w] \mapsto [dA - cB].$$

Rank one makes different choices proportional, so these maps glue. The hypotheses $cd(c+d) \neq 0$ exclude endpoints and the double point. The residual point has the same two projective components; its line of recombination is the same line, whose intersection with X has exactly the two distinct reduced points. Repeating the construction returns $[w]$. This proves regularity and the involution assertion on the stated reduced stratum.

Finally the ambient projection has a one-dimensional vertical tangent at $[w]$, represented by B modulo $\mathbb{C}w$. The affine tangent space to the Pluecker cone at w_R is $\bigwedge^3 R \wedge W$, as follows by differentiating a wedge of four basis vectors of R . This proves (S4.1). Equivalently set $a = 1$, $b = 1 + t$; each restricted equation has derivative $\ell_{F,w}(1, 1)$. All these derivatives vanish exactly when $H_w = 0$, or when its generator vanishes at $[1 : 1]$. The rank-two case cannot have this property. The vertical tangent is nonzero because A, B are independent. This proves the last claims. $\square$

Corollary S4.2 (Candidate bounds without a genericity certificate). *On $X^\times$, away from the flag-line locus, the two-component invariant has fibres of length at most two. Failure of the rank-two certificate can represent either a double point, a genuine second candidate, or a discarded pure endpoint. Only the flag-line case allows a positive-dimensional fibre. For an abstract failure-scheme isomorphism these are necessary candidate alternatives after one common projective change of coordinates.*

Proof. Remove the two endpoints from the three closed intersections in the theorem, and apply Lemma 5.10 for the last assertion. No converse from equality of component pairs to isomorphism of failure schemes is used. $\square$

The theorem describes every fibre configuration on $X^\times$. Theorem 5.15 below determines their incidence with the smooth Jacobian open: only the rank-two case occurs. No ambient irreducible-component or codimension classification is implicit in the fibre classification itself. It separates positive-dimensional flag ambiguity from tangent degeneracy and from two distinct candidates; these distinctions are invisible in the single rank-failure condition.

Example S4.3 (A nonempty flag-line stratum). In the monomial notation of Appendix S14 put

$$A = b_1 \wedge b_2 \wedge b_3 \wedge b_5, \quad B = b_1 \wedge b_2 \wedge b_3 \wedge b_4.$$

These are highest-weight vectors of weights $(4, 3, 1, 0)$ and $(5, 1, 1, 1)$; hence $A \in E_{175}$, $B \in E_{35}$. Every $aA + bB$ is the decomposable vector $b_1 \wedge b_2 \wedge b_3 \wedge (ab_5 + bb_4)$. Thus the flag-line stratum is nonempty in $X^\times$, and this example supplies an entire $\mathbf{G}_m$ -fibre. It is not an example on G_4° : the first three quadrics have the common factor x_1 , and the fourth quadric has a zero on $x_1 = 0$.

S4.2. Algebraic rank strata and Fitting ramification. The pencil classification admits a global scheme-theoretic formulation. On $X^\times$, let $\mathcal{L} = \mathcal{O}_X(-1)$, and let $\mathcal{A} = P\mathcal{L}$, $\mathcal{B} = Q\mathcal{L}$ be its two image line bundles in $E \otimes \mathcal{O}_X$. Set $\mathcal{M} = \mathcal{A} \oplus \mathcal{B}$. The tautological line $\mathcal{L} \subset \mathcal{M}$ is the diagonal under the isomorphisms induced by P and Q . Restricting the Pluecker quadrics gives a morphism

$$\mathcal{I}_{\text{Pl},2} \otimes \mathcal{O}_{X^\times} \xrightarrow{\rho} \mathcal{K} := \ker(\text{Sym}^2 \mathcal{M}^* \longrightarrow (\mathcal{L}^*)^{\otimes 2}). \quad (\text{S4.2})$$

The target has rank two. Multiplication identifies it with $(\mathcal{M}/\mathcal{L})^* \otimes \mathcal{M}^*$. The map lands in this kernel because each quadric vanishes on the original Pluecker point. A local generator of $\mathcal{L}$ identifies ρ with the family $(b - a)\ell_{F,w}(a, b)$.

Lemma S4.4 (Residual Cartier sections over the rank-one scheme). *Endow $T = D_1 \setminus D_0$ with the determinantal locally closed scheme structure, not its reduction. On a bundle chart the restricted Pluecker ideal on $\mathbb{P}(\mathcal{M}_T)$ is generated by*

$$(b - a)(ca + db), \quad (c, d) = \mathcal{O}_T.$$

Let $T_{\text{sep}} \subset T$ be the open where the residual section avoids both endpoints and the diagonal. In these coordinates it is $T \setminus V(cd(c + d))$. The closed pencil intersection over this open is the disjoint union of the effective Cartier sections

$$s = [1 : 1], \quad r = [d : -c].$$

All these statements commute with arbitrary base change, including to a nonreduced base. The residual map $\tau : T_{\text{sep}} \rightarrow T_{\text{sep}}$ is a regular involution and its fixed-point scheme is empty.

Proof. On $D_1 \setminus D_0$ some matrix entry of ρ is invertible. Its two-by-two minors vanish in the structure sheaf; row and column operations over that sheaf therefore put the matrix in rank-one normal form. The coefficient rows of the residual linear forms generate a line direct summand, represented by a unimodular pair (c, d) . Thus the displayed generator is an ideal-sheaf equality over T , not just a fibrewise equation.

A unimodular linear form on a projective-line bundle cuts out a Cartier section over any base. For example on an open where its second coefficient is a unit, its equation on the relevant affine chart is a unit multiple of a monic degree-one polynomial. Such a polynomial is a non-zero-divisor over an arbitrary coefficient ring. This proves the Cartier assertions for $b - a$ and $ca + db$. The two forms have determinant $-(c + d)$. Where $c + d$ is a unit their zero sections are disjoint, so their ideal sheaves are comaximal. Their product equals their intersection, and the Chinese remainder isomorphism identifies the divisor with $s(T_{\text{sep}}) \amalg r(T_{\text{sep}})$. The additional units c, d exclude the two pure endpoints. This construction and these unit conditions are unchanged by any base change.

The section r lies in the Grassmannian as a scheme: substitution in every restricted Pluecker equation gives zero. It has both Schur components nonzero. At this point the component frames

are changed from (Pw, Qw) to $(dPw, -cQw)$, an invertible diagonal change on T_{sep} . Writing $a = da'$, $b = -cb'$, its equation becomes

$$cd(b' - a')(da' + cb').$$

Thus the new residual coefficient pair is (d, c) , up to a unit; its section $[c : -d]$ in the new frames represents $cd(Pw + Qw)$, which is exactly s . Consequently its residual coefficient matrix again has rank one as a matrix over the base: the remaining linear factor is unimodular, and at least one generating coefficient is a unit. Thus r factors through $D_1 \setminus D_0$, including its determinantal scheme structure. Its residual section is s and is disjoint from it and from the endpoints, so the map factors through T_{sep} . Changing trivializations multiplies these equations by units and does not change their sections; the local morphisms therefore glue. Division by the other Cartier factor a second time gives $\tau^2 = \text{id}$ as morphisms. The disjointness of the sections gives an empty equalizer of τ and the identity, proving the fixed-point-scheme assertion. $\square$

Proposition S4.5 (The rank stratification and its finite part). *Let D_1 be the closed subscheme defined by the two-by-two minors of ρ , and D_0 the one defined by its entries. The rank strata*

$$X^\times \setminus D_1, \quad D_1 \setminus D_0, \quad D_0$$

have ranks two, one and zero, respectively. On the rank-one stratum the residual point is a regular section of $\mathbb{P}(\mathcal{M})$. Deleting its endpoint and diagonal loci gives a regular fixed-point-free involution. The restriction of π to $X^\times \setminus D_0$ is quasi-finite.

Put $Y = \mathbb{P}(E_{175}) \times \mathbb{P}(E_{35})$ and $\mathcal{R} = V(\text{Fitt}_0 \Omega_{X^\times/Y})$. We call $\mathcal{R}|_{X^\times \setminus D_0}$ the Fitting ramification locus of the quasi-finite part. This convention records the failure to be unramified, equivalently the nonvanishing of the relative differential module [33, Lemma 29.36.14]; it does not assert a globally finite or flat double cover. The same Fitting ideal is defined on all of $X^\times$. Its local formula is

$$(\ell_{F,w}(1, 1) : F \in \mathcal{I}_{\text{Pl},2}). \quad (\text{S4.3})$$

On the rank-one finite stratum it is exactly the double-point subscheme, defined locally by $c + d = 0$ when $\ell = ca + db$. It is empty on the rank-two stratum. In particular π is unramified on G_4° .

Proof. The determinantal definitions are invariant under changes of bundle frames. Where a rank-one matrix entry is invertible, its two-column coefficient vectors are all multiples of one unimodular vector (c, d) ; its image is a line subbundle of $\mathcal{K}$. Dividing by the equation of the diagonal gives the line of residual linear forms, whose zero is a regular section of $\mathbb{P}(\mathcal{M})$. This also works on the indicated scheme structures, not merely on their closed points. The residual section lies in X , because every restricted Pluecker quadric vanishes on it. The Cartier-section and involution assertions, with their nonreduced scheme structures, follow from Lemma S4.4. On the complement of D_0 , all geometric fibres have length at most two. The morphism is of finite type, hence quasi-finite there.

Let $U \subset \mathbb{P}(E)$ be the open where both components are nonzero. The projection $U \rightarrow Y$ is a $\mathbf{G}_m$ -bundle. In a local trivialization its relative differential module is generated by the torus differential. The conormal exact sequence for $X^\times \subset U$ presents $\Omega_{X^\times/Y}$ as the quotient of this line bundle by the relative derivatives of the Pluecker quadrics. At a point represented by w , put $a = 1, b = 1 + t$. The derivative of $(b - a)\ell_{F,w}(a, b)$ in the vertical direction is $\ell_{F,w}(1, 1)$. This proves (S4.3), including its scheme structure. At rank one one coefficient multiplier is a unit, so the ideal restricts to $(c + d)$. At rank two two independent linear forms cannot both vanish at the diagonal point. The last assertion now follows from Theorem 5.15. $\square$

For clarity, the lengths in the ambient closed pencil and in a fibre of π are different in the endpoint case:

Rank	Type	Closed pencil intersection	Fibre after endpoint deletion
2	reduced point	length 1	length 1
1	distinct secant	two reduced points	length 2
1	double point	length 2, nonreduced	length 2, nonreduced
1	pure endpoint	two reduced points	length 1
0	flag line	$\mathbb{P}^1$	$\mathbf{G}_m$

The first row is the whole of G_4° , a nonempty smooth open of $\mathrm{Gr}(4, 10)$ of dimension 24 and codimension zero in that Grassmannian. Every other row has empty intersection with G_4° . Thus there is no positive-codimension exceptional stratum in this geometric family whose dimension or generic singularity remains to be computed. The determinantal schemes may have further geometry on the binary-Jacobian boundary; the preceding assertions do not assign an expected codimension to those ambient schemes or make π into a global double cover.

S4.3. An alternative coordinate character calculation. The following direct calculation is independent of the character decomposition in Lemma 5.11; it is not needed for the main proof. Use an orthonormal basis and write $\rho = \sum_i x_i^2$.

Any possible image of ρ^2 , with its determinant twist, must transform by the determinant character of O_4 in $\wedge^3 \mathrm{Sym}^2 V$. There is no such vector. To see this directly, invariance with this character under the four coordinate reflections requires each variable to have odd total degree in any nonzero exterior monomial. Its total degree is six, so the multidegree must be a permutation of $(3, 1, 1, 1)$. For a fixed variable x_i of degree three, the possible monomials are

$$\omega_i = \bigwedge_{j \neq i} x_i x_j, \quad x_i^2 \wedge (x_i x_j) \wedge (x_k x_l) \quad (\{i, j, k, l\} = \{1, 2, 3, 4\}).$$

The coefficient of the second kind is zero: the transposition of x_k, x_l fixes that monomial but has determinant minus one. Only a combination of the ω_i remains. Applying the infinitesimal rotation $x_i \partial / \partial x_j - x_j \partial / \partial x_i$, the coefficient of $x_i^2 \wedge (x_i x_k) \wedge (x_i x_l)$ comes, up to sign, only from ω_i . Invariance forces its coefficient to vanish, for every i . This proves the assertion.

S5. BOUNDARY STRUCTURE AND FAMILY THEOREMS

The following statements organize the independent boundary and family results proved in the subsequent appendices. They retain the full primary and specialization theory while separating it from the logical spine of the inverse theorem.

S5.1. Sharp global laws. The general exponent has a sharper geometric form on the open G_4° . Theorem S12.2 proves

$$\widehat{\mathcal{N}}_R^5 = 0, \quad \widehat{\mathcal{N}}_R^4 \neq 0 \text{ at every point of projection corank three or four.}$$

The reason is global rather than semicontinuous: the determinant-four identity in the full matrix coordinate ring restricts to every Schur chart. The needed coordinate-ring Pieri projection is calculated with the Fischer form; its coefficient is $1/35$, not an unspecified straightening coefficient. The lower bound at corank three is the Schur-symbol exclusion proved below. The coefficient-space tables are retained in full, but Proposition S12.3 determines necessary rank conditions for their occurrence on the smooth, basepoint-free geometric open.

There is also a global geometric description beyond determinant depth. Let Z_R be the image of the incidence vector bundle $\mathrm{Hom}(V, \mathcal{H})$ over the Jacobian K3 surface, and let D_2 be the rank-at-most-two matrix variety. Theorem S12.4 identifies these as the unique minimal supports of the first and second layers. The incidence resolution has fibres given by linear sections of the same K3: four contact points, a plane quartic of genus three, and the entire surface. Theorem S12.6

realizes this sequence in one two-parameter family and computes its seven reduced components. Proposition S12.7 computes the intrinsic graded residue-field algebras across the same corank transitions. These statements do not assert a complete list of all embedded transverse primes of the third and fourth layers.

The relative theorem is proved by primary modules, rather than by an inference from open-containment constructibility. A diagonal injection into filtered primary quotients excludes every untracked associated point; associated-element injections give the opposite inclusion. A regular element on the quotient by a coherent torsion module prevents new packet torsion after specialization. Both kinds of injections survive by flattening their actual cokernels. In the proper quotient-induced family, the classical Bruns–Vasconcelos identity [25] further gives $dJ = (d) \cap P^{a+1}$, an exact exponent and explicit embedded lengths (Theorem S12.9). The classical minor identity itself is not a novelty claim of this paper.

The argument has two scales. The first is universal: determinant completion bounds the principal nilpotence exponent in every symmetric power. The second is relative: once such an exponent is bounded, the entire colon tower is finite and can be stratified simultaneously. Combining the two produces a uniform theorem on the parameter space of surjections $\gamma : \text{Sym}^r V \twoheadrightarrow S$: after finite stratification, all residual colons, geometric associated-point packets, embedded multiplicities, and multiplication ranks are compatible with arbitrary base change. The bounded principal-colon theorem proved below makes kernel formation commute with arbitrary base change before any associated-point argument is invoked. Actual primary quotients then exhaust the geometric associated set. Orbit-ideal descent and a power-certified construction on the original base produce the closed packets; a relative torsion module controls their embedded multiplicities. For the $(1, 4, 6)$ problem this mechanism is supplemented by exact rational-function-field calculations on transverse stabilizer slices and by explicit boundary degenerations.

S5.2. The intrinsic graded boundary packet. Write $\widehat{\mathcal{N}}_R$ for the nilradical of $\mathcal{O}_{\widehat{D}_R}$. The schemes

$$\mathcal{Z}_j(\widehat{D}_R) = V(\text{Ann}(\widehat{\mathcal{N}}_R^j))$$

and the products among $\widehat{\mathcal{N}}_R^j / \widehat{\mathcal{N}}_R^{j+1}$ are invariant under every abstract isomorphism of $\widehat{D}_R$. On a Schur chart the reduced equation is $d = \det T$ and

$$\mathcal{I}_{\widehat{D}_R} = dJ.$$

The residual-colon theorem of Section S9 identifies

$$\widehat{\mathcal{N}}_R^j / \widehat{\mathcal{N}}_R^{j+1} \simeq \mathcal{L}^{\otimes j} \otimes \mathcal{O}_{W_j}, \quad W_j = V((d) + (J : d^{j-1})). \quad (\text{S5.1})$$

The coordinate expression changes from chart to chart; the modules and schemes in (S5.1) do not.

Theorem S5.1 (Boundary structure). *For $R \in G_4^\circ$ the following hold.*

- (1) *On every projection-rank chart,*

$$d^5 \in J, \quad \widehat{\mathcal{N}}_R^6 = 0.$$

This is the $(e, r) = (4, 2)$ case of the universal determinant completion theorem, whose exponent $\binom{e+r-1}{r-1}$ is sharp in the universal class.

- (2) *The multiplication of the positive graded pieces is the canonical quotient multiplication*

$$\mathcal{O}_{W_i} \otimes \mathcal{O}_{W_j} \longrightarrow \mathcal{O}_{W_{i+j}}$$

with the expected conormal-line twist. No additional scalar multiplication constants occur.

- (3) *If the pure block A_H is injective at projection corank two, the mixed block has nine orbit types. On the generic coefficient stratum of every orbit, the complete nonzero graded supports through W_3 , their Hilbert functions, radicals, associated supports, and vertex-primary corrections are given in Theorems S6.4 and S11.2. Genericity is proved over rational function fields on transverse stabilizer slices.*
- (4) *If A_H drops rank at projection corank two, every generic rank type (a, b) has an explicit primary signature. Besides the Segre quadric, the only positive-dimensional generic supports that occur are reduced divisors of two or four lines in one ruling. The remaining layers are vertex-primary; in the $(a, b) = (1, 3)$ row a length-three embedded vertex accompanies the two ruling lines.*
- (5) *At projection corank three, $d^3 \notin J$ and $d^4 \in J$; the index is exactly five. The upper bound is the restriction of the full-matrix identity, as proved in Theorem S12.2. The lower bound is the functorial Schur-symbol exclusion.*
- (6) *At projection corank four,*

$$d^3 \notin J, \quad d^4 \in J.$$

The second membership is proved from the actual Cauchy–Pieri multiplication in $\mathbb{C}[\text{End}(V)]$: the product of the $(4, 4, 4, 0)$ - and (4) -components has nonzero $(4, 4, 4, 4)$ -projection. Since that line is spanned by d^4 , the local nilpotency index is exactly five.

Thus the phrase *boundary atlas* refers to intrinsic graded data, not to the chosen coefficient matrices. The Macaulay matrices form a finite presentation atlas. Any two such presentation atlases have a common refinement and compute the same intrinsic primary signature.

S5.3. Primary specialization. The preceding finite calculations are promoted to family statements in Section S11. The key homological step is explicit. If $B = P_S/J$, $C_j = B/d^{j-1}B$, and $I_j = d^{j-1}B$, we first flatten B and C_j . The exact sequence $0 \rightarrow I_j \rightarrow B \rightarrow C_j \rightarrow 0$ then makes I_j flat, and $0 \rightarrow K_j \rightarrow P_S \rightarrow I_j \rightarrow 0$ remains exact after arbitrary base change. Only after the equality $(K_j)_s = (J_s : d_s^{j-1})$ has been established do we apply relative assassins. Finite primary models and the two opposite associated-point inclusions give exact geometric support packets. A regular element on the quotient by a coherent torsion module excludes new torsion after specialization, and a finite power filtration computes the local lengths. The closed packets descend as ideals with an explicit cocycle in characteristic zero. In arbitrary characteristic their ideals are constructed downstairs first and their geometric components are certified by finite power inclusions on a splitting cover. The relative torsion module is also constructed downstairs. Noetherian induction then handles the complement.

The same section writes explicit one-parameter degenerations of the mixed block. For example an off-Segre kernel point specializes to a Segre point; a secant kernel line specializes to a tangent or either ruling line; rank-two kernels specialize to the two rank-one types; and both rank-one types specialize to the zero mixed map. The graded schemes in the two fibres are computed, so the specialization records the birth of vertex-primary and ruling components rather than merely changing orbit labels.

S5.4. The simple-contact locus. The general atlas contains the earlier primary theorem as its split form. We retain the following statement because it is the geometric entry point to the collision analysis.

Theorem S5.2 (Corank-two primary structure on the simple-contact open). *There is a nonempty invariant open $G_4^\dagger \subset G_4^\circ$ such that, for each $R \in G_4^\dagger$, a dense open set of rank-two images $H \subset V$*

has, after an étale extension and smooth auxiliary coordinates,

$$\mathcal{I}_{\widehat{D}_R} = (\delta) \cap P_1^2 \cap P_2^2 \cap P_3^2 \cap P_4^2 \cap \mathfrak{m}^5. \quad (\text{S5.2})$$

The rank-two stratum is an additional associated support and $\widehat{\mathcal{N}}_R^3 = 0 \neq \widehat{\mathcal{N}}_R^2$.

Repeated roots, the containment locus, the decomposable-kernel wall, and all remaining mixed-kernel orbits are treated without splitting the contact quartic.

S5.5. Variation and finite ambiguity.

Theorem S5.3 (Variation and finite ambiguity). *For $e = 4$, the image of*

$$R \longmapsto (Y_R, \mathcal{O}_{Y_R}(1))$$

in quartic K3 moduli has dimension nine. On a nonempty invariant open the induced map from projective equivalence classes of relation spaces to its image is generically finite.

The nine-dimensional count is classical Reye geometry, not a novelty claim. The polarized K3 alone is not identified with the original web. Theorem 5.1 instead uses the deepest nonreduced normal cone to recover every smooth Jacobian web, and Corollary 5.17 separates the finite packet. The degree of the map defined by the K3 alone is not asserted to be one. The sharp exponent, the two minimal supports, and the explicit cross-corank geometry of Section S12 remain independent global consequences of the primary-boundary analysis.

S6. UNIVERSAL DETERMINANT COMPLETION AND THE INTRINSIC BOUNDARY ATLAS

The residual-colon filtration of Theorem S9.1 is intrinsic at every projection rank. We first isolate the determinant-completion mechanism in arbitrary dimension and symmetric degree. We then compute the projection-corank-two boundary on genuine generic coefficient strata, including all pure-block rank drops, and finally analyze projection coranks three and four. The finite coefficient matrices used below form a presentation atlas; the schemes W_j , their associated points, generic lengths, and graded multiplication form the intrinsic boundary packet.

Keep the notation of Theorem S8.1. Thus, at a point of projection rank r , put

$$H = \text{im}(\mathcal{K} \rightarrow V), \quad W = V/H, \quad c = \dim W = 4 - r,$$

choose a Schur chart

$$M = I_H \oplus T, \quad d = \det T,$$

and write

$$\mathcal{I}_{\widehat{D}_R} = dJ, \quad J = I_6 \left[A_H, B_H(1_H \otimes T), C_H \text{Sym}^2 T \right]. \quad (\text{S6.1})$$

The local nilradical is generated by the class of d .

Theorem S6.1 (Universal symmetric-power determinant completion). *Let V be a complex vector space of dimension e , let $r \geq 1$, and let*

$$\gamma : \text{Sym}^r V \rightarrow E$$

be a surjection, with $s = \dim E$. On $\text{End}(V)$ let M be the generic endomorphism, put $d = \det M$, and set

$$J_{\gamma,r} = I_s(\gamma \text{Sym}^r M).$$

Then

$$d^{\kappa(e,r)} \in J_{\gamma,r}, \quad \kappa(e,r) = \binom{e+r-1}{r-1}. \quad (\text{S6.2})$$

The exponent is sharp in the universal class: if $E = \text{Sym}^r V$ and γ is an isomorphism, then $J_{\gamma,r} = (d^{\kappa(e,r)})$.

For $(e, r) = (4, 2)$, $\kappa(4, 2) = 5$. Thus the global depth bound below is a specialization of a dimension-free determinant-completion principle, rather than an isolated identity for the Hilbert function $(1, 4, 6)$.

Theorem S6.2 (Finite all-corank depth and multiplication). *For every $R \in G_4^\circ$ and every projection-rank chart,*

$$d^5 \in J. \quad (\text{S6.3})$$

Consequently

$$\widehat{\mathcal{N}}_R^6 = 0$$

on the whole Grassmannian. In particular

$$W_j = \emptyset \quad (j \geq 6),$$

and the special fibre of the intrinsic normal-cone specialization has at most five positive graded layers.

For $i, j \geq 1$, $i + j \leq 5$, the multiplication map

$$\frac{\widehat{\mathcal{N}}^i}{\widehat{\mathcal{N}}^{i+1}} \otimes \frac{\widehat{\mathcal{N}}^j}{\widehat{\mathcal{N}}^{j+1}} \longrightarrow \frac{\widehat{\mathcal{N}}^{i+j}}{\widehat{\mathcal{N}}^{i+j+1}}$$

is, under the identifications

$$\frac{\widehat{\mathcal{N}}^j}{\widehat{\mathcal{N}}^{j+1}} \simeq \mathcal{L}^{\otimes j} \otimes \mathcal{O}_{W_j},$$

the tensor twist of the canonical quotient multiplication

$$\mathcal{O}_{W_i} \otimes_{\mathcal{O}_{\widehat{\Delta}_R}} \mathcal{O}_{W_j} \longrightarrow \mathcal{O}_{W_{i+j}}. \quad (\text{S6.4})$$

In particular it is surjective. Thus no additional multiplication constant is hidden after the schemes W_j are known.

The coefficient-space tables below concern surjective quadratic quotients on their stated generic coefficient strata. Their intersection with the smooth, basepoint-free web open is subject to Proposition S12.3. In particular the ambient index-six rows are not asserted to occur on that geometric open. Theorem S12.2 strengthens the global bound there to nilpotency index five.

We next sharpen the corank-two analysis. Assume A_H is injective. Put

$$E_H = S_R / \text{im } A_H, \quad \beta_H = \overline{B}_H : H \otimes W \longrightarrow E_H, \quad \chi_H = \overline{C}_H : \text{Sym}^2 W \longrightarrow E_H.$$

Here $\dim H = \dim W = 2$ and $\dim E_H = 3$. Let

$$Q = \mathbb{P}(H \otimes W)_{\text{rk}=1} \simeq \mathbb{P}^1 \times \mathbb{P}^1$$

be the Segre quadric.

Definition S6.3 (Mixed-kernel orbit type). The *mixed-kernel orbit type* of (R, H) is the rank $b = \text{rank } \beta_H$, together with the incidence type of $\mathbb{P}(\ker \beta_H)$ with Q .

For $b = 3$ this is a point off Q or on Q . For $b = 2$ it is a secant line, a tangent line, an H -ruling $\mathbb{P}(h_0 \otimes W)$, or a W -ruling $\mathbb{P}(H \otimes w_0)$ contained in Q . For $b = 1$, the one-dimensional image of $\beta_H^* : E_H^* \rightarrow (H \otimes W)^*$ is generated by a tensor of rank two or rank one; equivalently the kernel plane cuts Q in a smooth conic or in a pair of ruling lines. The final type is $b = 0$.

These are exactly the nine $GL(H) \times GL(W) \times GL(E_H)$ -orbits for the mixed block. The first two are the mixed-regular and decomposable-kernel cases already encountered in Section S9; the remaining seven are higher-dimensional mixed kernels or failures of mixed surjectivity.

For a homogeneous ideal $K \subset P = \mathbb{C}[a, b, c, d]$, write

$$H_{P/K}(n) = \dim_{\mathbb{C}}(P/K)_n.$$

For each orbit, the *generic coefficient stratum* is the dense open obtained from a transverse stabilizer slice by deleting the explicit denominator and leading-coefficient divisors in Theorem S11.1. Equivalently it is the locus on which the rational-function-field Groebner bases used there retain their initial ideals. The isolated integral matrices listed in the proof are base points of those slices; they are not used as a substitute for genericity.

Theorem S6.4 (Corank-two mixed-kernel atlas). *Assume A_H is injective. Then the first positive graded support is determined without any mixed-regularity assumption:*

$$\mathcal{O}_{W_1} \simeq P/(\delta, g_0, \dots, g_4), \quad \delta = ad - bc, \quad (\text{S6.5})$$

where, on $P/(\delta)$ with $T = uv^t$, the degree-four classes span

$$h_H(u) \text{Sym}^4(\mathbb{C}_v^2).$$

Thus, if $h_H \neq 0$,

$$\text{Proj } W_1 = V(h_H) \times \mathbb{P}_v^1 \subset Q,$$

whereas $h_H = 0$ gives $W_1 = V(\delta)$.

On the generic coefficient stratum of each mixed-kernel orbit, the remaining graded supports and the exact local nilpotency index are as follows.

b	mixed-kernel type	index	W_2	W_3
3	point off Q	3	length 1	empty
3	point on Q	3	length 3, $H = (1, 2)$	empty
2	secant line	3	length 8, $H = (1, 4, 3)$	empty
2	tangent line	3	length 8, $H = (1, 4, 3)$	empty
2	H -ruling $\mathbb{P}(h_0 \otimes W)$	4	length 15, $H = (1, 4, 6, 4)$	length 1
2	W -ruling $\mathbb{P}(H \otimes w_0)$	3	$H(n) = n + 1$ for $n \geq 4$	empty
1	dual tensor rank 2	4	$H(n) = 2n + 2$ for $n \geq 3$	length 1
1	dual tensor rank 1	4	$H(n) = 3n + 3$ for $n \geq 3$	length 3
0	zero mixed map	4	$V(\delta)$	$V(\delta)$

For $b = 0$, surjectivity of the original quotient multiplication forces χ_H to be an isomorphism and

$$J = (\delta^3); \quad (\text{S6.6})$$

hence the last row is an exact identity, not only a generic one.

In the H -ruling and $b \leq 1$ rows one has $h_H = 0$. On the W -ruling row the contact quartic is nonzero on the transverse open and has the root forced by w_0 . For $b = 1$, the projectivization of W_2 has degree two in the dual-rank-two case and degree three in the dual-rank-one case. The length-one and length-three third layers are respectively the reduced rank-two vertex and the length-three vertex thickening already appearing on the decomposable-kernel wall.

The complement of the generic coefficient stratum is itself part of the atlas, rather than an omitted exceptional set. We make this finite refinement explicit.

Definition S6.5 (Macaulay valuation atlas). Let $P_c = \mathbb{C}[t_{ij} : 1 \leq i, j \leq c]$, $d = \det T$, and let J be the residual ideal on a fixed projection-rank chart. For $1 \leq k \leq 5$, let

$$\mathbf{M}_k(J)$$

be the coefficient matrix whose columns are all products

$$m q, \quad q \in \mathcal{G}(J), \quad \deg m + \deg q = ck,$$

written in the monomial basis of $(P_c)_{ck}$, where $\mathcal{G}(J)$ is the displayed set of maximal minors in (S6.1). Let v_k be the coefficient vector of d^k . Define

$$\varepsilon_k(J) = \text{rank}[\mathbf{M}_k(J) \mid v_k] - \text{rank } \mathbf{M}_k(J) \in \{0, 1\}.$$

Then

$$\varepsilon_k(J) = 0 \iff d^k \in J.$$

The locally closed loci obtained by fixing the ranks of the matrices M_k , their augmented matrices, and the analogous matrices for the kernels

$$K_j = (J : d^{j-1}), \quad 1 \leq j \leq 5,$$

are called the *Macaulay valuation atlas*. All of their equations are minors of explicit finite coefficient matrices.

Theorem S6.6 (Bounded principal-colon primary stratification). *Let S be noetherian, let A be a finitely presented S -algebra which is flat over S , let $J \subset A$ be a finitely generated ideal, and let $f \in A$. Assume that $f^N \in J$ for some $N \geq 1$. Put*

$$B = A/J, \quad K_q = (J : f^q), \quad M_q = A/(f, K_q), \quad 0 \leq q < N.$$

Then S admits a finite locally closed stratification $S = \coprod_{\alpha} S_{\alpha}$ with the following properties.

For every α , all the assertions below are preserved by an arbitrary base change $T \rightarrow S_{\alpha}$. Here J on a stratum means the image ideal, and each K_q is formed anew over that stratum. No commutation across different strata is asserted.

- (1) *Formation of J and of every K_q commutes with base change. The modules*

$$B, \quad C_q = A/(J, f^q), \quad I_q = f^q B, \quad M_q$$

are flat over S_{α} .

- (2) *For each q there are finitely many closed relative support packets $Z_{q,\nu} \subset \text{Spec } A_{S_{\alpha}}$, flat after a further refinement, such that for every geometric point $\bar{s} \rightarrow S_{\alpha}$ the associated points of $(M_q)_{\bar{s}}$ are exactly the generic points of the irreducible components of the nonempty $(Z_{q,\nu})_{\bar{s}}$. Geometric splitting of a packet is allowed. Dimensions, incidence, and minimal-versus-embedded status are constant on the stratum.*

- (3) *If p is one of these geometric associated points, the canonical embedded multiplicity*

$$\mu_p(M_q) = \text{length}_{\mathcal{O}_{\text{Spec } A_{\bar{s}},p}} H_{p, \mathcal{O}_{\text{Spec } A_{\bar{s}},p}}^0((M_q)_p)$$

is constant along the corresponding packet.

- (4) *Any prescribed finite collection of maps between the M_q may be included in the same refinement so that the relevant source, target, image and cokernel modules are flat, and their defining exact diagrams commute with arbitrary base change. When grading is present, degreewise numerical ranks are given by Lemma S11.12.*

Thus a bounded principal power converts the a priori infinite primary problem into a finite, base-change-compatible invariant.

Corollary S6.7 (Universal primary finiteness for symmetric-power quotients). *Fix $e, r \geq 1$ and $1 \leq s \leq \dim \text{Sym}^r V$, and fix a target vector space E of dimension s . Let $\mathcal{U}$ be the open parameter space of surjections $\gamma : \text{Sym}^r V \twoheadrightarrow E$. On a coefficient affine chart write*

$$A = \mathcal{O}_{\mathcal{U}}[t_{ij}], \quad \deg t_{ij} = 1, \quad \deg \mathcal{O}_{\mathcal{U}} = 0, \quad d = \det T, \quad J = I_s(\gamma \text{Sym}^r T), \quad N = \binom{e+r-1}{r-1}.$$

There is a finite locally closed stratification of $\mathcal{U}$ on which all the colons $K_q = (J : d^q)$, $0 \leq q < N$, commute with arbitrary base change within a stratum. The graded modules

$$M_q = A/(d, K_q), \quad L_j = M_{j-1}(-ej), \quad 1 \leq j \leq N,$$

are flat. Their geometric associated-support packets, containment posets, minimal and embedded status, and canonical local torsion lengths are constant. Their Hilbert functions, and therefore their Hilbert polynomials, are constant on each connected final stratum.

The finite family of natural multiplication maps meant here is

$$\mu_{i,j} : L_i \otimes_A L_j \longrightarrow L_{i+j}, \quad i, j \geq 1, \quad i + j \leq N. \quad (\text{S6.7})$$

These maps are surjective homogeneous maps; their sources, targets, and images are flat and their ranks in every degree commute with base change and are constant. A prescribed finite list of additional maps between fixed coefficient-degree pieces can also be included in the refinement. No simultaneous rank assertion for an unspecified infinite family of such additional maps is intended.

Proof. The universal determinant-completion identity gives $d^N \in J$. Apply Theorem S6.6 to the finite list of homogeneous modules M_q . The primary models and relative torsion construction in Propositions S11.5 and S11.6 give the associated-support and multiplicity assertions. Apply Lemma S11.12 to obtain the Hilbert functions and polynomials. This is an additional graded consequence, not a clause of an ungraded flattening theorem.

The colon ideals form an increasing chain. Thus, with the indicated shifts, $L_i \otimes_A L_j$ is the quotient $M_{\max(i-1, j-1)}$, and the multiplication in (S6.7) is the canonical quotient to M_{i+j-1} . Its source and target are among the already flat modules and its cokernel is zero. Lemma S11.12 applies in every degree. Finally, any additionally prescribed finite list of maps between finite coefficient-degree modules is handled by flattening its finite list of images and cokernels. Affine coefficient charts admit a common finite refinement. $\square$

Definition S6.8 (Primary signature and primary components). In the universal statements a *primary signature* is the package of residual modules, associated-support packets, their containment poset and local torsion lengths, graded Hilbert functions, and the finite family (S6.7). This package is intrinsic. A primary decomposition means an actual intersection of primary ideals or primary submodules. Such noncanonical decompositions are constructed on finite splitting covers by Proposition S11.5; they need not descend as individually labelled components and are not asserted to be canonical or explicitly classified throughout the universal family.

The theorem is independent of the special $(1, 4, 6)$ model. In the present paper the universal determinant-completion theorem supplies $N = 5$ on every projection-rank chart. Consequently the residual colon algebra, its associated-point packets, and all graded multiplication maps admit one finite primary stratification before any orbit-specific computation is made. Because of Theorem S6.2, no matrix with $k > 5$ occurs. We refine these rank strata simultaneously by the flattening strata of the finite modules

$$\mathcal{M}_j = P_c / ((d) + K_j), \quad 1 \leq j \leq 5,$$

by the flattening strata of the cokernels of (S6.4), and by the associated-point stratification in the following proposition.

Proposition S6.9 (Finite intrinsic primary-signature refinement). *On every finite-type coefficient chart for (A_H, B_H, C_H) , the presentation atlas admits a finite locally closed refinement such that the following statements hold for every geometric fibre.*

- (1) *Formation of each residual colon $K_j = (J : dj^{-1})$, $1 \leq j \leq 5$, commutes with base change, and*

$$\mathcal{M}_j = P_c / ((d) + K_j)$$

is flat over the stratum with constant Hilbert polynomial.

- (2) *There are finitely many relative closed supports $Z_{j,\nu}$ whose nonempty fibrewise generic points are exactly $\text{Ass}((\mathcal{M}_j)_s)$. Their dimensions, minimal-versus-embedded status, incidence relations, and generic lengths are constant on the stratum.*
- (3) *The cokernels of the graded multiplication maps are flat, so their ranks are constant and commute with base change.*

Consequently the radicals, associated supports, generic multiplicities, nilpotency index, Hilbert polynomials, and multiplication ranks of all positive graded layers have constant intrinsic primary signature on each final stratum.

The refinement is not asserted to be canonical as a collection of coordinate strata. The primary signature is intrinsic; different Schur charts, generating sets, or monomial orders admit a common finite refinement and compute the same signature.

This refinement is used below for rank drops. The proof in Section S11 is the application of Theorem S6.6: it flattens both P_S/J and $P_S/(J, d^{j-1})$, deduces flatness of the image of multiplication by d^{j-1} , and only then invokes relative assassins. Thus coefficient collisions are recorded by actual specialization of the intrinsic graded algebra, not by relabelling a primary decomposition.

There is also a useful degree bound that detects many strata before any Macaulay matrix is formed. Put

$$a = \text{rank } A_H, \quad q = 6 - a,$$

and, after quotienting by $\text{im } A_H$,

$$b = \text{rank } \overline{B}_H.$$

Every nonzero residual q -minor uses at most b linear mixed columns and therefore has transverse degree at least

$$\ell(a, b) = 2q - b. \tag{S6.8}$$

Proposition S6.10 (Rank-drop depth bounds at corank two). *At projection corank two,*

$$\min\{k : d^k \in J\} \geq \kappa_0(a, b) := q - \left\lfloor \frac{b}{2} \right\rfloor. \tag{S6.9}$$

Together with $d^5 \in J$, this leaves only the following finite possibilities when A_H drops rank:

a	q	possible $(b; k)$
2	4	$(4; 2, 3, 4, 5), (3; 2; 3, 4, 5), (1; 4, 5),$
1	5	$(4; 3, 4, 5), (3; 2; 4, 5),$
0	6	$(4; 4, 5), (3; 5).$

Here $k = \min\{j : d^j \in J\}$, so the nilpotency index is $k + 1$. The omitted values of b cannot occur under surjectivity of the original quadratic quotient, because the quadratic block contributes at most three independent target directions. Each comma-separated choice of k is cut out exactly by $\varepsilon_k(J) = 0$ and $\varepsilon_{k-1}(J) = 1$. In particular the extreme rank-drop type $(a, b) = (0, 3)$ has exact nilpotency index six.

At projection corank three a stronger representation-theoretic exclusion reduces the atlas to one binary choice.

Theorem S6.11 (Projection corank three). *For $R \in G_4^\circ$, at every projection-corank-three point,*

$$d^3 \notin J, \quad d^4 \in J. \tag{S6.10}$$

The local nilpotency index is exactly five, and $\varepsilon_4(J) = 0$ identically on this geometric open. For arbitrary surjective coefficient data outside this open, the universal bounds $d^3 \notin J$, $d^5 \in J$ and the membership test for d^4 remain valid; the smooth-web theorem eliminates that binary alternative on G_4° .

At the deepest Schubert stratum the remaining condition is identified geometrically.

Theorem S6.12 (Projection corank four and the Jacobian quartic). *Let $R \in G_4^\circ$. At a projection-corank-four point, $H = 0$, $T \in \text{Mat}_{4 \times 4}$, and*

$$J = I_6(\gamma_R \text{Sym}^2 T).$$

Then

$$d^3 \notin J, \quad d^4 \in J, \quad d = \det T. \quad (\text{S6.11})$$

Consequently the local nilradical has exact index five.

More precisely, as a $GL(V)$ -module,

$$\bigwedge^6 \text{Sym}^2 V \simeq \mathbb{S}_{(5,4,2,1)} V \oplus \mathbb{S}_{(4,4,4,0)} V. \quad (\text{S6.12})$$

The absence of $(\det V)^3$ in (S6.12) gives the first nonmembership in (S6.11). Under the canonical duality

$$\bigwedge^6 \text{Sym}^2 V \simeq (\det V)^5 \otimes \left(\bigwedge^4 \text{Sym}^2 V \right)^*,$$

the second summand in (S6.12) is dual, up to determinant twist, to

$$\det V \otimes \text{Sym}^4 V,$$

and the corresponding covariant of $\bigwedge^4 R$ is the Jacobian quartic F_R . Since $F_R \neq 0$ on G_4° , the determinant line $(\det V)^4$ occurs in the degree-sixteen part of J , which is exactly the second membership in (S6.11).

Theorems S6.2, S6.4, S6.11, and S6.12, together with the generic primary signature and specialization theorems of Section S11, form the boundary classification used here. The residual-colon formulas are coordinate-free identities; the presentation matrices are finite computational devices. At corank two the generic supports and their primary corrections are geometrically identified, including pure-block rank drops. At higher corank the exact depth statements remain intrinsic and are coupled to the finite primary-signature refinement.

S7. FIXED TENSORS ACROSS PROJECTION CORANK TWO

We now work with $e = 4$ and $p = 6$. All calculations below concern a fixed quotient tensor γ_R , while its generating plane varies. They are identities of ideals, before taking radicals.

Definition S7.1. A pair (R, H) , where $R \in G_4^\circ$ and $H \subset V$ is a plane, is called admissible for the corank-two calculation if the following conditions hold. The map $A_H = \gamma_R|_{\text{Sym}^2 H}$ is injective. For a complement W to H , put $Q_H = S_R / \text{im } A_H$ and let

$$B_H : H \otimes W \longrightarrow Q_H, \quad C_H : \text{Sym}^2 W \longrightarrow Q_H$$

be induced by multiplication. The map B_H is surjective and its one-dimensional kernel is generated by a rank-two tensor in $H \otimes W$. Finally, the line of hyperplanes containing H meets Y_R in four distinct points.

The first-order map B_H is intrinsically a map from $H \otimes (V/H)$: changing a lift of a vector in V/H adds an element of $\text{Sym}^2 H$, which is zero in Q_H . Thus these conditions are independent of the chosen complement. The last condition is transverse intersection with a projective line, not smoothness of a parameter-selected polynomial without a geometric interpretation.

Lemma S7.2 (Nonemptiness of the corank-two open). *The admissible pairs form a nonempty open subset of $G_4^\circ \times \text{Gr}(2, V)$. Its projection is a nonempty invariant open $G_4^\dagger$. For each R in that image, the admissible H form a dense open subset of $\text{Gr}(2, V)$.*

Proof. Injectivity and surjectivity are open rank conditions. After taking the line kernel, tensor rank two is the nonvanishing of its determinant. On this locus the pencil restriction of the determinant is a binary quartic; nonzero discriminant is open. These conditions are invariant. Projection from a product over $\mathbb{C}$ is open, and $\text{Gr}(2, V)$ is irreducible, giving the density claim once nonemptiness is proved.

For C_* in (8.2), take $H = \langle x_0, x_1 \rangle$ and $W = \langle x_2, x_3 \rangle$. In the square-first, cross-second basis, $\gamma = (-C_* \mid I_6)$. The three columns for x_0^2, x_0x_1, x_1^2 have rank three. After quotienting their image, the remaining maps are

$$B_H = \begin{pmatrix} -1/2 & -1 & 1 & 0 \\ -1/2 & -1 & 0 & 1 \\ -3/2 & -2 & 0 & 0 \end{pmatrix}, \quad C_H = \begin{pmatrix} -9/2 & 0 & -1/2 \\ -5/2 & 0 & 1/2 \\ -13/2 & 1 & 7/2 \end{pmatrix}. \quad (\text{S7.1})$$

The four columns of B_H correspond to $x_0x_2, x_0x_3, x_1x_2, x_1x_3$; the three of C_H correspond to x_2^2, x_2x_3, x_3^2 . The kernel vector is $(-4, 3, 1, 1)$, whose 2×2 determinant is -7 . For $ux_2 + vx_3$, the final square column is $C_H(u^2, 2uv, v^2)^t$. The determinant of that column and the two mixed columns is

$$h(u, v) = \frac{u^4 - 8u^3v - u^2v^2 - 4uv^3 - 9v^4}{2}, \quad \text{disc } h(u, 1) = -\frac{632301}{4} \neq 0. \quad (\text{S7.2})$$

Its leading coefficient is nonzero, so there is no root at infinity. This determinant is the restriction determinant on $\text{Sym}^2(H + \mathbb{C}(ux_2 + vx_3))$ after eliminating A_H , hence defines the pencil intersection with Y_R . Lemma 8.1 places this witness in G_4° . $\square$

Lemma S7.3 (The residual matrix). *Near a plane K whose projection has admissible rank-two image H , there are smooth auxiliary parameters and four transverse coordinates a, b, c, d in which $\mathcal{I}_{\widehat{D}_R} = \delta J$, with*

$$\begin{aligned} T &= \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad \delta = ad - bc, \quad \mathbf{m} = (a, b, c, d), \\ L(T) &= \begin{pmatrix} a & b & 0 & 0 \\ c & d & a & b \\ 0 & 0 & c & d \end{pmatrix}, \quad T^{(2)} = \begin{pmatrix} a^2 & ab & b^2 \\ 2ac & ad + bc & 2bd \\ c^2 & cd & d^2 \end{pmatrix}, \\ J &= I_3[L(T), C T^{(2)}]. \end{aligned} \quad (\text{S7.3})$$

Here C is a 3×3 matrix depending only on the auxiliary parameters, not on a, b, c, d .

Proof. Use Proposition 4.1 and an invertible projection two-minor to write $M = \text{diag}(I_2, T)$. The basis of its first image plane and a complement can be chosen independently of the Schur block T : the Gaussian row operations depend on the first two columns, and the remaining column operations remove their components from the last two columns. This proves the asserted separation of parameters.

The quadratic columns divide into $\text{Sym}^2 H$, $H \otimes W$, and $\text{Sym}^2 W$. Eliminate the injective rank-three A_H block. The residual matrix is $[B_H(I_H \otimes T), C_H T^{(2)}]$, and its maximal minors generate the residual ideal: the eliminated identity block removes a free direct summand of the cokernel. A rank-two generator of $\ker B_H$ can be put in the form $h_1 \otimes w_2 - h_2 \otimes w_1$ by invertible changes of bases. The quotient is then the symmetrization map from two two-dimensional spaces, with the displayed matrix $L(T)$. These basis changes involve only the auxiliary parameters. Finally $\det M$ is a unit times δ . This proves the ideal identity, not just its leading term. $\square$

Let

$$h(u_1, u_2) = \det \left[\begin{pmatrix} u_1 \\ u_2 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ u_1 \\ u_2 \end{pmatrix}, C \begin{pmatrix} u_1^2 \\ 2u_1u_2 \\ u_2^2 \end{pmatrix} \right]. \quad (\text{S7.4})$$

It is the binary quartic of the pencil intersection. After an étale extension of the auxiliary coefficient ring, its four roots split as distinct line subbundles $\ell_i \subset W$. Write P_i for the ideal of $\text{im } T \subset \ell_i$; it is generated by the two entries of the composite $\mathbb{C}^2 \xrightarrow{T} W \rightarrow W/\ell_i$.

Lemma S7.4 (A rank-one-cone computation). *Under the simple-root hypothesis, in the local coefficient ring of Lemma S7.3 one has*

$$I := \bigcap_{i=1}^4 P_i = (\delta, g_0, \dots, g_4), \quad J = I \cap \mathfrak{m}^3 = \delta \mathfrak{m} + (g_0, \dots, g_4). \quad (\text{S7.5})$$

The degree-four polynomials g_j can be chosen with restrictions $h(u_1, u_2)v_1^{4-j}v_2^j$ on $T = uv^t$.

Proof. Give a, b, c, d degree one and the auxiliary coefficients degree zero. The coordinate ring of the rank-one cone is the diagonal subring

$$A_0[a, b, c, d]/(\delta) \simeq \bigoplus_{r \geq 0} \text{Sym}^r(A_0^2)_u \otimes \text{Sym}^r(A_0^2)_v, \quad T = uv^t, \quad (\text{S7.6})$$

where A_0 denotes the auxiliary ring. This follows directly from $a = u_1v_1$, $b = u_1v_2$, $c = u_2v_1$, $d = u_2v_2$ and the single relation $ad - bc$; the monomials on either side give the same basis. In degree r , vanishing on the four distinct u -lines means divisibility by h in the u variables. The resulting module is zero for $r < 4$, and for $r \geq 4$ it is

$$h \text{Sym}^{r-4}(A_0^2)_u \otimes \text{Sym}^r(A_0^2)_v.$$

The simple-root discriminant is a unit, so the elementary divisibility and interpolation assertions used here hold over A_0 as well as over its residue field. Multiplication of binary forms shows that this ideal is generated in diagonal degree four by $hv_1^{4-j}v_2^j$. Lifting those generators proves the first identity.

The four 3-minors of the linear block $L(T)$ generate $\delta \mathfrak{m}$, by direct expansion. Modulo δ , its four columns are

$$v_1B(h_1u), \quad v_2B(h_1u), \quad v_1B(h_2u), \quad v_2B(h_2u),$$

and the three quadratic columns are $v_1^2C(u^2)$, $v_1v_2C(u^2)$, and $v_2^2C(u^2)$. Here u^2 has coefficient vector $(u_1^2, 2u_1u_2, u_2^2)^t$. A minor using one column from each of the two linear groups and one quadratic column is h times a degree-four monomial in v ; these span all five such monomials. Thus they furnish the lifts g_j . A minor with at least two quadratic columns vanishes modulo δ , and its degree is at least five, so it already lies in $\delta \mathfrak{m}$. All other minors have been accounted for. This proves $J = \delta \mathfrak{m} + (g_j)$ exactly.

Finally I is generated by its degree-two element δ and the g_j of degree four. Its intersection with $\mathfrak{m}^3$ deletes only the scalar multiple of δ in degree two, proving the second equality. Localizing in the auxiliary parameters or taking the indicated étale extension preserves these explicit identities. $\square$

Theorem S7.5 (The corank-two primary structure). *For an admissible pair the ideal δJ has the decomposition (S5.2). It is irredundant. In the local failure ring its nilradical $\widehat{\mathcal{N}}$ satisfies*

$$\widehat{\mathcal{N}}^3 = 0, \quad \widehat{\mathcal{N}}^2 \neq 0, \quad \text{Ann}(\widehat{\mathcal{N}}^2) = \mathfrak{m}/(\delta J). \quad (\text{S7.7})$$

Consequently the rank-two stratum is intrinsic to this local scheme as the annihilator support of $\widehat{\mathcal{N}}^2$.

Proof. Work on a connected regular étale auxiliary chart, so that its coefficient ring is a domain. Each P_i is a prime generated by two independent linear coordinates. Its ordinary square is P_i -primary; the same argument applies to $\mathfrak{m}^5$. The ideal (δ) is prime. At the generic point of P_i , the determinant has order one, because the other row after choosing ℓ_i as a coordinate line is not zero. Therefore

$$(P_i^2 : \delta) = P_i, \quad (\mathfrak{m}^5 : \delta) = \mathfrak{m}^3.$$

The second equality also follows immediately from the homogeneous order two of δ and the polynomial associated graded ring of $\mathfrak{m}$. Intersecting with (δ) and cancelling δ now gives

$$(\delta) \cap \bigcap_i P_i^2 \cap \mathfrak{m}^5 = \delta \left(\bigcap_i P_i \cap \mathfrak{m}^3 \right) = \delta J$$

by Lemma S7.4.

At the generic point of the determinant divisor all the other ideals are units, so (δ) is necessary. At the generic point of P_i the other P_j and $\mathfrak{m}$ become units, and the local ideal is $\delta P_i = (\delta) \cap P_i^2$; its two components are necessary. To see that $\mathfrak{m}^5$ is necessary, note that δ^2 lies in (δ) and in every P_i^2 , but is not in δJ : $\delta \notin J$ because its degree is two. On the other hand $\mathfrak{m}\delta^2 \subset \delta J$ since $\delta \mathfrak{m} \subset J$. This also proves that the annihilator of the class of δ^2 is exactly $\mathfrak{m}$; a nonzero auxiliary coefficient cannot kill that degree-four class.

The radical of the failure ideal is (δ) , so the nilradical is generated by the class of δ . Its square is the nonzero class just considered and its cube vanishes, since $\delta^2 \in \delta \mathfrak{m} \subset J$. This proves (S7.7). Over the auxiliary ring the square is free of rank one on the rank-two stratum. Intrinsically its transition functions are those of $\mathcal{I}_{\Delta}^2$ restricted to that stratum; thus it is $\mathcal{O}_{\Sigma}(-2\hat{\Delta})$ on the admissible open Σ . The generic transverse length is one. $\square$

Proof of Theorem S5.2. Lemma S7.2 supplies the asserted nonempty open class. Lemmas S7.3 and S7.4 and Theorem S7.5 supply the fixed-tensor calculation on every indicated frame chart. The graph coordinates are smooth auxiliary variables, so no transverse component has been discarded. The individual P_i are allowed to permute under monodromy; the intersection and the intrinsic annihilator in (S7.7) descend. Four local branches need not be four distinct global irreducible components. This is why the primary decomposition was stated étale locally rather than with globally labelled roots. $\square$

S8. RESIDUAL PRESENTATIONS AND THE RANK-TWO CONTACT BOUNDARY

We first record the projection-rank block presentation inherited from the full Fitting formula, and then isolate the rank-two structure to which it gives rise. The distinction between a presentation theorem and a structure theorem is important: the block formula below is exact at every projection rank, whereas the associated-prime, collision, and nilpotent-filtration theorems are proved on the loci explicitly stated. Section S9 pushes the structure theorem through all mixed-regular contact collisions, through the containment locus, and across the first non-mixed-regular tensor-rank wall.

For $0 \leq r \leq 3$, let $\Sigma_R^{(r)} \subset X_R$ be the locally closed stratum on which the projection $\mathcal{K} \rightarrow V$ has rank exactly r . If $K \in \Sigma_R^{(r)}$ has image $H \subset V$, choose a complement $V = H \oplus W$, where $c = \dim W = 4 - r$. On a frame chart around K , an invertible r -minor and Schur reduction put the projection block in the form

$$M = \begin{pmatrix} I_H & 0 \\ 0 & T \end{pmatrix}, \quad T \in \text{Mat}_{c \times c}.$$

Decompose the quotient multiplication tensor according to

$$\text{Sym}^2 V = \text{Sym}^2 H \oplus (H \otimes W) \oplus \text{Sym}^2 W$$

and write the corresponding blocks as A_H, B_H, C_H .

Theorem S8.1 (Projection-rank residual block presentation). *For every $R \in G_4^\circ$, every $0 \leq r \leq 3$, and every $K \in \Sigma_R^{(r)}$, one has on the preceding frame chart, before taking radicals,*

$$\mathcal{I}_{\widehat{D}_R} = (\det T) I_6 [A_H, B_H(1_H \otimes T), C_H \operatorname{Sym}^2 T]. \quad (\text{S8.1})$$

The formula is independent of any injectivity, mixed-rank, tensor-rank, or contact-transversality assumption. Thus every projection-rank stratum has the same exact block presentation. No claim of a primary or associated-prime classification on the non-mixed-regular corank-three or corank-four boundary is made by this formula alone; the separate structure theorems below derive additional information where the residual algebra can be determined.

Proof. Proposition 4.1 gives on every frame chart

$$\mathcal{I}_{\widehat{D}_R} = (\det M) I_6 (\gamma_R \operatorname{Sym}^2 M).$$

The Schur operations placing M in the displayed block form are invertible and therefore change neither the maximal-minor ideal nor the local scheme. On $\operatorname{Sym}^2 H \oplus (H \otimes W) \oplus \operatorname{Sym}^2 W$ the induced map $\operatorname{Sym}^2 M$ is respectively

$$I_{\operatorname{Sym}^2 H}, \quad 1_H \otimes T, \quad \operatorname{Sym}^2 T.$$

Since $\det M$ is a unit times $\det T$, substitution proves (S8.1). $\square$

We now specialize this exact all-corank formula to $r = 2$. Write $\Sigma_R = \Sigma_R^{(2)}$. Then $c = 2$, and after choosing bases

$$T = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad \delta = \det T = ad - bc.$$

Decompose the quotient multiplication tensor according to

$$\operatorname{Sym}^2 V = \operatorname{Sym}^2 H \oplus (H \otimes W) \oplus \operatorname{Sym}^2 W$$

and write

$$A_H : \operatorname{Sym}^2 H \rightarrow S_R, \quad B_H : H \otimes W \rightarrow S_R, \quad C_H : \operatorname{Sym}^2 W \rightarrow S_R.$$

Theorem S8.2 (Exact presentation on every rank-two chart). *In the preceding coordinates one has, before taking radicals,*

$$\mathcal{I}_{\widehat{D}_R} = (\delta) I_6 [A_H, B_H(1_H \otimes T), C_H \operatorname{Sym}^2 T]. \quad (\text{S8.2})$$

The formula is valid at every projection-rank-two point; no injectivity, mixed-rank, tensor-rank, or simple-contact hypothesis is required.

Proof. Proposition 4.1 gives

$$\mathcal{I}_{\widehat{D}_R} = (\det M) I_6 (\gamma_R \operatorname{Sym}^2 M).$$

For $M = I_H \oplus T$, the induced map on the three summands of $\operatorname{Sym}^2(H \oplus W)$ is respectively

$$I_{\operatorname{Sym}^2 H}, \quad 1_H \otimes T, \quad \operatorname{Sym}^2 T.$$

Substitution gives (S8.2). All row and column operations used to reach the block form are invertible, so the maximal-minor ideal is unchanged. $\square$

Thus the exceptional rank-two locus is included in the exact presentation. Determining radicals, associated primes, and nilpotent layers of its residual factor is a separate problem. The mixed-regular case and its first tensor-rank boundary are solved below and in Section S9.

Definition S8.3. A pair (R, H) is *mixed-regular* if A_H is injective and, after quotienting S_R by its image, the induced map

$$\overline{B}_H : H \otimes (V/H) \longrightarrow S_R / \text{im } A_H$$

is surjective with one-dimensional kernel generated by a rank-two tensor. No hypothesis is imposed on the multiplicities of the four ramification contacts.

These are precisely the first three conditions in Definition S7.1; the discriminant condition has been removed. On this locus the same Gaussian elimination as in Lemma S7.3 is valid and yields

$$\mathcal{I}_{\widehat{D}_R} = \delta J, \quad J = I_3[L(T), C T^{(2)}],$$

with $L(T)$ and $T^{(2)}$ as in (S7.3).

Lemma S8.4 (Root-free residual normal form). *Over the local auxiliary coefficient ring of a mixed-regular pair,*

$$J = \delta \mathbf{m} + (g_0, \dots, g_4), \quad \mathbf{m} = (a, b, c, d), \quad (\text{S8.3})$$

where each g_i is homogeneous of transverse degree four. Modulo (δ) , under the Segre parametrization $T = uv^t$, their classes span

$$h_H(u) \cdot \text{Sym}^4(\mathbb{C}_v^2).$$

Here h_H is the binary quartic obtained by restricting the ramification equation to the pencil of hyperplanes containing H . The identity is valid with repeated roots, and also when the discriminant of h_H vanishes.

Proof. The four 3-minors of the linear block $L(T)$ generate $\delta \mathbf{m}$, exactly as in Lemma S7.4. Modulo δ , write $T = uv^t$. The two pairs of linear columns are scalar multiples, by v_1, v_2 , of the two mixed columns determined by u , while the three quadratic columns are scalar multiples, by $v_1^2, v_1 v_2, v_2^2$, of $C(u^2)$. A minor containing one column from each linear pair and one quadratic column is therefore $h_H(u)$ times a degree-four monomial in v ; the five choices span $\text{Sym}^4(\mathbb{C}_v^2)$.

Every minor containing at least two quadratic columns vanishes on the rank-one cone, hence is divisible by the prime equation δ . Its transverse degree is at least five, so the quotient by δ lies in $\mathbf{m}$. The remaining minors are the already listed linear and mixed ones. This proves (S8.3). No factorization of h_H was used. When its roots are distinct, factorizing h_H recovers Lemma S7.4. $\square$

Theorem S8.5 (Depth two across contact collisions). *At every mixed-regular projection-rank-two point, the radical of $\mathcal{I}_{\widehat{D}_R}$ is (δ) and the nilradical $\widehat{\mathcal{N}}$ of the local failure ring satisfies*

$$\widehat{\mathcal{N}}^3 = 0, \quad \widehat{\mathcal{N}}^2 \neq 0, \quad \text{Ann}(\widehat{\mathcal{N}}^2) = \mathbf{m}/(\delta J).$$

Consequently $\mathcal{Z}_2(\widehat{D}_R)$ is the rank-two stratum on the mixed-regular locus, even when the contact quartic has multiple roots. Moreover

$$\widehat{\mathcal{N}}^2|_{\Sigma_R} \simeq \mathcal{I}_{\Delta_R}^2|_{\Sigma_R}.$$

Proof. By Lemma S8.4, $J \subset \mathbf{m}^3$ and $\delta \mathbf{m} \subset J$. Since $\delta \in \mathbf{m}^2$, we have $\delta^2 \in J$, hence $\delta^3 \in \delta J$. On the other hand $\delta \notin J$ by transverse degree, so $\delta^2 \notin \delta J$. Since $\delta J \subset (\delta)$ and a power of δ belongs to δJ , the radical is exactly (δ) . Thus the nilradical is generated by the class of δ and has exact index three.

For the annihilator, cancellation in the regular local coefficient ring gives

$$(\delta J : \delta^2) = (J : \delta).$$

The inclusion $\mathfrak{m} \subset (J : \delta)$ follows from $\delta\mathfrak{m} \subset J$. Conversely, use the transverse grading: J has no component of transverse degree two, whereas δ has degree two. If the transverse degree-zero part of r were nonzero, the degree-two part of $r\delta$ could not lie in J . Hence $r \in \mathfrak{m}$, proving $(J : \delta) = \mathfrak{m}$.

Finally δ is a local equation of the Schubert divisor $\hat{\Delta}_R$. Therefore the generator δ^2 of the square of the nilradical transforms as the square of the conormal line. This gives the displayed line bundle after restriction to the rank-two stratum. $\square$

We next globalize the four contacts without labelling them. Put

$$G = \text{Gr}(2, V)$$

and let $\mathcal{Q}$ be its universal rank-two quotient bundle. The quartic equation F_R of $Y_R \subset \mathbb{P}(V^*)$ restricts to every line of hyperplanes containing H . Equivalently it defines a global binary quartic

$$h_R \in H^0(G, \text{Sym}^4 \mathcal{Q}). \quad (\text{S8.4})$$

Define the contact incidence

$$\mathcal{R}_R = \{(H, [\alpha]) \in G \times Y_R : H \subset \ker \alpha\}. \quad (\text{S8.5})$$

Proposition S8.6 (The global contact cover). *The projection $\mathcal{R}_R \rightarrow Y_R$ is a $\mathbb{P}^2$ -bundle; in particular $\mathcal{R}_R$ is smooth and irreducible. Over the open set $G_R^{\text{fin}} \subset G$ on which the binary quartic (S8.4) is not identically zero, the map*

$$\mathcal{R}_R \longrightarrow G$$

is finite flat of degree four. There is a globally defined discriminant divisor

$$\mathfrak{D}_R = V(\text{Disc}(h_R)), \quad \text{Disc}(h_R) \in H^0(G, (\det \mathcal{Q})^{12}).$$

Whenever this section is not identically zero, $[\mathfrak{D}_R] = 12H$ in the Pluecker class group. On the finite-flat locus G_R^{fin} , the support of the branch divisor is the restriction of $\mathfrak{D}_R$; away from $\mathfrak{D}_R$ the degree-four cover is finite etale. The monodromy action on the four sheets over that etale locus is transitive.

Proof. For $[\alpha] \in Y_R$, the fibre of $\mathcal{R}_R \rightarrow Y_R$ consists of the two-planes in the three-dimensional hyperplane $\ker \alpha$, hence is $\text{Gr}(2, \ker \alpha) \simeq \mathbb{P}^2$. These fibres form the relative Grassmannian of the rank-three hyperplane bundle on Y_R , proving smoothness and irreducibility.

For fixed H , the other projection has fibre $Y_R \cap \mathbb{P}(H^\perp)$, the zero scheme of the restricted quartic. On the locus where the restriction is nonzero this is a relative effective Cartier divisor of degree four in the $\mathbb{P}^1$ -bundle $\mathbb{P}(\mathcal{Q}^*)$, hence finite flat of degree four.

The discriminant of a binary quartic is homogeneous of degree six in its coefficients and is a relative invariant of determinant weight twelve. Applied to the section of $\text{Sym}^4 \mathcal{Q}$ in (S8.4), it is therefore a section of $(\det \mathcal{Q})^{12}$. Since $\det \mathcal{Q}$ is the Pluecker line bundle up to the constant factor $\det V$, its zero divisor has class $12H$ whenever it is nonzero. On G_R^{fin} the fibre is a degree-four Cartier divisor on a projective line, so its non-etale locus is exactly the multiple-root locus, i.e. the restriction of the discriminant locus. The total incidence is irreducible; therefore the monodromy action of the etale restriction on a geometric fibre is transitive. $\square$

For R in the nonempty open $G_4^\dagger$ of Lemma S7.2, the discriminant in Proposition S8.6 is not identically zero. Let $G_R^{\text{mr}} \subset G$ denote the mixed-regular locus of Definition S8.3. The contact sheets themselves split etale locally over $G_R^{\text{fin}} \setminus \mathfrak{D}_R$. Their interpretation as the four primary ideals P_i of Theorem S7.5, however, also requires the independent mixed-regular rank conditions. Thus the exact locus for that primary decomposition is

$$G_R^{\text{mr}} \cap (G_R^{\text{fin}} \setminus \mathfrak{D}_R),$$

not the discriminant complement by itself. On this intersection the P_i are the four etale contact sheets and their permutation is the monodromy of the canonical cover (S8.5).

The collision divisor has the five nonzero fibre types corresponding to the partitions

$$1 + 1 + 1 + 1, \quad 2 + 1 + 1, \quad 2 + 2, \quad 3 + 1, \quad 4$$

of the degree of a binary quartic, together with the separate containment locus $h_H \equiv 0$. Theorem S8.5 does not change across these collision types. What changes is the primary splitting of the contact part, so we do not force the simple-root intersection formula onto the boundary.

The ordinary fifth power of the rank-two Schubert ideal is of course globally defined. The stronger assertion that it remains the actual $\mathfrak{m}$ -primary component of the failure ideal through every mixed-regular collision, including the zero-quartic containment locus, is proved in Theorem S9.2; it is not inferred merely from the existence of the global ideal. The same theorem gives the exact collided primary types and separates the finite contact geometry from the containment geometry.

Corollary S8.7 (Rank-two part of Theorem S5.1). *The contact-incidence and collision assertions of Theorem S5.1 hold on the stated mixed-regular locus.*

Proof. Theorem S8.5 identifies depth two and its line bundle, while Proposition S8.6 globalizes the contact incidence and discriminant. The global residual-colon filtration, the simultaneous identification of the first two graded pieces, the collided primary components, the containment locus, and the decomposable-kernel wall are proved in Section S9. $\square$

S9. THE STRATIFIED NILPOTENT ALGEBRA AT PROJECTION CORANK TWO

The root-free formula of Lemma S8.4 contains more scheme structure than the nilpotency-index statement extracted from it in the preceding local calculations. We record that structure before passing to the exceptional mixed-kernel boundary.

Theorem S9.1 (Global residual-colon filtration). *Let $X = X_R$, let $\widehat{\Delta}_R \subset X$ be the reduced Schubert divisor, and put*

$$\mathcal{J}_R^{\text{res}} = (\mathcal{I}_{\widehat{D}_R} : \mathcal{I}_{\widehat{\Delta}_R}) \subset \mathcal{O}_X.$$

Then

$$\mathcal{I}_{\widehat{D}_R} = \mathcal{I}_{\widehat{\Delta}_R} \mathcal{J}_R^{\text{res}}. \quad (\text{S9.1})$$

For every $j \geq 1$,

$$\text{Ann}_{\mathcal{O}_{\widehat{D}_R}}(\widehat{\mathcal{N}}_R^j) = \frac{(\mathcal{J}_R^{\text{res}} : \mathcal{I}_{\widehat{\Delta}_R}^{j-1})}{\mathcal{I}_{\widehat{D}_R}}. \quad (\text{S9.2})$$

Define the closed subscheme of the reduced divisor

$$W_j = V_{\widehat{\Delta}_R} \left((\mathcal{J}_R^{\text{res}} : \mathcal{I}_{\widehat{\Delta}_R}^{j-1}) + \mathcal{I}_{\widehat{\Delta}_R} \right). \quad (\text{S9.3})$$

If

$$\mathcal{L} = \mathcal{I}_{\widehat{\Delta}_R} / \mathcal{I}_{\widehat{\Delta}_R}^2$$

is the conormal line of the Schubert divisor, then there are canonical isomorphisms

$$\widehat{\mathcal{N}}_R^j / \widehat{\mathcal{N}}_R^{j+1} \simeq \mathcal{L}^{\otimes j} \otimes_{\mathcal{O}_{\widehat{\Delta}_R}} \mathcal{O}_{W_j}. \quad (\text{S9.4})$$

Consequently

$$\text{gr}_{\widehat{\mathcal{N}}_R} \mathcal{O}_{\widehat{D}_R} \simeq \mathcal{O}_{\widehat{\Delta}_R} \oplus \bigoplus_{j \geq 1} \mathcal{L}^{\otimes j} \otimes \mathcal{O}_{W_j}, \quad (\text{S9.5})$$

with the multiplication induced by multiplication of powers of $\mathcal{I}_{\widehat{\Delta}_R}$. The sum is finite.

These formulas hold on the whole Grassmannian, including every projection corank and every rank-drop locus. On a Schur chart with local equation $d = \det T$ and residual ideal J , they become

$$\text{Ann}(N^j) = (J : d^{j-1})/(dJ), \quad N^j/N^{j+1} \simeq A/((d) + (J : d^{j-1})). \quad (\text{S9.6})$$

In particular the nilpotency index at the point is

$$1 + \min\{k \geq 0 : d^k \in J\}. \quad (\text{S9.7})$$

Proof. Proposition 4.1 shows locally that $\mathcal{I}_{\widehat{D}_R} = dJ$, where d is a local equation of the Cartier divisor $\widehat{\Delta}_R$. Since d is a non-zero-divisor in the regular local ring of X ,

$$(dJ : d) = J.$$

Thus J is the local expression of the intrinsic colon $\mathcal{J}_R^{\text{res}}$, and the local identities glue to (S9.1).

In $A/(dJ)$ the nilradical is the image of (d) , because the reduction is $A/(d)$. Therefore

$$\text{Ann}(N^j) = (dJ : d^j)/(dJ) = (J : d^{j-1})/(dJ),$$

which proves (S9.2). For the graded piece, the map

$$A \longrightarrow N^j/N^{j+1}, \quad r \longmapsto rd^j$$

has kernel

$$(d) + (J : d^{j-1}).$$

Indeed

$$rd^j \in d^{j+1} + dJ \iff rd^{j-1} \in d^j + J,$$

and the latter is equivalent to writing $r = ds + t$ with $td^{j-1} \in J$. This proves the local form of (S9.4). On overlaps local equations of the Cartier divisor differ by units, and d^j transforms by the j -th tensor power of the conormal transition function. Hence the local identifications glue to (S9.4) and (S9.5). Nilpotence of the nilradical makes the sum finite, and the same cancellation gives (S9.7). $\square$

The theorem converts the globally defined annihilator filtration into an exact residual-colon filtration. What remains is to identify the colon schemes W_j geometrically on the successive rank strata. For projection corank one, $W_1 = E_R$ and all higher W_j are empty. The next two results compute W_1, W_2 through the entire mixed-regular rank-two boundary and then across its first tensor-rank wall.

Work on a mixed-regular rank-two chart. Let A_0 be the regular auxiliary coefficient ring, put

$$P = A_0[a, b, c, d], \quad \mathfrak{m} = (a, b, c, d), \quad \delta = ad - bc,$$

and localize, or complete, at the point under consideration. Thus

$$\mathcal{I}_{\widehat{D}_R} = \delta J, \quad J = \delta \mathfrak{m} + (g_0, \dots, g_4),$$

where the g_i have transverse degree four. Define the *unsplit contact ideal*

$$I_H = (\delta, g_0, \dots, g_4). \quad (\text{S9.8})$$

Modulo δ , the Segre parametrization $T = uv^t$ identifies the homogeneous coordinate ring with

$$P/(\delta) \simeq \bigoplus_{n \geq 0} \text{Sym}^n(\mathbb{C}_u^2) \otimes \text{Sym}^n(\mathbb{C}_v^2),$$

and the image of I_H in degree n is zero for $n < 4$ and

$$h_H \text{Sym}^{n-4}(\mathbb{C}_u^2) \otimes \text{Sym}^n(\mathbb{C}_v^2) \quad (n \geq 4). \quad (\text{S9.9})$$

Theorem S9.2 (Saturated contact ideal and the fixed vertex component). *At every mixed-regular rank-two point one has the exact identities*

$$J = I_H \cap \mathfrak{m}^3 \quad (\text{S9.10})$$

and

$$\mathcal{I}_{\widehat{D}_R} = \delta I_H \cap \mathfrak{m}^5. \quad (\text{S9.11})$$

If $h_H \not\equiv 0$, the ideal I_H is $\mathfrak{m}$ -saturated. Put $\mathcal{C}_H = V(I_H)$ for the affine contact cone. Its projectivization on the Segre quadric

$$Q = \mathbb{P}_u^1 \times \mathbb{P}_v^1 \subset \mathbb{P}^3$$

is the effective Cartier divisor

$$C_H^{\text{proj}} = V(h_H) \times \mathbb{P}_v^1 \quad (\text{S9.12})$$

of class $(4, 0)$. Consequently δI_H is also $\mathfrak{m}$ -saturated, and $\mathfrak{m}^5$ in (S9.11) is an actual irredundant $\mathfrak{m}$ -primary component, not merely a globally defined ideal.

If

$$h_H = \prod_{\nu=1}^s \ell_\nu^{e_\nu}, \quad \sum_\nu e_\nu = 4,$$

over an etale coefficient extension, and P_ν denotes the height-two contact prime corresponding to ℓ_ν , then

$$\text{Ass}(P/\mathcal{I}_{\widehat{D}_R}) = \{(\delta), P_1, \dots, P_s, \mathfrak{m}\}. \quad (\text{S9.13})$$

At the generic point of P_ν , choose a second regular parameter q_ν so that $P_\nu = (\delta, q_\nu)$. The P_ν -primary part of δI_H has the exact two-parameter normal form

$$Q_{\nu, e_\nu} = (\delta^2, \delta q_\nu^{e_\nu}, q_\nu^{e_\nu+1}). \quad (\text{S9.14})$$

For $e_\nu = 1$ this is P_ν^2 ; for a collided contact it is the corresponding non-ordinary primary thickening.

If $h_H \equiv 0$, then

$$I_H = (\delta), \quad J = \delta \mathfrak{m}, \quad \mathcal{I}_{\widehat{D}_R} = \delta^2 \mathfrak{m} = (\delta^2) \cap \mathfrak{m}^5. \quad (\text{S9.15})$$

In particular the rank-two vertex component remains $\mathfrak{m}^5$ on the containment locus.

Proof. The equality (S9.10) is a graded calculation. The ideal I_H is generated in transverse degrees two and four, its only degree-two generator being δ . Intersecting with $\mathfrak{m}^3$ therefore removes exactly the scalar multiple of δ and leaves $\delta \mathfrak{m} + (g_0, \dots, g_4) = J$.

Since the initial form of δ has degree two and is nonzero in the polynomial associated graded ring,

$$(\mathfrak{m}^5 : \delta) = \mathfrak{m}^3.$$

Cancellation in the regular ring gives

$$\delta I_H \cap \mathfrak{m}^5 = \delta(I_H \cap (\mathfrak{m}^5 : \delta)) = \delta(I_H \cap \mathfrak{m}^3) = \delta J,$$

which proves (S9.11).

Assume first $h_H \neq 0$. Formula (S9.9) is precisely the section ideal of $\mathcal{O}_Q(-4, 0) \subset \mathcal{O}_Q$, cut out by the nonzero section h_H of $\mathcal{O}_Q(4, 0)$. Hence it is saturated and defines the Cartier divisor (S9.12). The inverse image $I_H \subset P$ is therefore $\mathfrak{m}$ -saturated. The product δI_H is saturated as well: if $\mathfrak{m}^N f \subset \delta I_H$, choose a coordinate not divisible by the prime element δ to see that $\delta \mid f$; after cancelling δ , saturation of I_H applies. Thus $\mathfrak{m} \notin \text{Ass}(P/\delta I_H)$. On the other hand $\delta^2 \in \delta I_H$ has transverse degree four and does not belong to $\mathfrak{m}^5$, so the $\mathfrak{m}^5$ factor in (S9.11) is irredundant.

The divisor C_H^{proj} has components $e_\nu(\{\ell_\nu = 0\} \times \mathbb{P}^1)$. Hence P/I_H is Cohen–Macaulay of pure codimension two and its associated primes are exactly the P_ν . Multiplication by δ gives an exact sequence

$$0 \longrightarrow P/I_H \xrightarrow{\delta} P/\delta I_H \longrightarrow P/(\delta) \longrightarrow 0.$$

It follows, and localization shows equality, that the associated primes of $P/\delta I_H$ are $(\delta), P_1, \dots, P_s$. Intersecting with the irredundant $\mathfrak{m}$ -primary ideal $\mathfrak{m}^5$ gives (S9.13).

At the generic point of P_ν , the Cartier divisor has equation $q_\nu^{e_\nu}$, so

$$(I_H)_{P_\nu} = (\delta, q_\nu^{e_\nu}), \quad (\delta I_H)_{P_\nu} = (\delta^2, \delta q_\nu^{e_\nu}).$$

The ideal

$$(\delta^2, \delta q_\nu^{e_\nu}, q_\nu^{e_\nu+1})$$

is P_ν -primary, and its colon by δ is $(\delta, q_\nu^{e_\nu})$. Therefore

$$(\delta) \cap Q_{\nu, e_\nu} = \delta(\delta, q_\nu^{e_\nu}),$$

which proves the asserted generic primary normal form.

If $h_H = 0$, every g_i vanishes modulo δ . Since it has transverse degree four, $g_i = \delta q_i$ with $q_i \in \mathfrak{m}^2$. Thus $I_H = (\delta)$ and $J = \delta \mathfrak{m}$. Finally

$$(\delta^2) \cap \mathfrak{m}^5 = \delta^2(\mathfrak{m}^5 : \delta^2) = \delta^2 \mathfrak{m},$$

again by the polynomial associated graded ring. This is (S9.15). $\square$

The preceding theorem gives the requested specialization law without labelling roots. A collision changes the Cartier multiplicities in C_H and hence the primary type (S9.14); it does not change the vertex component. The zero quartic is the unique point at which the Cartier contact divisor ceases to be a divisor and fills the whole Segre quadric.

Theorem S9.3 (Associated graded nilpotent algebra across coranks one and two). *On a mixed-regular rank-two chart let*

$$A = P/(\delta J), \quad N = \sqrt{0_A}.$$

Then $N = (\delta)/(\delta J)$, $N^3 = 0 \neq N^2$, and

$$N/N^2 \simeq P/I_H, \tag{S9.16}$$

$$N^2 \simeq P/\mathfrak{m}. \tag{S9.17}$$

Equivalently, with $\mathcal{L}$ the conormal line of the Schubert divisor, the associated graded algebra has the intrinsic form

$$\text{gr}_N \mathcal{O}_{\widehat{D}_R} \simeq \mathcal{O}_{\widehat{\Delta}_R} \oplus (\mathcal{L} \otimes \mathcal{O}_{C_H}) \oplus (\mathcal{L}^{\otimes 2} \otimes \mathcal{O}_{\Sigma_R}) \tag{S9.18}$$

on the mixed-regular neighbourhood, with multiplication induced by the incidence of the affine contact cone C_H with the rank-two vertex Σ_R . Moreover

$$\text{Ann}_A(N) = J/(\delta J), \quad \text{Ann}_A(N^2) = \mathfrak{m}/(\delta J), \tag{S9.19}$$

so

$$\mathcal{Z}_1 = V(I_H \cap \mathfrak{m}^3), \quad \mathcal{Z}_2 = V(\mathfrak{m}).$$

On the punctured determinant cone, i.e. on the adjacent projection-corank-one locus, $\mathcal{Z}_1$ is exactly the ramification support E_R . Thus C_H is its scheme-theoretic closure away from the vertex, while $\mathcal{Z}_1$ records in addition the canonical third-order vertex thickening forced by the Fitting ideal.

Proof. The radical of δJ is (δ) , so the nilradical is generated by the class of δ . The isomorphism (S9.16) follows by cancelling one factor:

$$N/N^2 = \frac{(\delta)}{(\delta^2) + (\delta J)} \simeq \frac{P}{(\delta) + J} = \frac{P}{I_H}.$$

For the square, multiplication by δ^2 identifies its kernel with

$$(\delta J : \delta^2) = (J : \delta) = \mathfrak{m}$$

by Theorem S8.5; hence (S9.17). The transition functions of the local generator δ are those of the conormal line, giving (S9.18).

Cancellation also gives

$$(\delta J : \delta) = J,$$

while the preceding colon calculation gives the second formula in (S9.19). Away from the vertex $\mathfrak{m}$ becomes the unit ideal, so $J = I_H$. On the smooth rank-one part of the determinant divisor, the corank-one normal form $t(t, f)$ identifies this contact equation with the ramification equation. Hence the restriction is precisely E_R . $\square$

We finally cross the first non-mixed-regular wall. This is the rank-one tensor wall in the one-dimensional mixed kernel, explicitly requested by the referee.

Definition S9.4. Assume A_H is injective and $\overline{B}_H$ is surjective with one-dimensional kernel generated by a decomposable tensor $h_1 \otimes w_1$. Choose h_2, w_2 completing bases and put

$$q_1 = \overline{B}_H(h_1 \otimes w_2), \quad q_2 = \overline{B}_H(h_2 \otimes w_1).$$

We call the point *transverse decomposable-kernel regular* if

$$\det[q_1, q_2, \overline{C}_H(w_1^2)] \neq 0.$$

This condition is independent of the chosen completing vectors. It says that the contact root forced by the decomposable kernel is simple in the direction transverse to the kernel wall.

Theorem S9.5 (The decomposable mixed-kernel wall). *At a transverse decomposable-kernel regular rank-two point there are bases for which*

$$L_{\text{dec}}(T) = \begin{pmatrix} c & d & 0 & 0 \\ 0 & 0 & a & b \\ 0 & 0 & c & d \end{pmatrix}. \quad (\text{S9.20})$$

Let

$$J_{\text{dec}} = I_3[L_{\text{dec}}(T), C T^{(2)}], \quad K_{\text{dec}} = (c, d, a^2, ab, b^2).$$

Then

$$(J_{\text{dec}} : \delta) = K_{\text{dec}}. \quad (\text{S9.21})$$

Hence the local failure ring still has nilradical of exact index three, but its second intrinsic depth support is no longer reduced:

$$\mathcal{Z}_2 = V(K_{\text{dec}}). \quad (\text{S9.22})$$

Its support is the rank-two vertex, and its transverse local length is three. Thus crossing the tensor-rank-two wall changes the scheme structure of $\mathcal{Z}_2$ before it changes its set-theoretic support.

Proof. The stated normal form is obtained by taking $\ker \overline{B}_H = \mathbb{C}(h_1 \otimes w_1)$ and using

$$q_1, q_2, q_3 = \overline{B}_H(h_2 \otimes w_2)$$

as target basis. Applying $1_H \otimes T$ gives (S9.20). Its maximal minors generate

$$I_3(L_{\text{dec}}) = \delta(c, d).$$

After replacing the first quadratic target column by its component modulo $\langle q_1, q_2 \rangle$, the transversality assumption allows us to normalize $\overline{C}_H(w_1^2) \equiv q_3$. The minors containing one column from each mixed group and this quadratic column give

$$\delta a^2, \quad \delta ab, \quad \delta b^2 \in J_{\text{dec}}.$$

Together with $I_3(L_{\text{dec}})$ this proves $\delta K_{\text{dec}} \subset J_{\text{dec}}$, hence $K_{\text{dec}} \subset (J_{\text{dec}} : \delta)$.

For the reverse inclusion use the transverse grading. The degree-three part of J_{dec} is exactly $\delta(c, d)$; every minor using a quadratic column has degree at least four. Modulo K_{dec} , every class is represented by

$$\lambda + \mu a + \nu b.$$

If $r\delta \in J_{\text{dec}}$, the degree-two term rules out $\lambda \neq 0$, while the degree-three identity

$$(\mu a + \nu b)\delta \in \delta(c, d)$$

forces $\mu = \nu = 0$. Thus $r \in K_{\text{dec}}$, proving (S9.21).

Since $\delta \in (c, d)$, one has $\delta^2 \in \delta(c, d) \subset J_{\text{dec}}$, whereas $\delta \notin J_{\text{dec}}$ by degree. Therefore the nilradical of $P/(\delta J_{\text{dec}})$ has exact index three. Finally

$$\text{Ann}(N^2) = (\delta J_{\text{dec}} : \delta^2) = (J_{\text{dec}} : \delta) = K_{\text{dec}}.$$

The quotient by K_{dec} has basis $1, a, b$, proving the length assertion. $\square$

Proposition S9.6 (The containment locus is the finite line scheme). *For fixed $R \in G_4^\circ$, the locus*

$$\mathcal{C}_R = \{H \in \text{Gr}(2, V) : h_H \equiv 0\}$$

is canonically the Fano scheme $F_1(Y_R)$ of lines on the smooth quartic Y_R . It is finite and reduced. At its mixed-regular points the local failure ideal is (S9.15), and the contact morphism $\mathcal{R}_R \rightarrow \text{Gr}(2, V)$ has fibre $\mathbb{P}^1$, rather than a finite length-four fibre.

Proof. The line of hyperplanes containing H is $\mathbb{P}(H^\perp) \subset \mathbb{P}(V^*)$. Its quartic restriction is h_H , so $h_H = 0$ identically exactly when $\mathbb{P}(H^\perp) \subset Y_R$. This identifies $\mathcal{C}_R$ with $F_1(Y_R)$.

If $L \subset Y_R$ is such a line, the normal sequence in $\mathbb{P}^3$ gives

$$\deg N_{L/Y_R} = \deg N_{L/\mathbb{P}^3} - \deg N_{Y_R/\mathbb{P}^3}|_L = 2 - 4 = -2.$$

Since N_{L/Y_R} has rank one, $N_{L/Y_R} \simeq \mathcal{O}_{\mathbb{P}^1}(-2)$, and therefore $H^0(N_{L/Y_R}) = 0$. The Zariski tangent space of the Fano scheme vanishes at every point. The Fano scheme is projective and zero-dimensional; a zero-dimensional local algebra with zero tangent space is a field by Nakayama. Hence it is finite reduced. The final two assertions follow from Theorem S9.2 and the definition of the incidence fibre. $\square$

S10. PROOFS FOR THE INTRINSIC BOUNDARY ATLAS

We prove the assertions of Section S6. All calculations are made in the regular coefficient ring of a Schur chart. Since the identities are invariant under changes of frames, they descend by the same argument as the Fitting identities in Proposition 4.1.

S10.1. Universal determinant completion.

Proof of Theorem S6.1. Put $N = \dim \text{Sym}^r V$ and $s = \dim S$. Choose a vector space K of dimension $N - s$ and a linear map $\rho : \text{Sym}^r V \rightarrow K$ such that

$$G = (\gamma, \rho) : \text{Sym}^r V \xrightarrow{\sim} S \oplus K$$

is an isomorphism. For the generic endomorphism M ,

$$\det(\text{Sym}^r M) = (\det M)^{\kappa(e, r)}, \quad \kappa(e, r) = \frac{r}{e} \binom{e + r - 1}{r} = \binom{e + r - 1}{r - 1}.$$

The eigenvalue identity proves the formula on diagonal matrices and hence as a polynomial identity on $\text{End}(V)$.

Expand

$$\det(G \text{Sym}^r M) = \det(G) d^{\kappa(e,r)}$$

by Laplace expansion along the first s rows. Every summand is an s -minor of $\gamma \text{Sym}^r M$ times a complementary minor of $\rho \text{Sym}^r M$. Therefore the determinant belongs to $J_{\gamma,r}$. Since $\det G \neq 0$, $d^{\kappa(e,r)} \in J_{\gamma,r}$.

If γ is an isomorphism, the ideal of maximal minors is principal and generated by $\det(\gamma) \det(\text{Sym}^r M)$, so the exponent is sharp. $\square$

S10.2. The depth bound and graded multiplication.

Proof of Theorem S6.2. Apply Theorem S6.1 with $e = 4$, $r = 2$, $\gamma = \gamma_R$, and $M = I_H \oplus T$. Since $\kappa(4, 2) = 5$, $d^5 \in I_6(\gamma_R \text{Sym}^2 M) = J$. Now $\mathcal{I}_{\widehat{D}_R} = dJ$. Therefore $d^6 \in dJ$, so the nilradical generated by the class of d satisfies $N^6 = 0$. Formula (S9.4) then gives $W_j = \emptyset$ for $j \geq 6$.

For the multiplication statement work locally in $A = P/(dJ)$. The class of rd^i in N^i/N^{i+1} depends on r modulo

$$(d) + (J : d^{i-1}),$$

and multiplication sends

$$(rd^i)(sd^j) = rs d^{i+j}.$$

Because

$$(J : d^{i-1}) + (J : d^{j-1}) \subset (J : d^{i+j-1}),$$

this factors through the canonical restriction to W_{i+j} . The image of $1 \otimes 1$ is the generator d^{i+j} , hence the map is surjective. The transition functions of d supply the factor $\mathcal{L}^{\otimes(i+j)}$, proving (S6.4). $\square$

S10.3. The first layer at projection corank two. Assume throughout this subsection that A_H is injective. After eliminating its image, the residual ideal is the ideal of 3-minors of

$$\Psi_H(T) = \left[\beta_H(1_H \otimes T), \chi_H \text{Sym}^2 T \right], \quad \dim E_H = 3. \quad (\text{S10.1})$$

Lemma S10.1 (The contact symbol without mixed regularity). *Modulo (δ) , the residual ideal of (S10.1) is generated in degree four and equals*

$$h_H(u) \text{Sym}^4(\mathbb{C}_v^2)$$

under $T = uv^t$. In particular

$$J + (\delta) = (\delta, g_0, \dots, g_4)$$

with the notation of Theorem S6.4.

Proof. On the rank-one cone write $T = uv^t$. The four mixed columns are scalar multiples, with coefficients v_1, v_2 , of the two vectors

$$\beta_H(h_1 \otimes u), \quad \beta_H(h_2 \otimes u),$$

and the three quadratic columns are scalar multiples, with coefficients $v_1^2, v_1 v_2, v_2^2$, of the single vector $\chi_H(u^2)$. A 3-minor containing three mixed columns vanishes because the mixed span has dimension at most two. A minor containing at least two quadratic columns vanishes because those columns are proportional. Thus only minors with two mixed columns and one quadratic column survive. Their common u -factor is

$$\det[\beta_H(h_1 \otimes u), \beta_H(h_2 \otimes u), \chi_H(u^2)],$$

a binary quartic. The five possible v -monomials span $\text{Sym}^4(\mathbb{C}_v^2)$. By the same determinant comparison used in Lemma S8.4, this binary quartic is the restriction h_H of the Jacobian quartic to the pencil of hyperplanes containing H . This proves the assertion without a rank assumption on β_H . $\square$

The orbit classification in Definition S6.3 is elementary projective geometry. We record normal forms because they are also used in the colon calculation. Order the basis of $H \otimes W$ as

$$h_1w_1, h_1w_2, h_2w_1, h_2w_2.$$

Up to $GL(H) \times GL(W) \times GL(E_H)$, the mixed block can be taken to be one of the following matrices:

type	β	
3, off Q	$\begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$	
3, on Q	$\begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$	
2, secant	$\begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$	
2, tangent	$\begin{pmatrix} 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$	
2, H -ruling	$\begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$	
2, W -ruling	$\begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$	
1, dual rank 2	$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$	
1, dual rank 1	$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$	
0	0.	(S10.2)

For rank two, a projective kernel line meets the smooth quadric in two reduced points, one double point, or is contained in one of the two ruling families. For rank one, the dual generator has tensor rank two or one. These alternatives are invariant and exhaustive, proving the orbit statement.

Lemma S10.2 (Exact transverse models for the nine mixed-kernel orbits). *Let*

$$P = \mathbb{C}[a, b, c, d], \quad T = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad \delta = ad - bc.$$

For each normal form in (S10.2) there is a nonempty Zariski-open set of quadratic blocks χ on which the Hilbert functions of

$$P/((\delta) + (J : \delta^{j-1}))$$

are those displayed in Theorem S6.4.

Proof. The two rank-three rows are Theorem S9.3 and Theorem S9.5. For the remaining rows, the assertion is a homogeneous determinant calculation. We give integer witnesses and the resulting colons; this simultaneously proves nonemptiness of the relevant rank opens.

For the secant normal form take

$$\chi = \begin{pmatrix} 1 & 2 & -2 \\ -2 & -1 & 1 \\ -2 & 1 & -2 \end{pmatrix}.$$

A direct expansion of the 3-minors followed by cancellation by δ gives

$$(J : \delta) = (c^3, c^2d, cd^2, d^3, a^2 - c^2, ab - cd, b^2 - d^2, \\ ac - 2c^2, bc - 2cd, ad - 2cd, bd - 2d^2),$$

and $(J : \delta^2) = P$. The first ideal contains δ , and the quotient has Hilbert function $(1, 4, 3)$, hence length eight.

For the tangent normal form take

$$\chi = \begin{pmatrix} 1 & -2 & -1 \\ 0 & 1 & 0 \\ -1 & -2 & 2 \end{pmatrix}.$$

Then

$$(J : \delta) = (c^3, c^2d, cd^2, d^3, a^2 - 10c^2, ab - 10cd, b^2 - 10d^2, \\ ac + 2c^2, bc + 2cd, ad + 2cd, bd + 2d^2),$$

again with $(J : \delta^2) = P$. The same Hilbert function $(1, 4, 3)$ follows.

For the H -ruling normal form take

$$\chi = \begin{pmatrix} 1 & -1 & -1 \\ 1 & 1 & -1 \\ 2 & -1 & 0 \end{pmatrix}.$$

One obtains

$$(J : \delta) = (c^4, c^3d, c^2d^2, cd^3, d^4, 5ac^2 - c^3, 5bc^2 - c^2d, \\ 5acd - c^2d, 5bcd - cd^2, 5ad^2 - cd^2, 5bd^2 - d^3, \\ a^2 - ac, 2ab - ad - bc, b^2 - bd),$$

while

$$(J : \delta^2) = (a, b, c, d).$$

Here $\delta \notin (J : \delta)$, so $\delta^2 \notin J$; on the other hand $\delta \in (J : \delta^2)$, so $\delta^3 \in J$. After adjoining δ , the second-layer quotient has Hilbert function

$$(1, 4, 6, 4)$$

and length fifteen, while the third layer has length one.

For the W -ruling normal form take

$$\chi = \begin{pmatrix} 1 & -1 & 1 \\ 2 & 1 & 2 \\ 1 & 0 & -2 \end{pmatrix}.$$

Then

$$(J : \delta) = (ac, bc, ad, bd, c^3, c^2d, cd^2, d^3), \quad (J : \delta^2) = P.$$

This ideal already contains δ . The Hilbert function of the second layer is

$$1, 4, 6, 4, 5, 6, 7, 8, \dots,$$

hence its Hilbert polynomial is $n + 1$ from degree four onward.

For the dual-rank-two rank-one normal form take

$$\chi = \begin{pmatrix} -2 & 1 & -1 \\ 0 & 0 & -2 \\ -2 & 0 & -1 \end{pmatrix}.$$

After adjoining δ , a homogeneous basis for the second colon may be chosen as

$$(\delta, a^2c, ac^2, a^2d, abd, b^2d, acd, ad^2, bd^2).$$

Its Hilbert function is

$$1, 4, 9, 8, 10, 12, 14, 16, \dots,$$

so the Hilbert polynomial is $2n + 2$ from degree three onward. Moreover

$$(J : \delta^2) = (a, b, c, d),$$

and the third layer is the reduced vertex.

For the dual-rank-one normal form take

$$\chi = \begin{pmatrix} -2 & 2 & -1 \\ 1 & 1 & 1 \\ 2 & 1 & -2 \end{pmatrix}.$$

A homogeneous basis after adjoining δ is

$$(\delta, a^3 + 4a^2c + 3ac^2, a^2b + 4a^2d + 3acd, \\ ab^2 + 4abd + 3ad^2, b^3 + 4b^2d + 3bd^2),$$

with Hilbert function

$$1, 4, 9, 12, 15, 18, 21, \dots$$

The next colon is, after an invertible linear change,

$$(c^2, cd, d^2, a, b),$$

so the third layer has Hilbert function $(1, 2)$ and length three.

Finally, when $\beta = 0$, surjectivity of $[\beta, \chi]$ forces χ to be invertible. Hence

$$J = I_3(\chi \operatorname{Sym}^2 T) = (\det \chi) \det(\operatorname{Sym}^2 T) = (\delta^3),$$

because $\det(\operatorname{Sym}^2 T) = \delta^3$ in dimension two. This proves the last row exactly.

All displayed Hilbert functions are computed from the displayed homogeneous bases by counting standard monomials. The role of these integral points is only to provide convenient base points. Genericity is proved separately in Theorem S11.1: the stabilizer orbit of each normal form is supplemented by an explicit transverse affine slice, and the residual colons are computed over the rational function field of that slice. The resulting initial ideals give exactly the Hilbert functions displayed above. Hence the open conditions in the statement are genuine dense-open conditions on the whole orbit family, not constant-rank loci inferred from one fibre. $\square$

Proof of Theorem S6.4. Lemma S10.1 gives the first layer and its contact interpretation. The orbit list follows from (S10.2). Lemma S10.2 gives the second and third layers on each coefficient-transverse open. The nilpotency index is

$$1 + \min\{k : \delta^k \in J\}$$

by Theorem S9.1; the displayed colons give the values in the table.

If $b \leq 1$, two mixed columns never span two independent target directions, so the determinant in the proof of Lemma S10.1 vanishes identically and $h_H = 0$. If the kernel is the H -ruling $h_0 \otimes W$, one of the two mixed vectors vanishes for every u , so again $h_H = 0$. For the W -ruling, the mixed span drops only at $u = w_0$, giving the forced contact root but not identical vanishing on the transverse open. This proves the remaining assertions. $\square$

S10.4. Rank-drop depth and the finite coefficient atlas.

Proof of Proposition S6.10. Eliminate the rank- a block A_H and put $q = 6 - a$. The remaining target has dimension q . A maximal residual minor uses at most b independent mixed columns, each of transverse degree one. Every remaining column must come from the quadratic block and has transverse degree two. Therefore every nonzero generator of J has degree at least

$$b + 2(q - b) = 2q - b.$$

If $2k < 2q - b$, the homogeneous polynomial δ^k cannot lie in J , proving

$$k \geq q - \lfloor b/2 \rfloor.$$

Theorem S6.2 gives $k \leq 5$.

The quadratic block has only three source dimensions when $c = 2$. Surjectivity of the original quotient after eliminating A_H therefore implies $b + 3 \geq q$. Combining this with $b \leq 4$ gives exactly the table in the statement. The equivalence between $d^k \in J$ and the augmented-rank condition $\varepsilon_k(J) = 0$ is ordinary linear algebra in the degree $2k$ homogeneous component. In the type $(a, b) = (0, 3)$ the lower bound already gives $k \geq 5$, while $d^5 \in J$, so $k = 5$ exactly. $\square$

The same coefficient construction works for $c = 3, 4$. Because $d^5 \in J$, it terminates at $k = 5$. Colon ideals are kernels of the maps

$$(P_c)_m \xrightarrow{\cdot d^{j-1}} (P_c)_{m+c(j-1)} / J_{m+c(j-1)},$$

hence are obtained by finite syzygy elimination. Noetherianity gives finite homogeneous generating sets. Rank stratification of the resulting coefficient matrices, followed by the usual flattening stratification of the finitely presented graded quotients, gives the finite atlas in Definition S6.5. This is the finite algebraic replacement for an unspecified infinite colon tower.

S10.5. Projection corank three.

Lemma S10.3 (Functorial symbol containment under mixed-rank drop). *Fix the rank of A_H at projection corank three. In every homogeneous degree, the right $GL(W)$ -module generated by the residual maximal minors for a specialization of the mixed block is a quotient of the corresponding universal maximal-mixed-rank symbol module. After multiplication by coordinate functions the same is true degree by degree.*

Proof. After eliminating in A_H , a residual minor with u mixed columns and $q - u$ quadratic columns is the image of the natural alternating map

$$\bigwedge^u (H \otimes W) \otimes \bigwedge^{q-u} \text{Sym}^2 W \longrightarrow \bigwedge^q E_H.$$

The construction is polynomial and functorial in the mixed map $H \otimes W \rightarrow E_H$. Replacing that map by one of smaller rank factors the displayed map through the corresponding quotient of $H \otimes W$; it cannot create a new right $GL(W)$ -constituent. Tensoring with homogeneous coordinate functions and taking images preserves this quotient relation. $\square$

Lemma S10.4 (No cubic determinant power at corank three). *At projection corank three, $d^3 \notin J$.*

Proof. Here $\dim W = 3$ and $\deg d = 3$. First suppose A_H has rank one. After eliminating its image the residual target has dimension five. The maximal-rank mixed case has rank $\overline{B}_H = 3$. By Lemma S10.3, every mixed-rank drop is a quotient of the symbol module treated below, degree by degree. Thus a determinant representation absent from the universal symbol cannot be created by specialization.

The degree-seven, degree-eight and degree-nine symbols of the five-minors use respectively three, two and one mixed columns. As right $GL(W)$ -representations they are contained in

$$\bigwedge^3 W \otimes \bigwedge^2 \text{Sym}^2 W, \quad \bigwedge^2 W \otimes \bigwedge^3 \text{Sym}^2 W, \quad W \otimes \bigwedge^4 \text{Sym}^2 W.$$

The elementary Schur decompositions are

$$\begin{aligned} \bigwedge^2 \text{Sym}^2 W &= \mathbb{S}_{(3,1,0)} W, \\ \bigwedge^3 \text{Sym}^2 W &= \mathbb{S}_{(4,1,1)} W \oplus \mathbb{S}_{(3,3,0)} W, \end{aligned}$$

and

$$\bigwedge^4 \text{Sym}^2 W = \mathbb{S}_{(4,3,1)} W.$$

Consequently the degree-eight symbol is a sum of

$$\mathbb{S}_{(5,2,1)}, \mathbb{S}_{(4,4,0)}, \mathbb{S}_{(4,3,1)}, \mathbb{S}_{(4,2,2)},$$

and the degree-nine symbol is a sum of

$$\mathbb{S}_{(5,3,1)}, \mathbb{S}_{(4,4,1)}, \mathbb{S}_{(4,3,2)}.$$

The degree-seven symbol is $\mathbb{S}_{(4,2,1)}$. To obtain a degree-nine element of the ideal one may multiply the degree-seven symbol by degree-two coordinate functions, the degree-eight symbol by degree-one coordinate functions, or take the degree-nine symbol itself. None can contain

$$(\det W)^3 = \mathbb{S}_{(3,3,3)} W :$$

every displayed highest weight has first part at least four, and the Littlewood–Richardson rule only adds boxes. Since d^3 spans this determinant representation, it is not in J .

If $A_H = 0$, the residual target has dimension six. A degree-nine minor can occur only when the mixed block has rank three and uses all three mixed columns and three quadratic columns. Its right type is contained in

$$\det W \otimes \bigwedge^3 \text{Sym}^2 W = \mathbb{S}_{(5,2,2)} W \oplus \mathbb{S}_{(4,4,1)} W,$$

again with no $(3,3,3)$ summand. If the mixed rank drops, the minimal degree is larger than nine. This proves the assertion in all cases. $\square$

Proof of Theorem S6.11. Lemma S10.4 gives $d^3 \notin J$ for all the stated coefficient data. The polynomial restriction argument in Theorem S12.2, using only the corank-four theorem and this already proved exclusion, gives $d^4 \in J$ on G_4° . The nilpotency index is one plus the first determinant exponent, hence exactly five. Outside this open the universal exponent-five bound and the definition of ε_4 still give the stated membership test. $\square$

S10.6. Projection corank four. We use one standard plethysm calculation (see, for example, [18] for Schur functors and character methods). The identity may also be checked directly by the Weyl character formula:

$$\bigwedge^6 \text{Sym}^2 V = \mathbb{S}_{(5,4,2,1)} V \oplus \mathbb{S}_{(4,4,4,0)} V. \quad (\text{S10.3})$$

Both summands occur with multiplicity one.

Lemma S10.5 (Equivariance of the corank-four residual ideal). *Let $A = \text{Sym}(V^* \otimes V) = \mathbb{C}[\text{End}(V)]$ and $J = I_6(\gamma_R \text{Sym}^2 T)$ on the projection-corank-four chart. For the right action $T \mapsto Tg$ of $GL(V)$, the ideal J is stable. Hence every homogeneous piece $J_m \subset A_m$ is a $GL(V)$ -subrepresentation.*

Proof. For $g \in GL(V)$,

$$\gamma_R \text{Sym}^2(Tg) = \gamma_R(\text{Sym}^2 T)(\text{Sym}^2 g).$$

The last factor is invertible. Maximal minors are therefore changed only by an invertible linear combination of the columns, so their ideal is unchanged. Homogeneity gives the assertion degree by degree. $\square$

Lemma S10.6 (Effective Cauchy–Pieri multiplication in the matrix ring). *Let $A = \text{Sym}(V^* \otimes V) = \mathbb{C}[\text{End}(V)]$. In the Cauchy decomposition of A , put*

$$\lambda = (4, 4, 4, 0), \quad \mu = (4, 0, 0, 0), \quad \nu = (4, 4, 4, 4).$$

The actual polynomial multiplication

$$A_{12}[\lambda] \otimes A_4[\mu] \longrightarrow A_{16}$$

has nonzero projection to $A_{16}[\nu]$. This projection has multiplicity one on both the left and right Schur factors and, after the identifications

$$\mathbb{S}_\lambda V \simeq (\det V)^4 \otimes (\mathrm{Sym}^4 V)^*, \quad \mathbb{S}_\mu V = \mathrm{Sym}^4 V,$$

is a nonzero scalar multiple of the evaluation pairing.

Proof. The skew diagram ν/λ is a horizontal four-strip, so $c'_{\lambda,\mu} = 1$. This alone does not prove nonvanishing of polynomial multiplication. Lemma S12.1 calculates that the product $[123 \mid 123]^4 x_{44}^4$ has isotypic projection $(\det X)^4/35$. Its factors belong to the indicated Cauchy summands, so actual multiplication has nonzero projection. Multiplicity one on the left and right factors and Schur's lemma identify the resulting maps with nonzero multiples of the evaluation pairings under $\mathbb{S}_{(4,4,4,0)} V = (\det V)^4 \otimes (\mathrm{Sym}^4 V)^*$. They are nondegenerate. In particular an arbitrary nonzero Jacobian vector, not only the extremal vector used in the calculation, pairs nontrivially with a suitable degree-four coefficient. The coefficient $1/35$ is the Fischer isotypic normalization; no coefficient-one straightening identity is needed. $\square$

Lemma S10.7 (The Jacobian projection). *Under*

$$\bigwedge^6 \mathrm{Sym}^2 V \simeq (\det V)^5 \otimes \left(\bigwedge^4 \mathrm{Sym}^2 V \right)^*,$$

the summand $\mathbb{S}_{(4,4,4,0)} V$ is dual, up to the displayed determinant twist, to

$$\det V \otimes \mathrm{Sym}^4 V \subset \bigwedge^4 \mathrm{Sym}^2 V.$$

The projection

$$\bigwedge^4 \mathrm{Sym}^2 V \longrightarrow \det V \otimes \mathrm{Sym}^4 V$$

sends $q_1 \wedge \cdots \wedge q_4$, up to a nonzero scalar, to

$$\det \left(\frac{\partial q_i}{\partial x_j} \right).$$

Proof. Duality of Schur functors gives

$$(\det V)^5 \otimes \mathbb{S}_{(4,4,4,0)} V^* \simeq \det V \otimes \mathrm{Sym}^4 V.$$

The Jacobian determinant is alternating in the four quadrics and is $GL(V)$ -equivariant with exactly this target weight. It is nonzero, for instance on $x_1^2, x_2^2, x_3^2, x_4^2$. Multiplicity one in (S10.3) therefore identifies it with the stated projection. $\square$

Proof of Theorem S6.12. At $H = 0$ the ideal J is generated in degree twelve by the 6-minors of $\gamma_R \mathrm{Sym}^2 T$. Under right multiplication of T , their span is a quotient of the representation $\bigwedge^6 \mathrm{Sym}^2 V$. The polynomial d^3 spans $(\det V)^3 = \mathbb{S}_{(3,3,3,3)} V$, which does not occur in (S10.3). Hence $d^3 \notin J$.

Let

$$\lambda_R = \bigwedge^6 \gamma_R.$$

By Lemma S10.7, the component of λ_R on the $\mathbb{S}_{(4,4,4,0)}$ -summand is nonzero exactly when the Jacobian quartic F_R is nonzero. For $R \in G_4^\circ$, the quartic $Y_R = V(F_R)$ is smooth, so this component is nonzero.

Let $A = \mathbb{C}[\mathrm{End}(V)]$. The nonzero Jacobian component just identified gives a nonzero vector in the $\lambda = (4, 4, 4, 0)$ Cauchy component of J_{12} . By Lemma S10.6, multiplication by the $\mu = (4)$ component of A_4 has a nonzero $\nu = (4, 4, 4, 4)$ -projection. Because the induced pairing

$$\mathbb{S}_{(4,4,4,0)} V \otimes \mathrm{Sym}^4 V \longrightarrow (\det V)^4$$

is nondegenerate, one may choose the degree-four matrix coefficient to pair nontrivially with the specific nonzero Jacobian vector.

By Lemma S10.5, J_{16} is a right $GL(V)$ -subrepresentation of A_{16} . Over $\mathbb{C}$ it is completely reducible, so the equivariant projection onto any right isotypic summand preserves J_{16} . Apply the projection onto right type $\nu = (4, 4, 4, 4)$ to the product just constructed. Its image is nonzero by Lemma S10.6 and still belongs to J_{16} .

In the Cauchy decomposition of the full matrix coordinate ring, right type ν is paired with the same left type. Both $\mathbb{S}_{(4,4,4,4)}V$ and $\mathbb{S}_{(4,4,4,4)}V^*$ are one-dimensional determinant characters. Hence this entire Cauchy summand is the one-dimensional line

$$\mathbb{C}(\det T)^4 = \mathbb{C}d^4.$$

A nonzero element of this line lying in J_{16} therefore implies

$$d^4 \in J.$$

Thus ideal membership follows from three explicit ingredients: equivariance of the maximal-minor ideal, semisimple projection in characteristic zero, and the nonzero coordinate-ring Pieri coefficient. No projection is taken outside the ideal. Together with $d^3 \notin J$, this gives nilpotency index five. $\square$

S10.7. Specialization and associated supports. The coefficient matrices above have polynomial entries, but this fact alone does not justify fibrewise statements about embedded associated points or base change of colons. The required order of argument is carried out in Section S11. There we first flatten the finite presentations controlling each kernel $(J : d^{j-1})$ and its cokernel, then apply relative assassins, generic freeness at the associated generic points, and Noetherian induction. Proposition S6.9 follows from that argument.

The same section computes the generic corank-two primary signatures over rational function fields and gives explicit one-parameter orbit degenerations. Thus the specialization statements used by the paper are consequences of exact family ideals and relative associated-point theory, not of generic flatness alone.

S11. RELATIVE PRIMARY SIGNATURES AND SPECIALIZATION LAWS

This section supplies the two family arguments that are not consequences of a single exact witness: genericity on coefficient orbits and fibrewise control of associated points. It also extracts the geometric primary signatures of the corank-two layers and writes explicit orbit degenerations.

S11.1. Stabilizer slices and generic Groebner bases. Keep

$$P = \mathbb{C}[a, b, c, d], \quad T = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad \delta = ad - bc.$$

For a mixed normal form β , let $G_\beta \subset GL(H) \times GL(W) \times GL(E_H)$ be its stabilizer. If (X, Y, Z) lies in its Lie algebra, then

$$Z\beta - \beta(X \otimes 1 + 1 \otimes Y) = 0, \tag{S11.1}$$

and its infinitesimal action on the quadratic block is

$$\dot{\chi} = Z\chi - \chi \operatorname{Sym}^2 Y. \tag{S11.2}$$

These are linear equations over $\mathbb{Q}$ at the integral base points of Lemma S10.2.

Theorem S11.1 (Generic slice certification for the mixed-kernel atlas). *For each of the seven mixed-kernel rows not already covered by the mixed-regular and decomposable-kernel theorems, the Hilbert functions in Theorem S6.4 are the generic Hilbert functions on the entire orbit family.*

More precisely, at the integral base points used in Lemma S10.2, the ranks of the infinitesimal G_β -orbits in the nine-dimensional χ -space and one choice of transverse affine slice are

type	rank $T_\chi(G_\beta\chi)$	slice entries
secant	6	$\chi_{11}, \chi_{21}, \chi_{31}$
tangent	7	χ_{11}, χ_{21}
H -ruling	7	χ_{11}, χ_{13}
W -ruling	9	$\emptyset$
dual rank 2	9	$\emptyset$
dual rank 1	9	$\emptyset$
zero mixed map	9	$\emptyset$.

On the first three slices, with coordinates s_i , Groebner bases of all required colons over $\mathbb{Q}(s_i)$ give respectively

type	$\min\{k : \delta^k \in J\}$	H_{W_2}	H_{W_3}
secant	2	$(1, 4, 3)$	0
tangent	2	$(1, 4, 3)$	0
H -ruling	3	$(1, 4, 6, 4)$	(1) .

The only nonconstant denominators in reduced bases may be chosen to divide

$$s_2 - 2 \quad (\text{secant}), \quad s_0 + 2s_1 - 5 \quad (H\text{-ruling}),$$

and neither vanishes at the base point $s = 0$; the tangent slice requires no nonconstant denominator. In the remaining four rows the stabilizer orbit is already open in χ -space, so the exact base-point calculation is the generic calculation.

Proof. Solving (S11.1) and applying (S11.2) gives the ranks in the table. Adjoining the displayed coordinate directions raises the tangent rank to nine. Thus the morphism from the stabilizer times the affine slice to χ -space has surjective differential at the base point. Over characteristic zero it is therefore dominant and smooth on a neighbourhood of that point.

Compute the residual minors and successive colons in $\mathbb{Q}(s_i)[a, b, c, d]$. For the secant and tangent slices the second-layer generic initial ideal is

$$(c, d)^3 + (a, b)^2 + (a, b)(c, d),$$

which has Hilbert function $(1, 4, 3)$, and the next colon is the unit ideal. For the H -ruling slice a generic initial ideal of the second layer is generated by

$$c^4, c^3d, c^2d^2, cd^3, d^4, ac^2, acd, ad^2, bd^2, a^2, ab, b^2, bc,$$

with Hilbert function $(1, 4, 6, 4)$; the third-layer initial ideal is (a, b, c, d) . These initial ideals are unchanged after specialization away from the displayed denominator divisors.

For the four open-orbit rows the exact Groebner bases give the functions printed in Theorem S6.4. The calculation is reproduced, over exact rational function fields rather than floating-point samples, by `checks/generic_boundary_atlas.py`. Since the slice map is dominant, these are the generic initial ideals on the orbit families. $\square$

S11.2. Geometric primary signatures for the nine mixed-kernel types. Let $Q = V(\delta)$ be the affine Segre cone and let $\mathfrak{m} = (a, b, c, d)$ be its vertex. Write R_r for a reduced divisor consisting of r distinct lines in one ruling of Q . Write V_ℓ for a homogeneous $\mathfrak{m}$ -primary scheme of length ℓ . Finally, $R_r + E_\ell$ means a scheme whose saturation is R_r and whose saturation quotient is an embedded vertex module of length ℓ . This notation records associated supports and generic lengths; it does not choose a noncanonical embedded primary component.

Theorem S11.2 (Generic primary signatures for the mixed-kernel orbits). *On the generic coefficient stratum of each A_H -injective mixed-kernel orbit, the positive layers have the following primary signatures:*

<i>mixed-kernel type</i>	W_2	W_3
3, point off Q	V_1	$\emptyset$
3, point on Q	V_3	$\emptyset$
2, secant	V_8	$\emptyset$
2, tangent	V_8	$\emptyset$
2, H -ruling	V_{15}	V_1
2, W -ruling	$R_1 + E_5$	$\emptyset$
1, dual rank 2	$R_2 + E_3$	V_1
1, dual rank 1	R_3	V_3
0	Q	Q .

In particular, every associated support and every generic length in these layers is determined. The symbols R_r have r minimal ruling-line primes, each of generic length one. The only additional associated prime in $R_1 + E_5$ and $R_2 + E_3$ is the embedded vertex $\mathfrak{m}$.

Proof. The finite rows are homogeneous zero-dimensional schemes supported at the vertex, hence are $\mathfrak{m}$ -primary; their lengths are the Hilbert-function sums. For the W -ruling witness the second layer is

$$I = (ac, bc, ad, bd, c^3, c^2d, cd^2, d^3).$$

Its saturation is (c, d) , and $(c, d)/I$ has length five. Thus its associated primes are (c, d) and $\mathfrak{m}$, with an embedded vertex contribution of length five.

For the dual-rank-two witness, saturation gives

$$(a, b) \cap (c, d),$$

the union of two ruling lines, and the saturation quotient has length three. For the dual-rank-one witness, modulo δ , the second-layer ideal is exactly

$$\bigcap_{\lambda \in \{0,1,3\}} (a + \lambda c, b + \lambda d),$$

so it is a reduced union of three ruling lines and has no embedded point. The third layer is the length-three vertex thickening already computed in Lemma S10.2. The zero mixed-map identity $J = (\delta^3)$ gives $W_2 = W_3 = Q$.

Theorem S11.1 transports these exact signatures to the generic coefficient strata. Associated supports are preserved by the stabilizer action, and the relative-primary argument below prevents an isolated base-point calculation from being used beyond its certified open stratum. $\square$

S11.3. Pure-block rank drop at projection corank two. When A_H has rank $a < 3$, put $q = 6 - a$ and $b = \text{rank } \overline{B}_H$. For $b = 3, 2, 1$ the phrase *generic mixed kernel* means respectively an off-Segre kernel point, a secant kernel line, and a kernel plane whose dual tensor has rank two.

Theorem S11.3 (Generic primary signatures for every A_H -rank-drop type). *At projection corank two, every generic rank type allowed by Proposition S6.10 has the following complete list of*

nonzero positive layers:

a	b	nilpotency index	$(W_1, W_2, W_3, W_4, W_5)$
2	4	3	(Q, V_{22})
2	3	4	(Q, V_{27}, V_1)
2	2	4	(Q, R_4, V_8)
2	1	5	(Q, Q, Q, V_1)
1	4	5	(Q, Q, V_{11}, V_1)
1	3	5	$(Q, Q, R_2 + E_3, V_1)$
1	2	5	(Q, Q, Q, R_2)
0	4	6	(Q, Q, Q, R_2, V_1)
0	3	6	(Q, Q, Q, Q, V_1)

Here every Q is the reduced Segre quadric with generic length one; every R_r is a reduced union of r ruling lines with generic length one on each component; and every V_ℓ is $\mathfrak{m}$ -primary of the indicated length. In the unique row $(a, b) = (1, 3)$ carrying E_3 , the associated primes of that layer are the two ruling-line primes together with the embedded vertex.

Proof. Choose the generic normal forms of the mixed block described above and let χ vary. The stabilizer-slice codimensions in χ -space for

$$(a, b) = (2, 4), (2, 3), (2, 2), (2, 1), (1, 4), (1, 3), (1, 2), (0, 4), (0, 3)$$

are respectively

$$5, 4, 1, 0, 3, 1, 0, 0, 0.$$

On explicit complementary coordinate slices the successive colons are computed over the rational function fields of those slices. Their Hilbert functions are

a	b	non-quadric Hilbert functions
2	4	$(1, 4, 9, 8)$
2	3	$(1, 4, 9, 8, 5); (1)$
2	2	$4n + 4 \ (n \geq 4); (1, 4, 3)$
2	1	(1)
1	4	$(1, 4, 6); (1)$
1	3	$2n + 2 \ (n \geq 3); (1)$
1	2	$2n + 2 \ (n \geq 1)$
0	4	$2n + 2 \ (n \geq 1); (1)$
0	3	$(1).$

Every omitted earlier layer has Hilbert function $(n + 1)^2$, hence equals Q .

To identify the positive-dimensional supports, pull the layer ideals back to the rank-one parametrization $T = uv^t$, but now perform the factorization over the same rational-function fields as the Groebner calculation.

For $(a, b) = (2, 2)$, with slice parameter s , the second layer has common rank-one factor, up to a unit of $\mathbb{Q}(s)$,

$$u_0 u_1 \left((s + 9)u_0^2 + (3s + 6)u_0 u_1 - 7u_1^2 \right).$$

The quadratic discriminant is

$$9s^2 + 64s + 288.$$

On the open set $(s + 9)(9s^2 + 64s + 288) \neq 0$, its two roots are distinct and different from the two roots $u_0 u_1 = 0$. Thus the generic support is a reduced divisor of four distinct ruling lines. Its Hilbert polynomial agrees with the computed layer, so there is no generic embedded correction.

For $(a, b) = (1, 3)$, with slice parameter t , the third layer has common factor, again up to a unit of $\mathbb{Q}(t)$,

$$q(u) = 10u_0^2 + u_0u_1 - 4u_1^2.$$

Its discriminant is 161, so it gives two distinct ruling lines over the algebraic closure of $\mathbb{Q}(t)$. The exact generic-field calculation also computes the saturation. Writing $T = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, the saturated ruling ideal is

$$I_{\text{sat}} = (\delta, 10a^2 + ac - 4c^2, 10ab + ad - 4cd, \\ 10b^2 + bd - 4d^2).$$

The Groebner bases over $\mathbb{Q}(t)$ give

$$H_{M_3} = (1, 4, 9, 8, 10, 12, 14, \dots), \quad H_{P/I_{\text{sat}}} = (1, 4, 6, 8, 10, 12, 14, \dots).$$

The quotient I_{sat}/I_{M_3} is therefore concentrated in degree two and has length three. This is the claimed embedded vertex module, so the generic signature is $R_2 + E_3$.

For the two open-orbit rows $(1, 2)$ and $(0, 4)$, the previously displayed base-point factors remain generic by orbit transport:

$$u_0u_1, \quad 6u_0^2 + u_0u_1 + 4u_1^2,$$

and the second quadratic has nonzero discriminant. Thus every positive-dimensional rank-drop support in the table is certified on its generic coefficient field, rather than inferred from an isolated integral point. The exact stabilizer ranks, slice coordinates, Groebner leading monomials, denominator checks, and Hilbert counts are reproduced by `checks/generic_boundary_atlas.py`. Because each slice map is dominant at its base point, the calculation is at the generic point of the rank type rather than at a special witness. $\square$

S11.4. Relative assassins and base change for the colon tower. We prove the general bounded principal-colon theorem before applying it to the determinant d . The order of the argument is essential.

S11.4.1. Finite diagrams and primary models. We give an algebraic construction which distinguishes persistence of an associated point from exhaustion of the associated set. Every restriction to a base stratum in this construction is followed by formation of the indicated kernels on that stratum. We do not identify these kernels with the pullback of a kernel from an earlier, unflattened stratum.

Lemma S11.4 (Spreading a finite exact diagram). *Let D be a noetherian domain with fraction field K , let A be a finite type D -algebra, and fix a finite diagram of finite A_K -modules and maps, including finitely many short exact sequences. After replacing D by D_g , the diagram has coherent models for which all the specified short exact sequences remain exact after every base change over D_g . The same assertion holds for a diagram over a finite extension of K , after a finite faithfully flat cover of a nonempty open of $\text{Spec } D$.*

Proof. Choose presentations, maps, and the finitely many identities expressing commutativity. Clear their denominators. Kernels and homology modules are finite over A . A finite module which vanishes after tensoring with K is killed by a nonzero element of D , so a further localization makes the specified sequences exact. Apply generic freeness to their finitely many terms and cokernels [27, Tag 051S]. In a short exact sequence with flat cokernel, tensoring with any base algebra preserves the injection; right exactness preserves the remaining terms. This proves the claim without any four-term kernel inference.

For a finite field extension L/K , choose finitely many generators of L , clear denominators in their monic equations, and let D' be the finite D_g -algebra which they generate. After another localization, D' is finite free of positive rank over D_g . It is therefore faithfully flat. Apply the

preceding construction over D' . A further shrinking upstairs descends to a shrinking downstairs: the image of its closed complement is closed under the finite morphism and misses the generic point. No separability is needed for this step. $\square$

Proposition S11.5 (Primary quotients on a splitting cover). *Let A be a finite type algebra over a noetherian domain D , and let M be a finite A -module. There is a nonempty open of $\text{Spec } D$ and a finite faithfully flat cover of that open on which there exist ideals $\mathfrak{p}_i$, modules Q_i , and maps*

$$M \longrightarrow Q_i, \quad M \longrightarrow \bigoplus_i Q_i, \quad A/\mathfrak{p}_i \longrightarrow M$$

with the following properties. On every geometric fibre the first maps are surjective, the last two displayed maps are injective, the rings $A/\mathfrak{p}_i$ are nonzero integral domains, and each Q_i is a nonzero $\mathfrak{p}_i$ -primary module. The primes may be taken distinct on every geometric fibre. Consequently

$$\text{Ass}(M_{\bar{s}}) = \{\mathfrak{p}_{i,\bar{s}}\}_i, \quad 0 = \bigcap_i \ker(M_{\bar{s}} \longrightarrow Q_{i,\bar{s}})$$

is an irredundant primary decomposition. All exact diagrams used to construct these maps are preserved by arbitrary base change on the chosen cover.

Proof. On the geometric generic fibre choose an irredundant primary decomposition $0 = \bigcap_i N_i$, and put $Q_i = M/N_i$, with $\text{Ass } Q_i = \{\mathfrak{p}_i\}$. Choose also an associated element for each $\mathfrak{p}_i$; it gives an injection $A/\mathfrak{p}_i \hookrightarrow M$. The diagonal map $M \hookrightarrow \bigoplus_i Q_i$ is injective.

We must spread the assertion that a quotient is primary, not merely its support. Choose n_i with $\mathfrak{p}_i^{n_i} Q_i = 0$, and set

$$F_{i,j} = (0 :_{Q_i} \mathfrak{p}_i^j), \quad 0 \leq j \leq n_i, \quad E_{i,j} = F_{i,j}/F_{i,j-1}.$$

Each $E_{i,j}$ is killed by $\mathfrak{p}_i$ and is torsion-free over the domain $A/\mathfrak{p}_i$. Indeed, if $h \notin \mathfrak{p}_i$ and $hv \in F_{i,j-1}$, then $h\mathfrak{p}_i^{j-1}v = 0$; multiplication by h is injective on the primary module Q_i , so $v \in F_{i,j-1}$. Clearing fraction-field denominators now gives injections

$$E_{i,j} \hookrightarrow (A/\mathfrak{p}_i)^{r_{i,j}}.$$

These injections, the filtration sequences, the diagonal map, the quotient maps, and the associated-element injections form a finite diagram. All its coefficients descend to a finite extension of the generic field. The prime quotients can be descended at the same time with geometrically integral generic fibres. After shrinking, their geometric fibres are integral and nonempty: apply generic flatness, spreading of geometric reducedness and irreducibility, and the fact that the image of a finite-presentation morphism containing the generic point contains a nonempty open [28, Sections 37.26–37.27]. Apply Lemma S11.4 to the whole finite diagram.

In a geometric fibre, every $E_{i,j}$ embeds in a free module over an integral prime quotient. It therefore has no associated prime other than $\mathfrak{p}_i$. The filtration gives $\text{Ass } Q_i \subset \{\mathfrak{p}_i\}$; the surviving quotient map and associated-element data imply nonvanishing whenever that index occurs. More explicitly, the diagonal injection gives the inclusion

$$\text{Ass } M_{\bar{s}} \subset \bigcup_i \text{Ass } Q_{i,\bar{s}} \subset \{\mathfrak{p}_{i,\bar{s}}\}_i.$$

The associated-element injections give the reverse inclusion. This is the exhaustion argument: an additional embedded prime anywhere in the fibre would contradict the diagonal injection into these filtered primary modules. It is not an open-neighbourhood containment argument.

For every generic noncontainment $\mathfrak{p}_i \not\subset \mathfrak{p}_j$, choose $h_{ij} \in \mathfrak{p}_i \setminus \mathfrak{p}_j$. Include multiplication by h_{ij} on $A/\mathfrak{p}_j$ and its cokernel in the finite diagram. Fibrewise injectivity preserves that noncontainment. Generic containments spread by finite generation. Thus distinctness and the containment poset persist. The kernels of the quotient maps have zero intersection by the diagonal injection. No

component can be omitted: otherwise the associated-prime inclusion for the remaining diagonal injection would exclude its distinct associated prime. This proves irredundance as well as primaryness. $\square$

S11.4.2. *A relative module for the local torsion length.*

Proposition S11.6 (No new packet torsion after refinement). *In the situation of Proposition S11.5, fix a tracked prime $\mathfrak{p}$. After shrinking the base there are an integer n , a coherent submodule $T \subset M$, and an element $h \in \mathfrak{p}$ such that, after every base change on this open,*

$$T = (0 :_M \mathfrak{p}^n) = H_{\mathfrak{p}}^0(M), \quad h : M/T \longrightarrow M/T \text{ is injective.}$$

At each generic point p of a geometric packet fibre,

$$T_p = H_{\mathcal{O}_p}^0(M_p).$$

There are nonnegative integers $r_0, \dots, r_{n-1}$, independent of that geometric fibre, for which

$$\text{length}_{\mathcal{O}_p} H_{\mathcal{O}_p}^0(M_p) = \sum_{j=0}^{n-1} r_j.$$

Here H_I^0 denotes elements killed by some power of the ideal I ; the notation does not denote a sheaf cohomology group.

Proof. On the split generic fibre the annihilator chain stabilizes. Choose n with $T = (0 :_M \mathfrak{p}^n) = H_{\mathfrak{p}}^0(M)$, and put $Q = M/T$. This quotient has no $\mathfrak{p}$ -power torsion. Equivalently, no associated prime of Q contains $\mathfrak{p}$. Prime avoidance therefore supplies $h \in \mathfrak{p}$ outside all the associated primes of Q . Multiplication by h is injective on Q . When $Q = 0$ this assertion is vacuous and no choice is needed.

Spread the sequence $0 \rightarrow T \rightarrow M \rightarrow Q \rightarrow 0$, the relation $\mathfrak{p}^n T = 0$, and the injection $h : Q \rightarrow Q$. Flatten Q and the cokernel of this injection, together with the other finite diagram terms. They remain exact after arbitrary base change. An element of the base-changed M killed by $\mathfrak{p}^a$ has image in Q killed by h^a . The image is zero, so the element belongs to T . The converse follows from $\mathfrak{p}^n T = 0$. This proves the asserted equality for the entire relative torsion module, not only for a submodule chosen at the generic point. Localization gives the displayed local equality.

Consider the finite filtration

$$T \supset \mathfrak{p}T \supset \dots \supset \mathfrak{p}^n T = 0, \quad G_j = \mathfrak{p}^j T / \mathfrak{p}^{j+1} T.$$

Include its exact sequences and the quotients $T/\mathfrak{p}^j T$ in the finite diagram. Their flatness makes this the same power filtration on every fibre. On the generic integral packet each G_j becomes free of rank r_j after inverting an element $g_j \notin \mathfrak{p}$. Spread such a free presentation and its inverse. Include multiplication by g_j on $A/\mathfrak{p}$ and its cokernel in the diagram. It remains injective, so g_j is nonzero at the generic point of every geometric packet fibre. Thus $(G_j)_p$ is an r_j -dimensional vector space over the residue field at p . Additivity of length gives the sum. All these operations were performed over the splitting cover, hence also after geometric field extension. In particular, inseparable residue field extension is not silently treated as an ordinary rational point. $\square$

S11.4.3. *Descent of the closed packets.* A packet is a closed subscheme on the original base whose geometric irreducible components are a full conjugacy orbit of tracked associated points. Its ideal, rather than its finite image as a set, is the object that must be constructed. We give an effective descent construction in characteristic zero and a power-certified construction in arbitrary characteristic. The latter retains the noetherian-base generality of Theorem S6.6; it does not require a packet to be geometrically reduced.

Lemma S11.7 (Effective Galois descent of an orbit ideal). *Let D be a noetherian domain with fraction field K of characteristic zero. Let A be a finite type D -algebra. After shrinking $\text{Spec } D$, a finite Galois extension L/K with group G can be modelled by a finite etale G -torsor D'/D . Put $B = A \otimes_D D'$. Suppose that a finite G -stable set of generic primes has closures $P_i \subset B$. For a full orbit O , set $I'_O = \bigcap_{i \in O} P_i$. There is a unique ideal $I_O \subset A$ with its natural inclusion in A such that*

$$I_O B = I'_O, \quad I_O = \{a \in A : a \otimes 1 \in I'_O\}. \quad (\text{S11.3})$$

If the B/P_i have nonempty geometrically integral fibres, a further shrinking makes every geometric fibre of $V(I_O)$ exactly the reduced union of the intended components. The orbit-ideal identity and the fibrewise intersection identity persist after every base change on this open.

Proof. Choose a finite D -subalgebra of L , include the conjugates of its generators, and invert one nonzero element of D . The resulting algebra D' is finite free of positive rank. The generic Galois isomorphism

$$D' \otimes_D D' \longrightarrow \prod_{g \in G} D', \quad b \otimes c \longmapsto (b g(c))_g$$

is an isomorphism after another shrinking: its kernel and cokernel are finite D -modules with zero generic fibre. The torsor identities, or equivalently this isomorphism together with faithful flatness, give a finite etale G -torsor. No normalization or excellence hypothesis on D is used.

Schematic closure commutes with each automorphism g , so $g(P_i) = P_{gi}$ and $g(I'_O) = I'_O$. On the double overlap $\text{Spec } B \times_{\text{Spec } A} \text{Spec } B$, identify the two pullbacks of I'_O on the component indexed by g by the restriction of this semilinear action. On a triple overlap the two successive identifications are multiplication of the group elements. They therefore equal the direct identification. This is the cocycle condition, as an identity of maps of ideal sheaves, not just of their supports. The inclusion $I'_O \hookrightarrow B$ is equivariant. Effective fpqc descent for quasi-coherent modules, together with full faithfulness, descends this inclusion to a submodule of A [29, Tag 023T and Section 35.6]. Multiplication preserves the submodule after the faithfully flat extension, hence before it; the descended submodule is an ideal. Faithful flatness also proves both identities in (S11.3).

To identify the fibres, include the actual exact sequence

$$0 \longrightarrow B/I'_O \longrightarrow \bigoplus_{i \in O} B/P_i \longrightarrow C_O \longrightarrow 0 \quad (\text{S11.4})$$

in the finite diagram and flatten C_O and its terms over D' . After arbitrary base change the diagonal map stays injective. Consequently the extended ideal I'_O is the intersection of the extended P_i ; this conclusion would not follow from flatness of the individual prime quotients alone. On a geometric fibre each P_i is prime, so this intersection is radical. Distinctness is protected by the noncontainment witnesses of Proposition S11.5. The torsor over an algebraically closed field is a disjoint union of copies of that field. Pulling the descended packet to any one of these copies identifies its fibre with precisely this intersection. Thus no additional component is introduced downstairs. All subsequent pullbacks of (S11.3) remain equalities; the flattened sequence gives the asserted intersection compatibility. $\square$

Lemma S11.8 (Packets without a separability hypothesis). *Let D , K , and A be as above, without a restriction on the characteristic. A full orbit of geometric generic associated points has a closed packet on $\text{Spec } A$, after shrinking the base, with exactly those geometric fibre components supplied by the split primary models. These packets and their intersections can be refined and flattened on the original base. Their component identification remains valid after every base change in the final stratum.*

Proof. Let $\mathfrak{q} \subset A_K$ be the contraction of a member of the orbit. Geometric components over $\mathfrak{q}$ form one conjugacy orbit: over the maximal separable subextension this is the usual transitivity

of primes in a Galois extension, and purely inseparable extension is a universal homeomorphism. The associated point formula for extension of the ground field identifies these components with the tracked geometric associated points [21, Section 31.7].

Construct the ideal on the original ring first:

$$Q = \ker(A \longrightarrow A_K/\mathfrak{q}).$$

This is the schematic-closure ideal and is finite because A is noetherian. Take a common finite splitting extension L/K and a finite faithfully flat model D'/D as in Lemma S11.4. On $B = A \otimes_D D'$, let P_i be the closures of the generic geometric primes in this orbit and put $I' = \bigcap_i P_i$. On the generic fibre

$$\sqrt{\mathfrak{q}A_L} = \bigcap_i (P_i)_L.$$

Finite generation therefore gives an integer $n \geq 1$ with $(I'_L)^n \subset \mathfrak{q}A_L \subset I'_L$. Clear the finitely many denominators in these two inclusions. After shrinking on the original base, they become actual ideal inclusions

$$(I')^n \subset QB \subset I'. \quad (\text{S11.5})$$

As before, a shrinking needed upstairs is made downstairs by deleting the finite image of its bad closed complement. Finite images are used only to shrink a base on which an ideal has already been constructed; they are not used as a substitute for descent.

Flatten the diagonal sequence $0 \rightarrow B/I' \rightarrow \bigoplus_i B/P_i \rightarrow C \rightarrow 0$, and arrange the nonempty geometrically integral fibres and distinctness of the P_i by the split primary models. For every geometric fibre Ω , the intersection identity and (S11.5) now give

$$\sqrt{(QB)_\Omega} = \bigcap_i (P_i)_\Omega. \quad (\text{S11.6})$$

Indeed, products of finitely generated ideals commute with extension of scalars, while the flattened diagonal sequence supplies exactly the intersection identity that is needed here. The right side is an intersection of distinct primes. A geometric point of $\text{Spec } D$ has a lift to $\text{Spec } D'$ over the same algebraically closed field: the nonzero finite fibre algebra has a residue field equal to that field. At this lift, the fibre of A and that of B are the same algebra. Hence (S11.6) identifies the components of the packet $V(Q)$ on the original base itself. This argument does not assert that a nonreduced inseparable fibre is reduced or that a rational local length is unchanged by inseparable extension.

Schematic closures glue on affine overlaps. In a finite affine cover, use also a finite affine refinement of the overlaps and one common shrinking. A closure whose generic restriction to a chart is empty has empty restriction after shrinking, by constructibility of the image of that finite-presentation restriction. Thus the chartwise component statements glue without an untracked boundary component. For any finite collection of packets, their scheme-theoretic intersections are defined downstairs by sums of their ideals. Flatten these finitely many quotient algebras and the packet algebras on the original base. Sums commute with base change, and all previous exact sequences and inclusions survive this further restriction. Stratifying their images and fibre dimensions fixes nonemptiness and dimensions of their incidences. This proves the assertion. $\square$

Proposition S11.9 (The torsion module on the original base). *Let $I \subset A$ be any one of the packet ideals just constructed and let M be a finite A -module. On a nonempty open of the integral base there are a finite submodule $T \subset M$ and an integer n such that, after every base change on this open,*

$$T = (0 :_M I^n) = H_I^0(M). \quad (\text{S11.7})$$

At the generic point p of each geometric component of the packet, $T_p = H_{p\mathcal{O}_p}^0(M_p)$. After common refinement with the split primary models, these geometric local lengths are constant and equal on the components of each packet.

Proof. On the generic fibre choose n for which $T_K = (0 :_{M_K} I_K^n) = H_{I_K}^0(M_K)$ and set $Q_K = M_K/T_K$. There is no associated prime of Q_K containing I_K ; otherwise an associated element would give nonzero I_K -torsion. Prime avoidance gives an element $h_K \in I_K$ regular on Q_K . If $Q_K = 0$, omit this map. Spread the submodule, the relation $I^n T = 0$, and

$$0 \rightarrow T \rightarrow M \rightarrow Q \rightarrow 0, \quad h : Q \longrightarrow Q.$$

Flatten the quotient Q and the cokernel of the second map. After every base change the first injection and regularity of h remain true. An element killed by a power of I has image in Q killed by a power of h , hence has zero image. This proves (S11.7) downstairs, with no descent assertion about a chosen primary decomposition.

At a minimal prime p of a geometric packet, the radical of $I\mathcal{O}_p$ is the maximal ideal $p\mathcal{O}_p$. Zeroth local torsion depends only on this radical, which proves the local equality. On the splitting cover apply the finite power filtrations and free generic presentations of Proposition S11.6 to every component. They give a constant integer for each component. Conjugate generic points have equal lengths, so the integers agree within an orbit. At a geometric lift over the same algebraically closed field, the local ring and module are literally the same as downstairs. Consequently their lengths agree. This supplies both the relative module and the length statement on the original base. $\square$

Lemma S11.10 (Finite geometric-assassin stratification). *Let $X \rightarrow S$ be of finite presentation with S noetherian, and let $\mathcal{F}$ be a finitely presented coherent module flat over S . There is a finite locally closed stratification with finitely many closed packets Z_ν on each stratum such that*

$$\text{Ass}(\mathcal{F}_{\bar{s}}) = \bigcup_{\nu} \{\text{generic points of the irreducible components of } (Z_\nu)_{\bar{s}}\}$$

for every geometric fibre. The packets may split geometrically. After further finite refinement they are flat, and their dimensions, incidences, minimal-versus-embedded status, and local torsion lengths at associated generic points are constant. On finite splitting covers of affine charts the packets are realized by the actual primary quotients of Proposition S11.5.

Proof. We first construct the required data on a dense open of an integral base. Choose a finite affine cover of X . Apply Propositions S11.5 and S11.6 on these charts, using one common finite extension of the generic field and one common shrinking. Associated points are local for an affine open cover. Consequently every fibrewise associated point is one of the primes in those finite primary models. This proves exhaustion on the whole fibre.

Take the schematic closures on the original base of the contractions of the tracked geometric generic points. Apply Lemma S11.8 to each conjugacy orbit, with one common splitting cover and refinement. It constructs the ideal of each packet downstairs and proves exact equality of its geometric component set using the power sandwich and a base-change-exact diagonal sequence. In characteristic zero, the stronger reduced orbit-ideal construction of Lemma S11.7 gives the explicit fpqc datum and its cocycle. On a geometric lift over the same algebraically closed field, the already proved associated-point equality upstairs is an equality for the original fibre. Thus the generic points of these descended packets are exactly the associated points, in both directions. The overlap construction in Lemma S11.8 proves the same statement across the affine charts.

The noncontainment witnesses in Proposition S11.5 fix the containment poset, hence minimal and embedded status, on the splitting cover. Conjugate generic points have equal dimensions and torsion lengths; these equalities persist by Proposition S11.6 and give the downstairs lengths by

Proposition S11.9. That proposition also constructs the coherent relative torsion module on the original base, not just on its cover. Generic flatness for the finitely many packet closures fixes their fibre dimensions. For intersections, apply the same finite-presentation dimension and image stratifications to the finitely many intersections of packet closures. This fixes nonemptiness and dimensions of their incidences as well. Finally flatten their structure sheaves.

There are finitely many irreducible components of the reduced base. Carry out the preceding construction on dense opens of those components, remove these opens, and repeat on the proper closed complement, with the modules and kernels re-formed there. Noetherian induction gives a finite stratification. Nilpotents in the base do not alter its geometric fibres; reduced locally closed strata suffice. The exact-diagram constructions hold after every base change within a final stratum. They make no assertion that a kernel or a torsion module commutes with a specialization crossing two different strata. $\square$

Remark S11.11 (Why a torsion construction is necessary). Over $D = \mathbb{C}[t]$ the module

$$M = D[x, y]/(x^2, x(y^2 - t)) = D[y] \oplus xD[y]/(y^2 - t)$$

is flat. The second summand is its finite torsion contribution along $(x, y^2 - t)$. For $t \neq 0$ it splits into two geometric embedded points of length one; at $t = 0$ they form one embedded point of length two. The quotient $D[y]$ is protected from new torsion by multiplication by $y^2 - t$. Flatness of the total module alone would neither identify this torsion summand nor keep the length of an individual geometric component constant across the collision.

S11.4.4. *Grading and the finite family of multiplication maps.*

Lemma S11.12 (Graded consequences of a finite flat diagram). *Let $A = D[x_1, \dots, x_m]$ have its standard grading, and let L be a finite graded A -module flat over a noetherian ring D . Every L_n is a finite locally free D -module. On a connected base its Hilbert function, and therefore its Hilbert polynomial, is constant. A prescribed finite family of homogeneous A -linear maps whose sources, targets, images, and cokernels are flat has base-change-compatible, constant rank in every degree on such a base.*

Proof. Every graded component is a direct summand of the underlying flat D -module and is finite over D . It is thus finite flat, hence locally free. Its rank is locally constant and is constant on a connected base. This applies simultaneously to all degrees because flatness of the whole graded module has already been obtained; it does not require an infinite sequence of independent refinements. Apply the same observation to the image and cokernel of each map, and base-change the two short exact sequences defining its image. $\square$

Proof of Theorem S6.6. Fix $0 \leq q < N$ and put

$$B = A/J, \quad C_q = B/f^q B = A/(J, f^q), \quad I_q = f^q B.$$

There are two short exact sequences

$$0 \longrightarrow I_q \longrightarrow B \longrightarrow C_q \longrightarrow 0, \quad 0 \longrightarrow K_q \longrightarrow A \xrightarrow{a \mapsto af^q} I_q \longrightarrow 0. \quad (\text{S11.8})$$

There is also

$$0 \longrightarrow J \longrightarrow A \longrightarrow B \longrightarrow 0.$$

By generic flatness, simultaneously for the finite list $q = 0, \dots, N - 1$, followed by Noetherian induction on the complement, refine S so that B and every C_q are flat. Then the first sequence in (S11.8) shows that every I_q is flat. Since A is flat, the second sequence remains exact after *arbitrary* base change. Since B is flat, the sequence defining J does as well.

Let $T \rightarrow S_\alpha$ be any base change. Exactness identifies

$$B_T = A_T/J_T, \quad (I_q)_T = f_T^q B_T,$$

and the kernel of $A_T \rightarrow (I_q)_T$ is consequently

$$(K_q)_T = (J_T : f_T^q).$$

This is the desired colon base-change identity. Notice that flatness of the final cokernel C_q alone is not being used to preserve a kernel; the image module I_q has first been proved flat.

Now refine once more so that every $M_q = A/(f, K_q)$, and the sources, targets and cokernels of the finite list of multiplication maps under consideration, are flat. The specified exact diagrams then commute with arbitrary base change. For graded coefficient algebras, Lemma S11.12 gives constant Hilbert functions and degree-wise ranks on connected strata. Apply Lemma S11.10 to each M_q , and take a common finite refinement. This produces the support packets, incidences, minimal/embedded status and embedded multiplicities asserted in the theorem. $\square$

Proof of Proposition S6.9. Apply Theorem S6.6 to the finite-type coefficient chart with

$$A = P_S, \quad f = d, \quad N = 5.$$

The hypothesis $d^5 \in J$ is Theorem S6.2, itself a specialization of the universal symmetric-power determinant-completion theorem. Include the finitely many graded multiplication maps in the same refinement. The theorem gives base change for every residual colon, flatness and constant Hilbert polynomial for every $\mathcal{M}_j$, the exact geometric fibrewise associated-point packets with constant embedded multiplicities, and constant multiplication ranks. These are precisely the three clauses of the proposition.

The construction of the coefficient strata is noncanonical, but its fibrewise output is the intrinsic graded algebra, its geometric assassin, and the canonical torsion lengths at associated generic points. Two presentation atlases therefore admit a common refinement carrying the same primary signature. $\square$

S11.5. Explicit orbit closures and graded specialization. Let

$$e_1 = h_1 w_1, \quad e_2 = h_1 w_2, \quad e_3 = h_2 w_1, \quad e_4 = h_2 w_2$$

be the standard basis of $H \otimes W$. The following families are written before applying a general quadratic block χ_t .

Theorem S11.13 (Corank-two orbit closure and specialization laws). *The closure relations needed to pass between all nine mixed-kernel types are realized by the following one-parameter families:*

(1) *rank three, off Q to on Q :*

$$\beta_t = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -t & 0 & 0 & 1 \end{pmatrix}, \quad \ker \beta_t = \langle e_1 + te_4 \rangle;$$

(2) *secant to tangent:*

$$\ker \beta_t = \langle e_1 + e_4, e_2 - te_3 \rangle,$$

for which $\delta|_{\mathbb{P}(\ker \beta_t)} = x^2 + ty^2$;

(3) *secant to the two ruling families:*

$$\langle e_1, e_2 + te_4 \rangle \rightsquigarrow \langle e_1, e_2 \rangle, \quad \langle e_1, e_3 + te_4 \rangle \rightsquigarrow \langle e_1, e_3 \rangle;$$

(4) *tangent to dual-rank-two:*

$$\beta_t = \begin{pmatrix} 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & t \\ 0 & 0 & 0 & 0 \end{pmatrix};$$

- (5) either ruling to dual-rank-one, by multiplying the second nonzero row in the corresponding rank-two normal form by t ;
- (6) either rank-one type to the zero map, by replacing β with $t\beta$.

For a general polynomial choice of χ_t , the generic fibre and special fibre lie in the generic coefficient strata of the indicated orbit types. The corresponding primary signatures specialize as follows:

<i>degeneration</i>	<i>change of positive layers</i>
<i>off $Q \rightarrow$ on Q</i>	$V_1 \rightarrow V_3$
<i>secant $\rightarrow$ tangent</i>	$V_8 \rightarrow V_8$
<i>secant $\rightarrow$ H-ruling</i>	$V_8 \rightarrow (V_{15}, V_1)$
<i>secant $\rightarrow$ W-ruling</i>	$V_8 \rightarrow (R_1 + E_5)$
<i>tangent $\rightarrow$ dual rank 2</i>	$V_8 \rightarrow (R_2 + E_3, V_1)$
<i>H-ruling $\rightarrow$ dual rank 1</i>	$(V_{15}, V_1) \rightarrow (R_3, V_3)$
<i>W-ruling $\rightarrow$ dual rank 1</i>	$(R_1 + E_5) \rightarrow (R_3, V_3)$
<i>dual rank 2 $\rightarrow$ 0</i>	$(R_2 + E_3, V_1) \rightarrow (Q, Q)$
<i>dual rank 1 $\rightarrow$ 0</i>	$(R_3, V_3) \rightarrow (Q, Q)$.

Thus ruling components and embedded vertex components are born in actual one-parameter families of Fitting schemes.

Proof. The determinant of a tensor in each displayed kernel gives the stated Segre incidence. For example

$$\det \left(xI_2 + y \begin{pmatrix} 0 & 1 \\ -t & 0 \end{pmatrix} \right) = x^2 + ty^2,$$

which is secant for $t \neq 0$ and tangent for $t = 0$. Likewise δ restricts to txy on each secant-to-ruling family. The row-scaling families have the stated rank-one limits by inspection.

The residual maximal minors are polynomial in t and in the coefficients of χ_t . Each source and target generic stratum is a nonempty open in its coefficient space. Choosing χ_0 in the target open and a general first-order perturbation $\chi_0 + t\chi_1$ avoids the source bad divisors over $\mathbb{C}(t)$. Hence the two fibres have the primary signatures in Theorem S11.2. Because the ideals come from one polynomial family, the changes displayed in the table are genuine specializations, not comparisons of unrelated normal forms. $\square$

There is an analogous pure-block degeneration. In a fixed target basis let

$$A_t = (f_1, f_2, tf_3) : \text{Sym}^2 H \longrightarrow S_R$$

and choose the remaining mixed and quadratic coefficients generally. For $t \neq 0$, the induced mixed block is generically the rank-three off-Segre type; at $t = 0$, A_H has rank two and the induced mixed block has rank four. Thus the first nontrivial layer specializes from V_1 to V_{22} . This gives an explicit A_H -injective-to-rank-drop family, while the full table of Theorem S11.3 describes the generic pure-block boundary reached after further rank drops.

S12. SHARP GLOBAL DEPTH AND GEOMETRIC BOUNDARY LAWS

The preceding calculations have a global consequence stronger than the uniform exponent five for arbitrary quadratic quotients. The relevant distinction is between the full coefficient space and the open of basepoint-free webs with smooth Jacobian quartic. We first make the Pieri projection quantitative, then use its polynomial nature to pass between all projection ranks. We subsequently identify two global minimal supports and construct an explicit cross-corank incidence family. None of these arguments replaces a primary decomposition by a list of Hilbert functions.

S12.1. A normalized determinant projection.

Lemma S12.1 (Fischer coefficient for the determinant-four line). *Let $X = (x_{ij})$ be a generic 4×4 matrix, put $D = (\det X)^4$, and let $p = [123 \mid 123]^4 x_{44}^4$. The Cauchy-isotypic projection of p onto the line $\mathbb{C}D$ is*

$$\text{pr}_{(4,4,4,4)}(p) = \frac{1}{35}D.$$

In particular the coordinate-ring product of the $(4, 4, 4, 0)$ and (4) Cauchy components has nonzero determinant-four projection.

Proof. On the polynomial ring use the Fischer Hermitian form $\langle x^\alpha, x^\beta \rangle = \alpha! \delta_{\alpha\beta}$. It is invariant under the unitary groups acting on the two matrix factors. Inequivalent Cauchy summands are therefore orthogonal. The monomials of D which use only the northwest 3×3 block and x_{44} are exactly the monomials of p , with the same coefficients. Indeed each of the four determinant factors must then use its x_{44} entry. Hence

$$\langle p, D \rangle = \langle p, p \rangle > 0.$$

Successive application of the classical Cayley identity gives

$$\|\det(X_n)^s\|^2 = \prod_{j=0}^{n-1} \frac{(s+j)!}{j!};$$

see [26]. The northwest variables and x_{44} are disjoint, so

$$\|p\|^2 = 24 \prod_{j=0}^2 \frac{(4+j)!}{j!} = 24883200, \quad \|D\|^2 = \prod_{j=0}^3 \frac{(4+j)!}{j!} = 870912000.$$

Their ratio is $1/35$. This calculates an isotypic projection; it does not assert a coefficient-one identity in an unspecified straightening normalization. $\square$

Theorem S12.2 (Sharp global exponent on the Jacobian open). *For every $R \in G_4^\circ$, the residual ideal on every Schur chart satisfies*

$$d^4 \in J.$$

Consequently

$$\widehat{\mathcal{N}}_R^5 = 0, \quad W_j = \emptyset \quad (j \geq 5).$$

At every projection-corank-three or projection-corank-four point the local nilpotency index is exactly five. Thus the global exponent five for the nilradical is sharp on G_4° .

Proof. The proof of Theorem S6.12, with Lemma S12.1 supplying its explicit nonvanishing, gives a polynomial identity

$$(\det X)^4 \in I_6(\gamma_R \text{Sym}^2 X) \quad \text{in } \mathbb{C}[x_{ij} : 1 \leq i, j \leq 4].$$

It is an identity in the coordinate ring of the entire matrix space, not an identity valid only in the local ring of its origin. On a Schur chart substitute $X = I_H \oplus T$. The maximal minors specialize to those of $[A_H, B_H(1_H \otimes T), C_H \text{Sym}^2 T]$, and $\det X$ specializes to $d = \det T$. Ideal membership is preserved by this ring map; no flatness hypothesis is involved. Hence $d^4 \in J$ on every chart. Since the failure ideal is dJ , the asserted nilpotence and vanishing of the fifth and subsequent graded layers follow.

The corank-three Schur-symbol exclusion of Lemma S10.4 gives $d^3 \notin J$. The corank-four exclusion in Theorem S6.12 gives the same nonmembership. Together with $d^4 \in J$ these prove exact local index five. The corank-four stratum is nonempty: S_R has dimension six and contains four-planes, whose projections to V are zero. $\square$

Proposition S12.3 (Admissibility of the ambient rank tables). *At projection corank two for $R \in G_4^\circ$, the ranks $a = \text{rank } A_H$ and $b = \text{rank } \bar{B}_H$ necessarily satisfy*

$$a \geq 2, \quad a + b \geq 5.$$

Thus the possible pairs on this open are contained in

$$\{(3, 3), (3, 2), (2, 4), (2, 3)\}.$$

The other rows of the coefficient-space tables remain valid statements about surjective quadratic quotients; they are not strata of the smooth, basepoint-free Jacobian open.

Proof. The dimension of $R \cap \text{Sym}^2 H$ is $3 - a$. If $a \leq 1$, the web contains two independent quadrics in the two H -variables. Along the line $\mathbb{P}(H^\perp)$, their gradients vanish. The Jacobian matrix therefore has rank at most two. Its determinant and all its first derivatives vanish there, contradicting smoothness of Y_R .

The dimension of $R \cap (\text{Sym}^2 H + H \otimes (V/H))$ is $(3 - a) + (4 - b)$. Consequently the restriction of the web to the line $\mathbb{P}((V/H)^*)$ has dimension $a + b - 3$. A vector space of binary quadrics of dimension at most one has a common zero over $\mathbb{C}$, contrary to basepoint freeness. Hence $a + b - 3 \geq 2$. The upper bounds on b are respectively three and four when $a = 3$ and $a = 2$, giving the displayed list. $\square$

S12.2. Two global minimal supports and a resolution. Regard Y_R as a subvariety of $\text{Gr}(3, V) = \mathbb{P}(V^*)$. Let $\mathcal{H}$ be the tautological rank-three bundle on it. Define

$$\tilde{Z}_R = \text{Tot Hom}(V, \mathcal{H}) = \{(H, T) : H \in Y_R, \text{im } T \subset H\}, \quad \pi : \tilde{Z}_R \longrightarrow \text{End}(V).$$

Let Z_R be its reduced image and let $D_2 \subset \text{End}(V)$ be the determinantal variety of matrices of rank at most two. These constructions refer to the fixed quotient γ_R , not to a choice of presentation of its colons.

Theorem S12.4 (Global minimal supports of the first two layers). *Let $R \in G_4^\circ$, and on the full matrix space put $J = I_6(\gamma_R \text{Sym}^2 T)$, $d = \det T$. Then*

$$\sqrt{(d, J)} = I(Z_R), \quad \sqrt{(d, J : d)} = I(D_2) = I_3(T).$$

The variety Z_R is irreducible of dimension fourteen, and π is a proper birational morphism from a smooth variety. The first layer has generic length one along its unique minimal support Z_R . For $R \in G_4^\dagger$, the second layer has generic length one along its unique minimal support D_2 . These assertions determine minimal supports and their generic lengths; they do not exclude additional embedded primes of either layer.

Proof. For a matrix of rank three with image H , the image of $\text{Sym}^2 T$ is $\text{Sym}^2 H$. The restriction of γ_R to this six-dimensional space has nontrivial kernel exactly when $R \cap \text{Sym}^2 H \neq 0$. If α defines H , contraction $\text{Sym}^2 V \rightarrow V$ by α has kernel $\text{Sym}^2 H$. The determinant of its restriction to R is, up to the fixed nonzero normalization, the Jacobian quartic evaluated at α . Thus the rank-three part of $V(J)$ is exactly the image of the rank-three part of $\tilde{Z}_R$.

Every matrix of rank at most two belongs to $V(J)$, since the symmetric square of its image has dimension at most three. It also belongs to Z_R : the projective space of hyperplanes containing its image has positive dimension and meets the quartic Y_R . Once such a hyperplane is chosen, the matrix is a limit of rank-three maps into that hyperplane. Invertible matrices do not belong to $V(J)$, since γ_R is surjective. This proves the first radical equality.

The total space $\tilde{Z}_R$ is a rank-twelve vector bundle over the smooth irreducible surface Y_R . It is smooth and irreducible of dimension fourteen. Its inclusion in $Y_R \times \text{End}(V)$ is closed, so its projection to matrix space is proper. A rank-three matrix determines its image hyperplane

uniquely; on this dense open the inverse is the regular image map to the Grassmannian. Hence π is birational.

We record the local scheme calculation at rank three, since it also controls the second layer. Write a Schur chart as $I_H \oplus t$. Smoothness of the Jacobian implies that A_H has rank five at a point of Y_R ; rank at most four would make every cofactor, and therefore every first derivative of its determinant, vanish. Eliminate an invertible five-minor. The residual ideal is generated by a local equation h of Y_R , the mixed entries tb_i , and a quadratic entry divisible by t^2 . At least one b_i is a unit. Otherwise a nonzero functional on the target cokernel, viewed as a quadratic form on V annihilating R , would vanish on $\text{Sym}^2 H + H \otimes (V/H)$, and would be proportional to α^2 . Then every quadric of R would vanish at $[\alpha]$, contradicting basepoint freeness. It follows that

$$J_{\text{loc}} = (t, h).$$

Thus the first layer is generically reduced on Z_R , whereas $(J : t)$ is the unit ideal at every rank-three point of Z_R . It is also the unit ideal off Z_R in rank three.

At a point of rank at most two, use the homogeneous transverse Schur coordinates. At corank two every residual minor has degree at least three, whereas d has degree two. At corank three the corresponding lower bounds are seven and three; at corank four they are twelve and four. Thus $d \notin J$ even after localization at the transverse origin. If $J : d$ contained a unit there, multiplication by its nonzero constant term would contradict this degree bound. Consequently the support of the second layer is exactly D_2 , proving the second radical equality. On the dense simple-contact, mixed-regular corank-two open supplied by $G_4^\dagger$, the inherited colon calculation gives the reduced transverse vertex for W_2 . Its generic length on the irreducible variety D_2 is therefore one. $\square$

Corollary S12.5 (Fibres of the boundary resolution). *For a matrix T of rank $r \leq 2$, the scheme-theoretic fibre is*

$$\pi^{-1}(T) = Y_R \cap \mathbb{P}((V/\text{im } T)^*).$$

For general matrices of rank two it consists of four reduced points; for general matrices of rank one it is a smooth plane quartic of genus three; over the zero matrix it is the quartic K3 surface Y_R .

Proof. The incidence equation is $\alpha T = 0$, which gives the displayed linear section scheme-theoretically. A general line meets the smooth quartic transversely in four points, and a general hyperplane section is a smooth plane quartic. Adjunction gives genus three in the latter case. At $T = 0$ the incidence equation imposes no condition. $\square$

S12.3. A cross-corank family with explicit components.

Theorem S12.6 (Contact points, plane quartics, and the K3 fibre). *Let $R \in G_4^\dagger$. Choose a general mixed-regular plane $H = \langle e_1, e_2 \rangle$, with a general basis, and extend it to a basis of V . On $B = \mathbb{A}_{u,v}^2$ set*

$$M(u, v) = \text{diag}(u, v, 0, 0).$$

There is a family of four-planes in $V \oplus S_R$ with this projection matrix. Its projection coranks are two on $uv \neq 0$, three on either punctured axis, and four at the origin.

Write $\alpha_1, \dots, \alpha_4$ for the dual homogeneous coordinates. The pullback of the resolution π is

$$\tilde{Z}_B = V(u\alpha_1, v\alpha_2) \subset B \times Y_R.$$

It has seven reduced irreducible components, no embedded components, and the following primary decomposition of ideal sheaves:

$$(u\alpha_1, v\alpha_2) = (u, v) \cap (u, \alpha_2) \cap (\alpha_1, v) \cap (\alpha_1, \alpha_2). \quad (\text{S12.1})$$

The last factor splits as the intersection of the ideals of the four constant contact-point sections. The other components are $\{0\} \times Y_R$, the $u = 0$ axis times $C_2 = Y_R \cap V(\alpha_2)$, and the $v = 0$ axis times $C_1 = Y_R \cap V(\alpha_1)$. The fibre Hilbert polynomials with respect to $\mathcal{O}_{Y_R}(1)$ are respectively

$$4 \quad (uv \neq 0), \quad 4n - 2 \quad (\text{a punctured axis}), \quad 2n^2 + 2 \quad ((u, v) = (0, 0)).$$

The family is flat on each of these strata, but not across them.

Proof. Choose a four-dimensional subspace $E \subset S_R$ and an isomorphism $\iota : V \rightarrow E$. The graphs

$$\mathcal{K}_{u,v} = \{(M(u, v)w, \iota(w)) : w \in V\}$$

are four-planes for every (u, v) , since their second component is injective. They define a morphism to the Grassmannian with the stated ranks. The resolution pulls back by the equations $\alpha M(u, v) = 0$, namely $u\alpha_1 = v\alpha_2 = 0$.

Generality of H makes the four points in $Y_R \cap V(\alpha_1, \alpha_2)$ distinct, and the general basis makes C_1, C_2 smooth plane quartics. They meet transversely at these four points. Locally at each of them, α_1, α_2 are regular parameters on the smooth surface. Equation (S12.1) is then the squarefree monomial identity for (ux, vy) . Away from the four points at least one of α_1, α_2 is a unit, and the same equality reduces to the elementary two-factor intersection. Thus the equality holds as ideal sheaves. The three surface components and four constant sections are distinct reduced irreducible components. The local squarefree description shows that there is no embedded component.

On the torus both linear equations remain; on a punctured axis only one remains; at the origin neither remains. This identifies the fibres. Their Hilbert polynomials are those of four points, a plane quartic of genus three, and a degree-four K3 surface. The restrictions to the displayed strata are products, proving their flatness. The change of Hilbert polynomial proves failure of flatness across them. $\square$

Proposition S12.7 (Intrinsic graded fibre algebra across coranks). *Pull back $\text{gr}_{\widehat{N}_R} \mathcal{O}_{\widehat{D}_R}$ along the preceding family of reduced points. After a local trivialization of the conormal line, its algebra on a geometric fibre is*

$$\kappa[\epsilon]/(\epsilon^3) \quad (uv \neq 0), \quad \kappa[\epsilon]/(\epsilon^5) \quad (uv = 0), \quad \deg \epsilon = 1.$$

Thus degrees three and four appear in an actual family passing from projection corank two to three to four.

Proof. Every positive graded piece is a cyclic quotient of the reduced local ring, with the conormal-line twist, and its products are the canonical quotient products of Theorem S6.2. Its residue-field fibre is therefore either zero or one-dimensional. The mixed-regular corank-two calculation gives local nilpotency index three on the torus. Theorem S12.2 gives index five on the punctured axes and at the origin. The nonvanishing pieces are consecutive, and quotient multiplication takes the chosen generator to its powers, giving the stated algebras. $\square$

Remark S12.8 (Two different base-change operations). The preceding proposition pulls back an already formed intrinsic graded algebra. It does not form the nilradical after pulling back the failure scheme to B . In fact $\det M(u, v) = 0$ identically, so the latter pullback loses its transverse nilpotent information. Likewise, the primary decomposition in (S12.1) is the primary decomposition of the incidence-resolution family, not a claim that all transverse primary components of W_3 and W_4 have been classified. This distinction is essential in a cross-stratum specialization, where kernel formation need not commute with pullback.

S12.4. An explicit infinite family of proper quotient maps. The next result gives an exact primary law and a sharp exponent in all symmetric degrees for quotients induced by a proper quotient of the underlying vector space. Its maximal-minor identity is classical: we use, and explicitly credit, Bruns–Vasconcelos [25, Theorem 1(2)]. The point here is to compute the resulting failure ideal, residual tower, and canonical embedded length in the notation of the present paper.

Theorem S12.9 (Primary law for quotient-induced symmetric powers). *Let $1 \leq k < e$, $r \geq 1$, and let $\rho : V \twoheadrightarrow W$ be a surjection with $\dim V = e$, $\dim W = k$. Set*

$$\gamma = \text{Sym}^r \rho, \quad A = \mathbb{C}[\text{End}(V)], \quad d = \det T, \quad P = I_k(\rho T), \quad a = \binom{k+r-1}{r-1}, \quad h = e - k + 1.$$

If $s = \dim \text{Sym}^r W$ and $J = I_s(\gamma \text{Sym}^r T)$, then

$$J = P^a = P^{(a)}, \quad (J : d^q) = P^{\max(a-q, 0)}, \quad \min\{q : d^q \in J\} = a.$$

Moreover

$$dJ = (d) \cap P^{a+1} \tag{S12.2}$$

is an irredundant primary decomposition. The nilradical of A/dJ has exact index $a + 1$. Its reduced determinant component has generic length one. Its only embedded associated prime is P , and the canonical local torsion length there is

$$\mu_P(A/dJ) = \binom{a+h-1}{h}.$$

For $1 \leq j \leq a$, the j -th positive graded layer is

$$A/(d, P^{a-j+1})(-je),$$

and its length at P is $\binom{a-j+h-1}{h-1}$.

Proof. Choose bases so that ρT consists of the first k rows of a generic $e \times e$ matrix. The maximal-minor identity of Bruns–Vasconcelos gives $J = P^a$ in characteristic zero. We recall why the powers of this maximal-minor ideal are symbolic, so that the primary assertions and colon calculation are explicit.

In the coordinate ring of a generic $k \times e$ matrix, the Cauchy summands are indexed by partitions λ of length at most k . A highest vector is the product of principal minors

$$v_\lambda = \prod_{j=1}^k \Delta_j^{\lambda_j - \lambda_{j+1}}, \quad \lambda_{k+1} = 0.$$

At the generic point of P , the minors Δ_j , $j < k$, are units, whereas Δ_k has order one. This follows also by eliminating an invertible $(k-1)$ -minor: the remaining maximal minors are the parameters of a regular local ring of dimension $h = e - k + 1$. Thus v_λ has order λ_k . The symbolic power is stable under the two general linear groups, so complete reducibility shows that it consists exactly of the summands with $\lambda_k \geq a$. Their highest vectors are divisible by Δ_k^a , and hence belong to P^a . The reverse containment is automatic. This proves $P^{(a)} = P^a$, and the same proof applies to every positive power. Adjoining the remaining rows is a polynomial extension and preserves these identities and primaryness.

In A_P , the full determinant d is a parameter: after the same elimination its linear term in the remaining maximal-minor parameters is nonzero, since the remaining generic rows can complete a rank- $(k-1)$ matrix to rank $e-1$. Hence

$$(P^a A_P : d^q) = P^{\max(a-q, 0)} A_P.$$

This also follows from the polynomial associated graded ring of the regular local ring. Contracting to A proves the asserted colons and exact exponent. In particular $(d) \cap P^{a+1} = d(P^{a+1} : d) = dP^a$. The determinant ideal is prime, P^{a+1} is P -primary, and their distinct radicals satisfy $(d) \subsetneq P$ because $h \geq 2$. Neither component can be removed, so the decomposition is irredundant and determines its associated primes.

Locally at P , write $\mathfrak{m} = PA_P$. Since d is a parameter and $h - 1 \geq 1$, the quotient $A_P/(d)$ has no finite-length torsion. Therefore the local torsion of $A_P/(d\mathfrak{m}^a)$ is

$$(d)/(d\mathfrak{m}^a) \simeq A_P/\mathfrak{m}^a,$$

of length $\binom{a+h-1}{h}$. Localizing at (d) makes P the unit ideal, giving generic determinant length one. The residual-colon formula gives the displayed graded layers. Modulo the parameter d , their lengths are the numbers of monomials of degree less than $a - j + 1$ in $h - 1$ variables, namely $\binom{a-j+h-1}{h-1}$. Their sum is the preceding embedded length. This identifies the canonical embedded torsion, not the noncanonical size of an arbitrarily chosen embedded primary component. $\square$

S13. DEFORMATION TO THE NORMAL CONE: STANDARD MECHANISM AND THE COMPUTED SPECIAL FIBRE

The passage from an ideal to its extended Rees algebra is the standard deformation-to-the-normal-cone construction; see [17, §5.1] and [19, Tag 062Z]. We use it here only as the canonical mechanism that packages the manuscript-specific graded algebra computed in Sections S9 and S10.

Proposition S13.1 (Intrinsic deformation to the normal cone). *Let $\widehat{\mathcal{N}}_R$ be the intrinsic nilradical of $\widehat{D}_R$. On an affine open $\text{Spec } A \subset \widehat{D}_R$, with nilradical $N \subset A$, put*

$$\mathcal{R}_N = A[t, Nt^{-1}] \subset A[t, t^{-1}].$$

These algebras glue and define a flat family

$$\mathfrak{D}_R \longrightarrow \mathbb{A}^1.$$

Over $\mathbb{G}_m$ it is $\widehat{D}_R \times \mathbb{G}_m$, while its special fibre is

$$\mathfrak{D}_{R,0} \simeq \text{Spec}_{\widehat{\Delta}_R} \text{gr}_{\widehat{\mathcal{N}}_R} \mathcal{O}_{\widehat{D}_R} = C_{\widehat{\Delta}_R/\widehat{D}_R}.$$

The construction is functorial for isomorphisms of the abstract failure scheme.

Proof. The extended Rees algebra is the standard construction. Multiplication by t is injective because $\mathcal{R}_N$ is a subring of $A[t, t^{-1}]$. Hence it is torsion-free over the PID $\mathbb{C}[t]$, and therefore flat. Inverting t gives $A[t, t^{-1}]$. Modulo t , the class of nt^{-j} , $n \in N^j$, depends only on $n \bmod N^{j+1}$, giving

$$\mathcal{R}_N/(t) \simeq \bigoplus_{j \geq 0} N^j/N^{j+1}.$$

Localization compatibility glues the construction. Since N is the ideal of the reduction $\widehat{\Delta}_R$ in $\widehat{D}_R$, the special fibre is the normal cone. The nilradical is intrinsic, so the construction is functorial. $\square$

The specific geometric content is not the standard construction in Proposition S13.1, but its bounded and stratified special fibre:

$$\text{gr}_{\widehat{\mathcal{N}}_R} \mathcal{O}_{\widehat{D}_R} = \mathcal{O}_{\widehat{\Delta}_R} \oplus \bigoplus_{j=1}^5 \mathcal{L}^{\otimes j} \otimes \mathcal{O}_{W_j},$$

with the quotient multiplication (S6.4). Projection corank two has the mixed-kernel atlas of Theorem S6.4; all A_H -rank drops have the finite valuation bounds of Proposition S6.10; projection

corank three has exact index five on G_4° , by Theorems S6.11 and S12.2; and projection corank four also has exact index five by Theorem S6.12. The five-or-six alternative for general coefficient data outside the smooth-web open is not a dichotomy on G_4° . On this open the summand with $j = 5$ in the displayed universal range is zero.

Flattening is used only to organize coefficient specializations of these explicitly presented finite modules. We use the standard flattening-stratification formalism in [20, Tag 052F]; no novelty is claimed for that formalism either.

Remark S13.2 (Novelty boundary). The standard ingredients in this section are the extended Rees algebra, deformation to the normal cone, and flattening stratification. The manuscript-specific statements are the Fitting factorization, the global determinant-power bound $d^5 \in J$, the nine mixed-kernel normal forms and their graded supports, the corank-three Schur-type exclusion, the corank-four Jacobian-covariant calculation, the collision primary laws, and the polarized reconstruction theorem. These roles are kept separate in the statements and in the referee response.

S14. EXACT SEPARATING PENCILS

The smooth integral witness C_* of (8.2) provides a direct arithmetic check inside the geometric locus. Its two residual linear forms have determinant $27787/432$. We retain below both this exact check and the five-coordinate square-web calculation. Neither is needed for the structural support proof of Theorem 5.15.

Choose a basis x_1, x_2, x_3, x_4 of V , and order the monomial basis of W as

$$(b_1, \dots, b_{10}) = (x_1^2, x_1x_2, x_1x_3, x_1x_4, x_2^2, x_2x_3, x_2x_4, x_3^2, x_3x_4, x_4^2).$$

For increasing indices put $e_{ijkl} = b_i \wedge b_j \wedge b_k \wedge b_l$. Let

$$w_* = e_{1,5,8,10}.$$

The Jacobian map

$$C : E \longrightarrow \det V \otimes \mathrm{Sym}^4 V, \quad q_1 \wedge \dots \wedge q_4 \longmapsto \det(\partial q_i / \partial x_j)$$

is equivariant, with the determinant factor understood in the chosen basis. It is nonzero, since $Cw_* = 16x_1x_2x_3x_4$. Its kernel is E_{175} .

Lemma S14.1 (An exact separating pencil). *The projection of w_* to E_{35} is*

$$Qw_* = \frac{1}{3}e_{1,5,8,10} - \frac{2}{3}e_{1,6,7,9} - \frac{2}{3}e_{2,3,6,10} + \frac{2}{3}e_{2,4,7,8} + \frac{2}{3}e_{3,4,5,9}. \quad (\text{S14.1})$$

On the pencil $aPw_ + bQw_*$, two Pluecker quadrics restrict to*

$$-\frac{2}{9}(b-a)(b+2a), \quad -\frac{4}{9}(b-a)^2. \quad (\text{S14.2})$$

Consequently $\mathcal{L}_ \cap \mathrm{Gr}(4, W)$ is the single reduced point $[w_*]$.*

Proof. Use the Fischer Hermitian forms: a degree- r monomial x^α has squared norm $\alpha!$, and the exterior basis has the induced product norm. The determinant factor has norm one. These forms are unitary invariant. Hence the adjoint C^* is unitary equivariant and, by complexification, is $GL(V)$ -equivariant. Schur's lemma gives $CC^* = c \mathrm{id}$. The only exterior basis vectors contributing to the monomial $x_1x_2x_3x_4$ are the five in (S14.1). Their Jacobian coefficients are respectively 16, -4 , -4 , 4, 4, and their squared norms are 16, 2, 2, 2, 2. The quartic monomial has squared norm one. It follows that

$$c = \frac{16^2}{16} + 4 \frac{4^2}{2} = 48, \quad Q = \frac{C^*C}{48}.$$

Applying this identity to w_* gives (S14.1).

Extend the Pluecker coordinates $z_{i_1 i_2 i_3 i_4}$ alternatingly, with repeated-index coordinates zero. For an ordered triple I and an increasing five-tuple $J = (j_1, \dots, j_5)$, use the quadratic relation

$$F_{I,J}(z) = \sum_{k=1}^5 (-1)^{k-1} z_{I,j_k} z_{J \setminus j_k}.$$

Substitution of (S14.1), together with $Pw_* = w_* - Qw_*$, gives (S14.2) for

$$\begin{aligned} I &= (1, 5, 8), & J &= (1, 6, 7, 9, 10), \\ I &= (1, 6, 7), & J &= (2, 3, 6, 9, 10), \end{aligned}$$

respectively. These two formulas can thus be checked with five nonzero coordinates. Their linear quotients after removing $b - a$ are independent. As ideal sheaves on $\mathbb{P}^1$, they generate $(b - a)$: the two remaining linear forms have no common projective zero. Every other restricted Pluecker quadric vanishes at $[1 : 1]$, so the full restricted ideal sheaf is also $(b - a)$. This proves both uniqueness and reducedness, including at the endpoints and at infinity. $\square$

Remark S14.2 (The same condition at the smooth integral witness). The smooth web C_* of (8.2) also satisfies the rank-two condition. For its Pluecker vector, in the monomial basis above, the same two quadrics restrict respectively to

$$\frac{(a-b)(339a+89b)}{24}, \quad \frac{37(a-b)(31a+17b)}{72}.$$

The determinant of their linear quotients by $b - a$ is $27787/432 \neq 0$. These identities follow by taking the four-by-four minors of the displayed integral coefficient matrix (8.2), applying $Q = C^*C/48$, and substituting in $F_{I,J}$. They are reproduced by `checks/revision132_exact.py`, together with equivariance for all sixteen matrix-unit generators of $\mathfrak{gl}_4$. The smoothness and basepoint-freeness of this same web are certified independently in Lemma 8.1. The smooth witness checks the geometric family directly. The five-coordinate example gives a transparent normalization check, while the support theorem proves the assertion for every smooth web.

S15. WEIGHTED DETERMINANTAL FLAGS AND THE CLASSICAL ORDER PROFILE

We record explicitly how the retained weighted-matrix calculation fits the classical determinantal theory. This comparison concerns the generic matrix only; the fixed-tensor theorem of §S7 was proved independently.

Let $A = (a_{ij})$ be a generic $q \times p$ matrix over a characteristic-zero field k , with $p \geq q$, and put $R_0 = k[a_{ij}]$, $I_0(A) = R_0$. For a nonincreasing sequence $\rho = (r_1, \dots, r_s)$ write

$$I^\rho = \prod_{\nu=1}^s I_{r_\nu}(A), \quad \gamma_j(\rho) = \sum_{\nu=1}^s \max(r_\nu - j + 1, 0).$$

The classical formula is

$$I^\rho = \bigcap_{j=1}^{r_1} I_j(A)^{(\gamma_j(\rho))}. \quad (\text{S15.1})$$

Here the superscript in parentheses denotes symbolic power. See De Concini–Eisenbud–Procesi [8], and its statement as Theorem 2.4 in Bruns–Conca [5], together with Proposition 2.2 there for the symbolic-order basis. The characteristic hypotheses and attribution are stated on printed pages 13–14 of that exposition.

Proposition S15.1 (Coefficientwise suspension). *In $R_0[t]$, put*

$$F = t \sum_{i=0}^q t^{q-i} I_i(A), \quad P_j = (t, I_j(A)).$$

Then

$$F = (t) \cap \bigcap_{j=1}^q P_j^{(q-j+2)}. \quad (\text{S15.2})$$

This is an irredundant primary decomposition. At a point with $t = 0$ where A has rank a , the nilradical of the corresponding local quotient has index $q - a + 1$.

Proof. In the regular local ring $(R_0)_{I_j}$, the symbolic order of an i -minor is $\max(i - j + 1, 0)$. Adjoining t gives the coefficient rule

$$\sum_{u \geq 0} t^u f_u \in P_j^{(b)} \iff f_u \in I_j^{(\max(b-u, 0))} \text{ for every } u. \quad (\text{S15.3})$$

For precision, the associated graded ring of the maximal ideal $(t, I_j(R_0)_{I_j})$ is a polynomial ring over its residue field; its homogeneous t -degrees are independent. The order of the displayed sum is therefore the minimum of $u + \text{ord}_{I_j}(f_u)$. Contracting the maximal-ideal power proves (S15.3), including the assertion that each contracted symbolic power is primary.

Intersecting with (t) deletes coefficient degree zero. For $1 \leq u \leq q$, put $k = q + 1 - u$. The remaining coefficient conditions on the right of (S15.2) are

$$f_u \in \bigcap_{j=1}^k I_j^{(k-j+1)} = I_k,$$

the one-factor case $\rho = (k)$ of (S15.1). For $u \geq q + 1$ there is no condition. These are exactly the coefficients of F , proving the equality.

For irredundancy, let d_{j-1} be a fixed $(j - 1)$ -minor, with $d_0 = 1$. The element $t^{q-j+1}d_{j-1}^2$ belongs to every component other than $P_j^{(q-j+2)}$. Indeed, its order along P_j is $q - j + 1$, while at P_k for $k < j$ its order is $q - j + 1 + 2(j - k) \geq q - k + 2$, and for $k > j$ its t -order suffices. The element d_q^2 witnesses the necessity of (t) . The associated primes are therefore exactly the displayed radicals.

After inverting an a -minor and Schur reducing, the localized coefficient ideal contains t^{q-a+1} , whereas t^{q-a} has coefficient one in degree $q - a$, where membership would require a one-minor of the residual matrix. That coefficient ideal is proper at the point. Thus the radical (t) has the asserted exact index. $\square$

The exponent $q - j + 2$ is therefore the scalar-variable suspension of the classical minor order $k - j + 1$, taken coefficientwise. This is a direct consequence of the classical profile plus (S15.3); it is not a new straightening theorem. The presentation $I_q[A, tI_q] = \sum_i t^{q-i} I_i(A)$ can also be viewed as substituting a scalar identity block in a larger generic matrix. Such substitution alone does not justify preservation of primary components or flatness. The coefficient proof is what supplies the exact intersection here, while the ideal $\delta \mathfrak{m} + (g_0, \dots, g_4)$ in §S7 requires its separate fixed-tensor analysis.

S16. AN EXPLICIT SYSTEM AND EXACT CERTIFICATE SPECIFICATIONS

The rational system used throughout is the following. It is the same relation system as in the preceding manuscript's exact smoothness example, now used for a differential and a higher-corank

calculation.

$$\begin{aligned}
q_0 &= x_0^2 + 2x_0x_1 - 2x_0x_2 + 2x_0x_3 + x_1x_2 + x_1x_3 + x_2x_3 \\
q_1 &= -x_0x_1 - x_0x_3 + x_1^2 - x_1x_2 - x_1x_3 - 2x_2x_3 \\
q_2 &= 2x_0x_1 - x_0x_2 - 2x_0x_3 + 2x_1x_2 + x_2^2 + x_2x_3 \\
q_3 &= x_0x_2 + x_1x_2 - 2x_2x_3 + x_3^2
\end{aligned}$$

For a compact complete specification of its quartic, list the 35 degree-four exponent tuples in descending lexicographic order, starting with $(4, 0, 0, 0)$ and ending with $(0, 0, 0, 4)$. The coefficients of $F = \det(\partial q_i / \partial x_j)$ in that order are the rows of

$$\begin{pmatrix} -8 & -4 & -20 & 16 & 20 & 0 & -28 \\ -2 & -36 & 40 & 20 & -50 & -20 & -26 \\ 72 & -6 & 0 & -66 & -6 & 30 & 4 \\ -14 & -4 & -6 & -12 & 28 & 18 & -56 \\ 28 & 6 & -18 & 8 & -2 & 16 & 2 \end{pmatrix}.$$

Thus this display, together with the fixed monomial order, specifies every coefficient, including zeros. Alternatively, the determinant of the displayed four quadrics is its coordinate-free recipe.

For any list of homogeneous forms $f_1, \dots, f_s$ define the degree- d Macaulay coefficient matrix $\mathcal{M}_d(f)$ as follows. Rows are all degree- d monomials in the same descending lexicographic order. Columns are the forms $x^\beta f_i$, with i outermost and all β of degree $d - \deg f_i$ in descending lexicographic order innermost. The entry in row α is the coefficient of x^α . All entries for this example are integers.

Matrix	Size	Certified rank	Minor modulo 101
$[F, G_{01}, \dots, G_{36}]$	35×25	25	41
$\mathcal{M}_5(q_0, q_1, q_2, q_3)$	56×80	56	44
$\mathcal{M}_9(F_{x_0}, F_{x_1}, F_{x_2}, F_{x_3})$	220×336	220	79

The differential minor consists of the first 25 rows and all columns; its full integer value is (8.4). The following deterministic rule specifies the other two minors without printing large matrices. Reduce the displayed coefficient matrix modulo 101. Scan columns from left to right. At the next pivot step choose the first remaining row with nonzero entry, exchange it with the next pivot row, normalize that row, and eliminate below. Skip a column if there is no nonzero remaining entry. Record the original row indices in the order selected and the pivot columns. The square submatrix of the *original*, unreduced matrix with exactly these row and column orders has the residue in the final column of the table. Full recorded index lists, along with the coefficient-generating program, are included in `evidence/K3_CERTIFICATES.json` and `checks/exact_k3.py`.

Every step is finite-field arithmetic on integer data. A nonzero residue of a specified square minor proves that integer minor is nonzero, so it proves a rank statement over $\mathbb{Q}$, not merely a rank in one numerical specialization. The Macaulay rank statements imply the homogeneous ideal containments used in Lemma 8.1. The exact integer determinant is also recomputed independently with rational-domain elimination. No floating-point rank, tolerance, or unverified smoothness inference is used.

For the corank-two quotient in (S7.1), one may reproduce its quotient map by multiplying the six-dimensional quadratic quotient by

$$\begin{pmatrix} 0 & -1/2 & -1 & 1 & 0 & 0 \\ 0 & -1/2 & -1 & 0 & 1 & 0 \\ 0 & -3/2 & -2 & 0 & 0 & 1 \end{pmatrix}.$$

This matrix annihilates the three columns of $\text{Sym}^2 H$ and has rank three. It yields the displayed B_H, C_H and the quartic (S7.2) by ordinary multiplication. The factor 2 in the coefficient of uv is essential: it records the square of $ux_2 + vx_3$, not a divided-power convention.

An additional exact regression test uses the normalized matrix (S7.3) with

$$C = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 1/2 & -3 \end{pmatrix}.$$

Then $h(u, v) = uv(u-v)(u-2v)$, so the four ideals are (a, b) , (c, d) , $(a-c, b-d)$, and $(a-2c, b-2d)$. Elimination computes both sides of (S5.2) over $\mathbb{Q}$, and mutual Gröbner reduction verifies their equality. It also checks that δ^2 is nonzero, that every element of $\mathfrak{m}\delta^2$ vanishes, and that $\delta^3 = 0$. This test checks a concrete instance; the general coefficient-ring proof is Lemma S7.4 and Theorem S7.5, not an extrapolation from that instance.

REFERENCES

- [1] E. Arrondo and I. Sols, *On congruences of lines in the projective space* (Chapter 6 written in collaboration with M. Pedreira), *Mém. Soc. Math. France (N.S.)* **50** (1992), 1–96. doi:10.24033/msmf.363.
- [2] E. Ballico, On the failure locus of higher order properties of embeddings in projective spaces, *Math. Nachr.* **163** (1993), 5–13. doi:10.1002/mana.19931630102.
- [3] E. Ballico, On the failure cycles for the quadratic normality of a projective variety, *Pacific J. Math.* **172** (1996), 307–313.
- [4] D. Bayer and D. Eisenbud, Ribbons and their canonical embeddings, *Trans. Amer. Math. Soc.* **347** (1995), 719–756. doi:10.1090/S0002-9947-1995-1273472-3.
- [5] W. Bruns and A. Conca, Gröbner bases and determinantal ideals, in *Commutative Algebra, Singularities and Computer Algebra*, NATO Sci. Ser. II Math. Phys. Chem. 115, Kluwer, 2003, 9–66; author preprint arXiv:math/0302058.
- [6] W. Bruns and U. Vetter, *Determinantal Rings*, Lecture Notes in Mathematics 1327, Springer, 1988.
- [7] F. Cossec, Reye congruences, *Trans. Amer. Math. Soc.* **280** (1983), 737–751. doi:10.1090/S0002-9947-1983-0716848-4.
- [8] C. De Concini, D. Eisenbud, and C. Procesi, Young diagrams and determinantal varieties, *Invent. Math.* **56** (1980), 129–165. doi:10.1007/BF01392548.
- [9] I. Dolgachev and S. Kondō, *Enriques Surfaces II*, Springer, 2025, chapter *Reye Congruences*, 67–141. doi:10.1007/978-981-96-1513-1_2.
- [10] I. Dolgachev and I. Reider, On rank 2 vector bundles with $c_1^2 = 10$ and $c_2 = 3$ on Enriques surfaces, in *Algebraic Geometry*, Lecture Notes in Mathematics 1479, Springer, 1991, 39–49. doi:10.1007/BFb0086262.
- [11] J.-M. Drézet, Primitive multiple schemes, *Eur. J. Math.* **7** (2021), 985–1045. doi:10.1007/s40879-020-00447-4.
- [12] D. Huybrechts, *Lectures on K3 Surfaces*, Cambridge Studies in Advanced Mathematics 158, Cambridge University Press, 2016. doi:10.1017/CBO9781316594193.
- [13] C. Ingalls and A. Kuznetsov, On nodal Enriques surfaces and quartic double solids, *Math. Ann.* **361** (2015), 107–133. doi:10.1007/s00208-014-1066-y.
- [14] J. Kollár, *Rational Curves on Algebraic Varieties*, *Ergebnisse der Mathematik und ihrer Grenzgebiete (3)*, vol. 32, Springer, 1996, Chapter IV, §5.
- [15] N. Manolache, Cohen–Macaulay nilpotent schemes, arXiv:math/0312514 (2003).
- [16] G. Martin, G. Mezzedimi, and D. C. Veniani, Nodal Enriques surfaces are Reye congruences, *J. Reine Angew. Math.* **808** (2024), 49–65. doi:10.1515/crelle-2023-0092.
- [17] W. Fulton, *Intersection Theory*, 2nd ed., *Ergebnisse der Mathematik und ihrer Grenzgebiete (3)*, vol. 2, Springer, 1998, §5.1.
- [18] W. Fulton and J. Harris, *Representation Theory: A First Course*, Graduate Texts in Mathematics 129, Springer, 1991.
- [19] The Stacks Project Authors, *The Stacks Project*, Section 31.20, “The normal cone of an immersion,” Tag 062Z, <https://stacks.math.columbia.edu/tag/062Z>.
- [20] The Stacks Project Authors, *The Stacks Project*, Section 38.21, “Flattening stratifications,” Tag 052F, <https://stacks.math.columbia.edu/tag/052F>.
- [21] The Stacks Project Authors, *The Stacks Project*, Section 31.7, “Relative assassin,” Tag 05AS, <https://stacks.math.columbia.edu/tag/05AS>.

- [22] The Stacks Project Authors, *The Stacks Project*, Section 37.18, “Flat modules and relative assassins,” Tag 0GSF; in particular Lemma 37.18.5, Tag 0GSJ.
- [23] The Stacks Project Authors, *The Stacks Project*, Section 38.18, “A criterion for purity,” Tag 05L2.
- [24] The Stacks Project Authors, *The Stacks Project*, Section 37.25, “Relative assassins,” in particular Lemma 37.25.5, Tag 05KR, <https://stacks.math.columbia.edu/tag/05KR>.
- [25] W. Bruns and W. V. Vasconcelos, Minors of symmetric and exterior powers, *J. Pure Appl. Algebra* **179** (2003), 235–240. doi:10.1016/S0022-4049(02)00115-9; arXiv:math/0111174.
- [26] S. Caracciolo, A. D. Sokal, and A. Sportiello, Algebraic/combinatorial proofs of Cayley-type identities for derivatives of determinants and pfaffians, *Adv. Appl. Math.* **50** (2013), 474–594. doi:10.1016/j.aam.2012.12.001; arXiv:1105.6270.
- [27] The Stacks Project Authors, *The Stacks Project*, Lemma 10.118.2, “Generic freeness,” Tag 051S, <https://stacks.math.columbia.edu/tag/051S>.
- [28] The Stacks Project Authors, *The Stacks Project*, Sections 37.26–37.27, “Reduced fibres” and “Irreducible components of fibres,” <https://stacks.math.columbia.edu/tag/0553>.
- [29] The Stacks Project Authors, *The Stacks Project*, Proposition 35.5.2 (Tag 023T) and Section 35.6 (Tag 0CDQ), fpqc and Galois descent of quasi-coherent modules, accessed September 23, 2026. <https://stacks.math.columbia.edu/tag/023T>.
- [30] A. Boralevi, A note on secants of Grassmannians, *Rend. Istit. Mat. Univ. Trieste* **45** (2013), 67–72. <https://rendiconti.dmi.units.it/volumi/45/011.pdf>.
- [31] E. Carlini, Reducing the number of variables of a polynomial, in *Algebraic Geometry and Geometric Modeling*, Springer, 2006, 237–247; author preprint arXiv:math/0507531, Proposition 1.
- [32] J. M. Landsberg and J. Weyman, On the ideals and singularities of secant varieties of Segre varieties, author manuscript (January 2007), Definition 1 and Theorem 3.1. <https://people.tamu.edu/~jml/1-07LWsecseg.pdf>.
- [33] The Stacks Project Authors, *The Stacks Project*, Section 29.36, Unramified morphisms, Tag 02G3, accessed September 23, 2026. <https://stacks.math.columbia.edu/tag/02G3>.
- [34] J. M. Landsberg and G. Ottaviani, *Equations for secant varieties via vector bundles*, arXiv:1010.1825v1 (2010), §6.1; subsequently merged into [35].
- [35] J. M. Landsberg and G. Ottaviani, Equations for secant varieties of Veronese and other varieties, *Ann. Mat. Pura Appl.* (4) **192** (2013), 569–606, doi:10.1007/s10231-011-0238-6; arXiv:1111.4567v1, §10.1.
- [36] J. Sheridan, Divisor varieties of symmetric products, *Int. Math. Res. Not. IMRN* **2022** (2022), no. 13, 9830–9863, doi:10.1093/imrn/rnaa313; arXiv:1906.05465v1, §2.4, Definitions 2.10–2.11.
- [37] S.-J. Cheng and W. Wang, Howe duality for Lie superalgebras, *Compositio Math.* **128** (2001), 55–94; arXiv:math/0008093v1, §2 and Theorem 3.5.
- [38] H. De Bie, D. Eelbode, and M. Roels, The harmonic transvector algebra in two vector variables, *J. Algebra* **473** (2017), 247–282; arXiv:1510.06566v3, §1, equations (1)–(3), and Theorem 2.1.
- [39] R. Kupferman and A. Shachar, A geometric perspective on the Piola identity in Riemannian settings, *J. Geom. Mech.* **11** (2019), 59–76; arXiv:1805.12365v1, equation (1.1) and §3.1.